The Boundaries of the Farm and the Death of Small Banks

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Abstract

How does firm consolidation affect the structure of banking markets? Using exogenous variation in subsidy payments, I show that subsidy payments nearly double farm concentration without increasing deposits, land values, investment, or production. As farms consolidate, large banks gain a greater share of local deposits and expand their branch networks, while small bank mergers increase. Local banking markets become more concentrated, and the resulting decrease in competition spills over into consumer credit markets, where households earn less on deposits and face higher borrowing costs. These findings show how consolidation in the real economy can alter banking markets and credit prices.

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1 Introduction

Firm consolidation has been widespread, with concentration increasing in more than 75% of U.S. industries over the past few decades. Prior research on bank consolidation has focused on drivers within banking markets, including deregulation, technological change, and economies of scale, as well as on the effects of bank consolidation on firms. However, banks do not operate independently of the firms they finance. Large and small banks specialize in lending to borrowers of different sizes, so changes in the size distribution of firms may affect which banks are best positioned to finance them and, in turn, the structure of local banking markets.[1] In this paper, I show how changes in borrower concentration affect the composition of local banking markets and spill over into higher prices for consumer financial products.

In product markets, buyer consolidation leads suppliers to consolidate. Whether borrower consolidation has similar implications for banks, the suppliers of credit, is less clear. As borrowers consolidate and expand in scale, banks with a comparative advantage in serving larger borrowers may gain market share, increasing local banking concentration. However, existing lending relationships may limit this reallocation. Incumbent lenders have informational advantages that make it costly for borrowers to switch lenders and for new banks to enter consolidating markets, so borrowers that expand may remain with the banks that initially financed them.[2]

I use the 1996 Federal Agriculture Improvement and Reform (FAIR) Act to study whether changes in the size distribution of borrowers affect local banking markets. The FAIR Act replaced price-dependent subsidies with fixed payments that followed a predetermined annual schedule and were based on each farm's average acreage planted in program crops (wheat, rice, cotton, corn, and other feed grains) between 1981 and 1985. This historical acreage, known as a farm's base acres, did not change with current planting decisions. Because larger farms mechanically operated more base acres, they received larger subsidy payments.

These payments provide a setting to study how government policies that allocate support according to firm size affect the distribution of firms and, in turn, financial intermediation. Farms have limited internal liquidity (Thakor, 2023), so predictable payments may help

[1]For literature studying industry consolidation, see Grullon, Larkin, and Michaely (2019) and Kwon, Ma, and Zimmermann (2024). For the drivers of bank consolidation, see Berger, Demsetz, and Strahan (1999); Jayaratne and Strahan (1996); and Kroszner and Strahan (1999). For its effects on firms, see Strahan and Weston (1998); Cetorelli and Strahan (2006); Huber (2021); and Saidi and Streitz (2021). For lending by bank size, see Petersen and Rajan (1994); Stein (2002); Boot and Thakor (2000); and Berger et al. (2005).

[2]For literature studying buyer and supplier consolidation, see Galbraith (1952); Fee and Thomas (2004); Becker and Thomas (2009); and Bhattacharyya and Nain (2011). For lending relationships, see Diamond (1984); Rajan (1992); Boot (2000); Dell'Ariccia (2001); and Brennecke, Jacewitz, and Pogach (2025).

finance the upfront costs of expansion. Larger farms also benefit from economies of scale in program crops, which decrease unit production costs and allow them to retain more of their subsidy payments (Duffy, 2009; Langemeier, 2013). These differences in payments and production costs between large and small farms may allow larger farms to acquire the land of smaller, higher-cost farms. A *Washington Post* article at the time documented that these payments encouraged owners of large farms to expand and outbid their smaller neighbors for land.[3] Given farms' reliance on local lenders, changes in the size distribution of farms may also affect the structure of local banking markets (Akhavein, Goldberg, and White, 2004).

I use a continuous difference-in-differences design based on variation in predicted subsidy payments across counties and over time. I construct predicted subsidy payments by combining each county's historical crop shares with national crop-level subsidy allocations. The historical crop shares initially reflect planting decisions made before farmers could anticipate the FAIR Act, and federal policy sets the national crop-level subsidy allocations. I find that subsidy payments increase local banking concentration. By the end of the program in 2014, a within-county one-standard-deviation increase in payments increases the bank deposits Herfindahl-Hirschman Index (HHI) by about 390 points, nearly twice the 200-point increase that triggers further review of a bank merger for potentially being anticompetitive (U.S. Department of Justice, Antitrust Division, 1995).

I then determine whether farm consolidation is the primary channel through which subsidy payments increase banking concentration. First, subsidy payments may encourage larger farms, which receive greater payments and benefit from lower production costs, to acquire land from smaller farms, increasing farm concentration. In addition to farm consolidation, I consider three alternative channels. Second, payments deposited in local banks may increase deposits per capita and the funds available for lending (Gilje, Loutskina, and Strahan, 2016). Third, payments may increase demand for eligible land, increasing land values and the collateral available to borrowers. Last, farmers may use payments to increase production or invest in equipment, increasing credit demand.

Farm concentration is the only one of these outcomes that increases persistently with subsidy payments. I measure county-level farm concentration using the median farm size weighted by harvested acres, or the "midpoint," from the Census of Agriculture (Roberts and Key, 2008). Subsidy payments nearly double the county midpoint over 15 years, increasing it by about 650 acres, or one square mile. Payments do not increase county land values or deposits per capita during the first decade of the program. They also do not increase

[3]See Federal Subsidies Turn Farms Into Big Business (https://www.washingtonpost.com/archive/politics/2006/12/21/federal-subsidies-turn-farms-into-big-business/733c4ae8-6064-4505-aa92-db34f4c11f23/).

production or investment in equipment. Instead, the payments increase farm concentration as the largest farms acquire additional farmland from smaller farms.

Borrower consolidation may change which banks are best positioned to serve the local market. Large banks tend to lend to larger, more transparent borrowers, while small banks specialize in relationship lending that relies on soft information (DeYoung, Hunter, and Udell, 2004; Berger et al., 2005). Changes in the distribution of borrowers may also affect bank entry. When lenders differ in their costs of acquiring information and extending credit, entrants target the borrowers they are relatively better positioned to serve (Gormley, 2014). At the same time, incumbent lenders retain an informational advantage over borrowers they have previously financed, which may limit entry and preserve their advantage among relationship borrowers (Dell'Ariccia, 2001).

I isolate the borrower consolidation channel in two ways. First, I use variation in terrain to test whether subsidy payments change the structure of local banking markets only where farm consolidation is physically feasible. In rugged counties, where expanding acreage is more difficult, subsidy payments are weakly related to farm consolidation and do not affect the structure of local banking markets. In flat counties, subsidy payments strongly predict increases in farm concentration and increase large banks' presence, with effects on branches and deposits about two to three times as large as those for small banks. The terrain difference remains unchanged after controlling for how the effects of payments differ by population, urbanization, and land values.

Second, I divide counties into three groups by how strongly payments predict farm consolidation (Acemoglu et al., 2022). If subsidy payments affect local banking markets through channels other than farm consolidation, they should affect the structure of local banking markets even in counties where they do not increase farm concentration. I find that subsidy payments do not increase banks' presence, number of branches, or deposits per capita in the counties where payments predict the least farm consolidation. In the counties where they predict the most, payments increase farm concentration and increase large banks' number of branches and deposits per capita, with effects that are about twice as large as those for small banks. In both tests, local banking markets change most where subsidy payments have the greatest effect on farm consolidation, consistent with payments affecting banks primarily through changes in borrower composition.

I next use subsidy payments as an instrument for farm concentration to study how borrower consolidation affects banking concentration. The prior results support the identifying assumption that payments affect banks only through farm concentration. A within-county one-standard-deviation increase in farm concentration of about 460 acres increases local banking concentration by 8%, comparable to one out of seven banks exiting a county. Ab-

sent subsidy-induced farm consolidation, local banking concentration would have been about 13% lower by 2014.

Borrower consolidation increases the presence and activity of large banks and leads small banks to consolidate through mergers. A within-county one-standard-deviation increase in farm concentration increases the probability that a large bank is present by about 5 percentage points. On the intensive margin, large banks' number of branches increases by 16% and deposits per capita by 33%, compared with 10% and 22% at small banks. Mergers among small banks increase by about 23%, whereas mergers between large banks decline.

To understand how large banks gain deposit market share and why small banks consolidate, I study their agricultural lending activity. If large banks enter these markets to serve larger borrowers, their lending should increase as farm concentration increases. Using Community Reinvestment Act (CRA) loan originations, which directly report large banks' agricultural lending at the bank-county level, I find that large banks increase lending to larger farms by about 30%, with the increase concentrated in the largest observable loan sizes.[4] To compare agricultural lending at both large and small banks, I also use county-level lending imputed from bank Call Reports (Key, Burns, and Lyons, 2019; Ifft et al., 2024). I find that a within-county one-standard-deviation increase in farm concentration increases large banks' agricultural lending by 46%, with real estate farm lending increasing by 40% and non-real estate farm lending by 46%. At small banks, only non-real estate farm lending increases, by 6%, which is about one eighth of the corresponding increase at large banks.

The limited increase in lending at small banks may be due to funding constraints. Unlike large banks, which have access to internal capital markets, small banks depend more heavily on local deposits to fund loans (Gilje, Loutskina, and Strahan, 2016; Cortés and Strahan, 2017). Because core deposits provide a relatively stable source of funding, small banks with a larger share of these deposits may be better able to finance larger agricultural loans. I use the pre-program core deposit share as a measure of funding capacity (Berlin and Mester, 1999; Cornett et al., 2011). Using the imputed Call Report data, I find that only small banks with a high core deposit share increase lending through larger agricultural loans, for both real estate and non-real estate farm lending. These results suggest that stable deposit funding allows small banks to compete for larger agricultural loans. In contrast, neither group of small banks increases lending in the smallest loan sizes, consistent with farm consolidation generating greater demand for larger agricultural loans without a broad increase in credit demand.

My results show that banks reorganize in response to farm consolidation, but information

[4]Beginning in 2005, banks with assets below about $1 billion were exempt from CRA reporting requirements, so the main CRA lending analysis covers large banks (Section B.3).

frictions and industry expertise may still play an important role in local banking markets. I therefore study how farm consolidation affects banks that specialize in agricultural lending. I classify a bank as an agricultural bank if farm loans exceed 5% of its loan portfolio in any of the preceding six years. A within-county one-standard-deviation increase in farm concentration increases the concentration of agricultural lending by 14%, nearly twice the increase in deposit market concentration. Large agricultural banks are 15 percentage points more likely to be present, more than three times the increase for large banks overall, and their deposits per capita increase by 75%, compared with 24% at small agricultural banks. These results suggest that industry expertise remains important, as large banks with existing agricultural lending experience benefit disproportionately from farm consolidation.

These changes in the competitive environment may spill over to the prices banks offer on consumer financial products. Buyer consolidation may affect supplier prices differently depending on supplier market power and the scope for price discrimination (Galbraith, 1952; Snyder, 1996; Chipty and Snyder, 1999; Inderst and Valletti, 2011). Kropp and Whitaker (2011) documents that farms with a greater share of base acres receive lower interest rates on their operating loans, potentially because the payments provide a risk-free source of income. For consumer credit products, banks in more consolidated local markets could increase prices by exercising market power or lower them to attract deposits and retail customers.

To test these spillover effects, I study interest rates offered on savings, checking, and time deposits (Drechsler, Savov, and Schnabl, 2017). Savings and checking interest rates decrease by 32 and 22 basis points, or about 26% and 14% relative to their unconditional means. When I compare the same bank in the same county over time, large banks lower interest rates across all three deposit products, but small banks increase interest rates on time deposits by 30 basis points. Time deposits fund commercial lending, so the increase is consistent with small banks competing for local funding to serve larger borrowers (Supera, 2021).

I also examine interest rates on auto loans, which are comparable across lenders, unrelated to farm lending, and subject primarily to local competition (Argyle, Nadauld, and Palmer, 2023). At the county level, auto loan interest rates increase by about 103 basis points, while mortgage interest rates do not increase. Because mortgage rates are closely tied to national secondary markets and are less sensitive to local competition (Fuster, Lo, and Willen, 2024), this result is consistent with the increase in auto loan rates reflecting a decrease in local competition instead of a general increase in borrowing costs. Large banks increase auto loan interest rates by about 97 basis points, and small banks lower them by about 37 basis points. Overall, these differences in consumer credit pricing across large and small banks are consistent with small banks competing for local funding and consumer borrowers and with

large banks exercising greater pricing power in more concentrated markets.

The same mechanism may operate in other industries experiencing firm consolidation in which bank debt is an important source of financing, such as construction, trucking, and small manufacturing. In these settings, consolidation may similarly change which banks are best positioned to finance the remaining firms. More broadly, these results suggest that the consolidation of any industry should be studied together with the consolidation of its upstream and downstream industries, because in equilibrium the market structures of these industries are jointly determined.

The remainder of the paper is structured as follows. Section 2 reviews the related literature, Section 3 provides institutional background on the fixed subsidy payment program, and Section 4 presents a conceptual framework. Section 5 describes the data and sample construction, and Section 6 outlines the identification strategy. Section 7 presents the results, and Section 8 concludes.

2 Literature Contributions

This paper contributes to several strands of literature by showing that subsidy payments that increase with firm size change the distribution of firm size and, in turn, the concentration and composition of local banking markets.

First, this paper contributes to the literature on product and credit markets by showing how consolidation among borrowers affects the banks operating in the same market. Existing research shows that leverage and bank behavior affect output, entry, and price competition among firms (Brander and Lewis, 1986; Chevalier, 1995; Cetorelli and Strahan, 2006; Bertrand, Schoar, and Thesmar, 2007; Huber, 2021; Saidi and Streitz, 2021). The effect of lenders on borrowers is unlikely to be the only connection between lender and borrower concentration. As borrowers increase in scale and complexity, their financing needs evolve, which may change both how much they borrow and from whom. I establish a causal connection in the other direction, showing that changes in borrower structure reorganize the local banking markets that finance these borrowers.

Within this literature, I build on work in industrial organization (IO) and banking showing that downstream consolidation drives upstream consolidation, as when increases in buyer concentration across industries lead to supplier consolidation (Becker and Thomas, 2009). Brennecke, Jacewitz, and Pogach (2025) applies this logic to banking, using a Bartik instrument based on national industry trends to show that decreases in small firm growth decrease deposit growth at small banks. Because the identification in Brennecke, Jacewitz, and Pogach (2025) aggregates variation across many industries and geographies, the resulting

estimates reflect the broad relationship between small firm activity and small bank health. However, this approach does not identify how the full distribution of banks changes within local markets following changes in borrower structure. Using a policy-driven shock to one place-based industry, I follow the composition of banks over time as borrowers consolidate. When small businesses expand and consolidate, the additional credit demand does not flow primarily to small banks. Instead, large banks enter affected markets and expand their deposits and branch networks, and households face less favorable terms on both deposits and consumer loans as local banking markets concentrate.

Second, this paper contributes to the literature on the effects of government transfers to firms. Existing work shows that transfers, tax incentives, and collateral shocks can relax financing constraints, increase investment, and change firms' demand for external finance (Fazzari, Hubbard, and Petersen, 1988; Gan, 2007; Chaney, Sraer, and Thesmar, 2012; Zwick and Mahon, 2017; Bahaj, Foulis, and Pinter, 2020; Garmaise and Natividad, 2024). Where support reaches firms through banks, the banks determine which firms receive it (Granja et al., 2022), and the support changes which firms employ workers (Barrot et al., 2024). I add to this literature by showing that government payments that increase with firm size may also change the structure of the borrower market, with consequences for the banks that allocate credit.

Third, this paper contributes to the literature on the boundaries of the firm (Coase, 1937; Williamson, 1975; Grossman and Hart, 1986; Hart and Moore, 1990). This literature studies which assets and activities are coordinated within a firm rather than through market transactions. In agriculture, the horizontal boundary of the firm is how much land and production a single farmer operates (Allen and Lueck, 1998). I show that subsidy payments concentrate land and production on larger farms, and that as farms become larger, banks that specialize in serving larger borrowers gain market share. Whereas Brickley, Linck, and Smith (2003) studies the boundaries of banking organizations, I show that farm expansion changes which banks lend in a county.

Fourth, this paper contributes to work on the structure of banking markets, which emphasizes supply-side drivers of consolidation such as deregulation, technology, and economies of scale in banking (Jayaratne and Strahan, 1998; Berger, Demsetz, and Strahan, 1999). I highlight a complementary demand-side channel. Large and small banks have different comparative advantages. Small banks rely on soft information and local ties (Petersen and Rajan, 1994; Berger and Udell, 1995; DeYoung, Hunter, and Udell, 2004; Berger, Bouwman, and Kim, 2017), and large banks specialize in standardized and scale-intensive lending (Boot and Thakor, 2000; Petersen and Rajan, 2002; Berger et al., 2005; Alessandrini, Presbitero, and Zazzaro, 2009). Banks also specialize by industry (Paravisini, Rappoport, and

Schnabl, 2023; Blickle, Parlatore, and Saunders, 2026). The gains at large banks suggest that expanding borrowers switch lenders as their scale changes the size and type of credit they demand. Building on Brickley, Linck, and Smith (2003), which documents that large banks concentrate in urban areas, where borrower scale can support centralized operations, I show that borrower consolidation can induce entry by large banks into markets where they previously had little presence.

Fifth, this paper contributes to research on the structure of banking markets and real economic outcomes. Prior studies show that bank market power affects firm entry, competition, and community development (Strahan and Weston, 1998; Cetorelli and Strahan, 2006; Garmaise and Moskowitz, 2006; Saidi and Streitz, 2021; Carlson, Correia, and Luck, 2022), and that higher banking concentration is associated with lower deposit rates, higher loan spreads, and higher markups (Berger and Hannan, 1989; Rice and Strahan, 2010; Erel, 2011; Drechsler, Savov, and Schnabl, 2017). Higher concentration also gives banks more stable funding and longer loan maturities (Li, Loutskina, and Strahan, 2023) and increases the pass-through of cost shocks to auto loan rates (Hankins, Momeni, and Sovich, 2026). My findings add a borrower-driven mechanism to this literature, showing that changes in the structure of borrower industries spill over into the interest rates banks offer households through changes in local banking concentration. Kropp and Whitaker (2011) finds that farms with greater base acre shares receive lower interest rates on operating loans. I complement this finding by showing that the reorganization of borrowers and lenders increases retail interest margins more broadly, increasing auto loan rates and decreasing deposit rates.

Last, this paper contributes to research using agriculture to study broader financial markets and to the agricultural finance literature. Agriculture is well suited for this purpose because credit relationships are place-based, farm financial data are systematically collected, and policy-driven shocks provide plausibly exogenous variation in credit demand (Butler and Cornaggia, 2011; Rajan and Ramcharan, 2015; Carlin and Mann, 2024). Shocks to farm income have been used to trace how banks move funds across markets and how local cash flows affect land values and bank failures (Bustos, Garber, and Ponticelli, 2020; Bergman, Iyer, and Thakor, 2020). Existing work documents how decoupled subsidy programs influence land allocation, land values, farm consolidation, and credit access (Burfisher and Hopkins, 2003; Roberts and Key, 2008; Bhaskar and Beghin, 2010; Gardner, Hardie, and Parks, 2010; Bekkerman, Belasco, and Watson, 2015; Ifft, Kuethe, and Morehart, 2015). I show that the farm consolidation these programs produce pushes credit demand toward large banks and changes the structure of local banking markets.

3 Institutional Background

The 1996 FAIR Act marked a sharp break from 60 years of farm support tied to commodity prices. Since the New Deal, federal policy had stabilized farm income through a combination of price floors, nonrecourse loans, and countercyclical deficiency payments (Effland, 2000). In exchange for this support, farms had to maintain base acreage allocations, meaning that payments were tied to the production of designated crops on enrolled land. The FAIR Act replaced this system with seven years of fixed annual subsidy payments determined solely by a farm's historical acres and yields, regardless of current production or prices (Kirwan, 2009). The program decoupled subsidy payments from planting decisions, current yields, and market prices, allowing farmers to respond to market signals rather than government incentives. The reform was intended to reduce production distortions and reliance on government support, and to comply with World Trade Organization (WTO) rules.[5]

Payments were determined by the number of base acres, defined as the average number of acres planted in program crops (wheat, corn, sorghum, barley, oats, rice, and upland cotton) between 1981 and 1985. Because larger farms had more base acres, they received larger payments. The FAIR Act also removed planting restrictions, so farms could change production without affecting payment eligibility.[6]

Between 1996 and 2014, the federal government disbursed about $89 billion, or about $4–6 billion per year, in direct payments. During the first three years of the FAIR Act, these fixed payments represented about 79% of subsidy payments distributed under the new farm policy (Williams-Derry and Cook, 2000). The payments were concentrated among the largest recipients. According to the Government Accountability Office (GAO), the top 25% of recipients collected 73% of all direct payments in 2011 (United States Government Accountability Office, 2012).

The payments were predictable and tied to the land, so some of their value could be capitalized into land prices. Kirwan (2009) finds that farmland rental rates increase by about 25 cents per marginal subsidy dollar. Because land prices capitalize expected rental income, this estimate implies that no more than about one quarter of each payment dollar would be reflected in land values.

Congress renewed the program in 2002 under the Farm Security and Rural Investment (FSRI) Act, expanding eligibility to oilseeds and peanuts, and allowing base acres to be updated using average plantings between 1998 and 2001. The program was renewed again

[5] See Statement on Signing the Federal Agriculture Improvement and Reform Act of 1996.

[6] Although the federal government limited payments to $180,000 annually per person, many large farms bypassed these limits by subdividing operations into multiple entities or assigning ownership shares to passive investors (United States General Accounting Office, 2004).

in 2008 without additional crops or base acre updates and ended with the 2014 Agricultural Act. Figure 1 shows a timeline outlining the policy changes.

[Insert Figure 1 Here]

There was substantial consolidation in farming during the period I study. Although the total number of farms nationwide was similar between 1992 and 2012, their size distribution changed. From 1992 to 2012, the number of large farms and farms with very low sales increased, and the number of small commercial and midsize farms declined (Burns and Kuhns, 2016). In my sample, the average number of farms per county remains in a narrow range of about 620 to 760 from 1992 to 2012, but the average midpoint farm size nearly doubles. The main change over this period is a reallocation of agricultural activity toward larger farms rather than a large change in the total number of farms.

4 Conceptual Framework

This section outlines the mechanisms through which borrower consolidation influences the structure of local banking markets. The argument proceeds in three steps. First, fixed subsidy payments relax farms' financial constraints and enable expansion, particularly among larger farms, increasing midpoint farm size and farm concentration. Second, as farms consolidate, credit demand becomes concentrated in larger loans. Third, this change in borrowers alters which banks are best positioned in the local market. Banks equipped to meet these financing needs enter and expand in these markets, and other banks exit, merge, or contract. These changes in the composition of borrowers and lenders may ultimately affect the pricing of consumer financial products in rural communities.

Commodity crop producers have strong incentives to expand. Advances in mechanization, chemical inputs, and crop management practices decrease the marginal cost of managing additional acres (Paul and Nehring, 2005; Duffy, 2009). High fixed costs and economies of scale also give larger farms lower average production costs than smaller farms (Hoppe and Banker, 2006). In competitive commodity markets where farms are price takers (Stiers, 2022), expanding acreage allows farms to spread fixed costs over more output and improve cost efficiency (Langemeier, 2013; Hoppe, 2014).

However, expansion requires substantial upfront capital for land acquisition, machinery, and seasonal operating inputs. With limited internal liquidity or collateral, many farms rely on external finance to cover these costs (Hubbard and Kashyap, 1992; Barry, Bierlen, and Sotomayor, 2000). Interviews with agricultural lenders also indicate that most farmers are cash constrained (Thakor, 2023).

Subsidy payments may help relax these constraints. The 1996 FAIR Act provided farms with predictable annual subsidy payments tied to historical base acres. Because base acres were correlated with farm size, larger farms systematically received larger total payments (Roberts and Key, 2008). These predictable transfers provided larger farms with more resources to finance expansion. Anecdotally, reports from the time suggest that large farms used these payments to go out "trolling for land."[7] Larger farms may have used these payments to pursue land acquisitions, accelerating the pace of consolidation.

Debt facilitates farm expansion. Between 1994 and 2017, total farm sector debt increased by 84% in real terms, reaching $390 billion in 2017 (U.S. Department of Agriculture Economic Research Service, 2026). Credit use also differs by farm size. Between 1996 and 2014, an average of 27% of farms with under $100,000 in sales carried debt, compared with 67% of farms with over $1 million in sales (U.S. Department of Agriculture, Economic Research Service and National Agricultural Statistics Service, 1996). These facts suggest that debt markets play an important role in financing agricultural investments, particularly among large farms.

Subsidy payments may also affect credit directly through borrower balance sheets, even holding farm size fixed. Direct payments affect farms' liquidity and repayment capacity (Kropp and Katchova, 2011). Theoretically, higher net worth mitigates moral hazard by improving borrower incentives and decreasing monitoring costs (Holmstrom and Tirole, 1997). Empirically, farms with more base acres receive lower interest rates on operating loans (Kropp and Whitaker, 2011), and farms with higher net worth invest more (Hubbard and Kashyap, 1992). Holding farm size fixed, this balance sheet channel generates predictions similar to those of an increase in collateral. Sections 7.1 and 7.2 test these predictions directly.

To guide the empirical tests, I develop a model in the Internet Appendix that relates farm scale, collateral, deposits, and production to local banking. The channels have different implications for loan size, bank choice, and small banks' funding capacity. When farms consolidate, large banks provide a greater share of lending. Small banks with greater core deposit shares are better able to retain these larger borrowers, and consolidation increases their incentives to merge and pool their deposit bases. For the same increase in farm size, required loan size increases more where farming is more capital intensive, so small banks become capacity constrained at smaller loan sizes and large banks gain a larger share of lending. By contrast, increases in collateral or production decrease borrowing costs and

[7]See Federal Subsidies Turn Farms Into Big Business (https://www.washingtonpost.com/archive/politics/2006/12/21/federal-subsidies-turn-farms-into-big-business/733c4ae8-6064-4505-aa92-db34f4c11f23/).

expand credit across loan sizes, most of all at small banks with less funding capacity. Deposit inflows instead give small banks more funds to lend, allowing them to retain larger borrowers that might otherwise move to large banks. Section 7 tests each implication in turn by examining the concentration of lending in larger loans and at large banks, the dependence on small banks' funding capacity, the role of farming costs, and whether the effects operate through collateral, deposits, or production.

5 Data Sources and Sample Construction

I construct a panel dataset linking county-level farm structure with banking outcomes and consumer financial product rates from 1992 to 2014. This section describes the banking outcomes, farm concentration measure, controls, and sample. Appendix B provides additional construction details.

5.1 Dependent Variables

The main banking outcomes measure local banking concentration, bank composition, and mergers. I measure local concentration using the bank deposits HHI from the Federal Deposit Insurance Corporation's (FDIC) Summary of Deposits (SOD), calculated as the sum of squared bank deposit shares within each county-year. The index ranges from 0 to 10,000, with 10,000 indicating a monopoly and lower values indicating greater competition.

I report many of the banking outcomes separately for small and large banks. I merge the SOD with the Federal Financial Institutions Examination Council's (FFIEC) Consolidated Reports of Condition and Income (Call Reports) and classify banks using a $1 billion threshold in inflation-adjusted assets. This threshold follows the standard definition of community banks and is close to the CRA asset threshold (Berger et al., 1998; DeYoung, Hunter, and Udell, 2004; Brennecke, Jacewitz, and Pogach, 2025; Federal Deposit Insurance Corporation, 2012). For each county, I calculate the number of branches and deposits per capita held by small and large banks using population estimates from the Bureau of Economic Analysis (BEA). Bank assets, deposits, agricultural loans, land values, and income per capita are expressed in 2020 dollars using the Consumer Price Index (CPI) from the Bureau of Labor Statistics (BLS).

To track consolidation activity, I use the Chicago Federal Reserve's Bank Merger database to construct three county-level outcomes aggregated over five-year intervals. The first, *Large Acquires Small*, measures the total number of mergers or acquisitions in which a large bank acquires or merges with a small bank. The second and third, *Small Acquires Small* and *Large Acquires Large*, count the total number of mergers or acquisitions between two small

banks and between two large banks, respectively. Because the Summary of Deposits is recorded each June 30, I measure each bank's size and the target's branch counties at the last Summary of Deposits before the transaction date. I count each transaction in every county where the target operated branches.

To examine how the composition of agricultural lending changes as farms consolidate, I use Call Report Schedule RC-C Part II, which reports outstanding farm loans by origination size. I test whether small banks' ability to supply larger loans depends on funding capacity, measured using the pre-program core deposit share and the loans-to-deposits ratio, both averaged over 1990 to 1995. The core deposit share is deposits other than large time deposits and brokered deposits, as a share of total deposits. A higher core deposit share indicates more stable funding, and a lower loans-to-deposits ratio indicates more funding available for additional lending (Acharya and Mora, 2015). Both measures are fixed at their pre-program values.

I also measure the importance of agricultural lending to each bank using real estate and non-real estate farm loans from Call Report Schedule RC-C. Real estate farm loans are secured by farmland, and non-real estate farm loans finance farm production and operating expenses, including seed, fertilizer, livestock, and equipment. I classify a bank as an agricultural bank if farm loans exceed 5% of total loans in any of the preceding six years. To construct county-level farm loan volumes, I allocate each bank's agricultural lending across the counties in which it operated a branch at the start of each five-year period. I allocate lending in proportion to each county's share of farm interest expenses in the most recent Census of Agriculture available at the start of the period. Real estate farm lending is weighted by real estate interest expenses, non-real estate farm lending by non-real estate interest expenses, and total agricultural lending by their sum. For example, if a bank operated branches in three counties with equal farm interest expenses five years earlier, I allocate one third of its lending to each county. I aggregate the resulting lending separately for small and large banks, following Key, Burns, and Lyons (2019) and Ifft et al. (2024). For each group, I measure growth in total, real estate, and non-real estate farm lending as the change in the group's lending as a share of the county's total agricultural lending in the prior period. This measure is each group's contribution to the change in county agricultural lending.

Although this method provides a proxy for county-level farm loan volumes, it has several limitations. First, the baseline allocation assumes that a bank lends only in counties where it operated a branch at the start of the period. As a robustness test, I also construct an alternative allocation that uses the bank's current branch counties and adds counties in which it originated farm loans reported under the CRA. For counties without a branch, the allocation weight declines with distance to the bank's nearest branch (Appendix B.1). Second, the

allocation assumes that banks lend in proportion to historical local credit demand, proxied by farm interest expenses, which may differ from current lending decisions. Nonetheless, the method provides evidence on the geographic distribution of agricultural lending. I also use CRA loan originations, which report each bank's farm loans of $500,000 or less directly by the county of the borrowing farm, allowing me to examine lending without relying on the allocation procedure (Appendix B.3).

To examine how borrower consolidation and changes in bank composition affect the interest rates banks offer on consumer deposits and loans, I use interest rate data from RateWatch. I focus on time deposits (12-month $10,000 certificates of deposit), savings deposits ($10,000 money market accounts), interest-bearing checking accounts, auto loans, and residential mortgages. For mortgages, I average the offered rates across the fixed rate, adjustable rate, and balloon products reported by each branch. I convert interest rates to real terms using the CPI. Because the FDIC measures SOD deposits as of June 30, I use each branch's average weekly quoted rate in June so that the RateWatch and SOD data refer to the same month.

To test whether farm consolidation changes access to federal credit for small farms, I also use data on direct loans issued by the U.S. Department of Agriculture's (USDA) Farm Service Agency (FSA). These loans target small, new, or socially disadvantaged farmers who cannot obtain commercial credit. I aggregate the number and volume of direct farm ownership loans, which finance farmland and farm property purchases or expansion, and direct operating loans, which finance production expenses and equipment, at the county-year level from 1994 to 2014.

5.2 Variables to Test Different Channels

The public Census of Agriculture reports only the number of farms within acreage categories (e.g., 1–9, 10–49, 50–69, 70–99, …, 1,000–1,999, and 2,000+ acres), so individual farm sizes are not observed. To compute traditional concentration measures such as the HHI, each farm in a bin would be assigned the same average acreage. For example, a 2,500-acre farm and a 25,000-acre farm in the "2,000+ acres" bin would contribute equally to the HHI calculation. Because the upper bins span wide acreage ranges, this method effectively treats all large farms as the same size. Doing so would tend to introduce bias by Jensen's inequality, because the sum of squared farm sizes exceeds the sum of squared average farm sizes. This bias would be largest in the larger farm size bins, making the HHI an unreliable measure of farm concentration.

Given the structure of the data, I follow standard practice in agricultural economics and measure borrower consolidation using the harvested acre-weighted median farm size

(the "midpoint") from the USDA's Census of Agriculture (MacDonald, Korb, and Hoppe, 2013). The midpoint is defined as the acreage at which half of a county's harvested cropland lies on larger farms and half on smaller farms. For example, if a county has 2,800 total harvested acres and the eight smallest farms harvest 1,000 acres in total, then the ninth-smallest farm is the one whose acreage spans the 1,400-acre halfway point and marks the midpoint. If that farm operates 700 acres, then the midpoint farm size is 700 acres. This approach is standard in the literature for two reasons. First, individual farm sizes are not publicly reported. Second, the production distribution is highly skewed, with a relatively small number of large farms operating much of the harvested acreage. The midpoint more accurately reflects changes in the scale of production than the simple median.

I also construct measures of the alternative channels through which subsidy payments may affect local banking markets. I measure land values using the market value of agricultural land and buildings per acre from the Census of Agriculture, expressed in real terms. I measure local deposits as county deposits per capita, using total deposits from the SOD and county population. For investment and production, I use four outcomes: the real value of county machinery, the number of recently manufactured tractors, a Törnqvist crop-yield index that aggregates per acre yield growth across the nine census crops with quantity data using revenue-share weights, and the share of county crop acres planted in the seven program crops covered by the 1996 FAIR Act.

5.3 Treatment Variable

I measure county-level predicted subsidy payments using historical base acres and national crop-level subsidy allocations. I obtain county-level base acres for each eligible crop from the USDA and combine them with the annual national crop-level subsidy allocation for that crop. For each crop, I multiply the county's share of national base acres by the national crop-level subsidy allocation and sum across crops to construct predicted county-level subsidy payments.

5.4 Controls

County-level farmland controls from the Census of Agriculture include idle cropland, pastureland, and unharvested cropland (Weber and Key, 2012). I measure local economic conditions using per capita income and the employment rate from the BEA. In additional analyses, I control for the lagged log number of bank branches and the log of total interest expenses on farm loans. To examine the role of crop insurance, I use the number of policies covering the relevant commodity crops from the USDA's Risk Management Agency (RMA). I also include the number of Farm Credit System (FCS) branches in each county, provided

by the Farm Credit Administration. To study the effect of countercyclical payments on my results, I include predicted countercyclical payments. For each crop, I multiply the county's share of national base acres by the national countercyclical payments on that crop in the same payment year and sum across crops. National countercyclical payments by crop are from the FSA.

I examine whether the effects of subsidy-induced consolidation vary with local geography. To characterize urban-rural status, I use county classifications from the National Center for Health Statistics (NCHS) Urban-Rural Classification Scheme from the 1990, 2000, and 2010 censuses. I classify large central metropolitan counties as urban, other metropolitan or micropolitan counties as metro, and noncore counties as rural. I also use the 2010 Terrain Ruggedness Index (TRI) from the USDA Economic Research Service (ERS) to test whether the effects differ in areas less conducive to large-scale farming. I classify counties in the top quartile of the national TRI distribution as rugged and all others as flat. Figure A.1 shows the distribution of U.S. counties by average terrain ruggedness across quartiles.

For each major commodity crop (barley, corn, oats, peanuts, rice, soybeans, sorghum, upland cotton, and wheat), I calculate average per acre economic costs from 1990 to 1995 using the Farm Costs and Returns Survey (FCRS) from the ERS. I use this pre-program period to measure production costs before the subsidy payments began. I construct a time-invariant county-level cost measure by weighting each crop's average cost by its share of county harvested acres in the 1992 Census of Agriculture. Counties with weighted average per acre economic costs above the nationwide median are classified as high-cost, and those below the median are classified as low-cost. Figure A.2 shows the distribution of high- and low-cost agricultural counties across the U.S.

Last, to identify counties exposed to the housing bust between 2007 and 2012, I use the annual county-level all-transactions House Price Index (HPI) from the Federal Housing Finance Agency (FHFA). The index is constructed from purchase and appraisal records for mortgages acquired or guaranteed by Fannie Mae and Freddie Mac. I classify a county as exposed to the housing bust if its HPI declines between 2007 and 2012 and its total predicted subsidy payments per acre of cropland over the sample period are below the national median. Figure A.3 compares agricultural land values in these counties with those in counties where predicted subsidy payments per acre of cropland are above the national median.

5.5 Sample Construction and Summary Statistics

My primary analysis uses a county-level panel observed at five-year intervals from 1994 to 2014. The USDA Census of Agriculture provides farm structure data every five years, which I merge with banking data from the FFIEC Call Reports and FDIC SOD for the

corresponding periods. I lag farm variables by two years to allow changes in agriculture to precede subsequent banking outcomes. I exclude 57 large central metropolitan counties from the main sample to focus on counties where agriculture may be a more important part of the local economy. To avoid overstating economic magnitudes, I standardize farm concentration using its within-county variation over time (Liu and Winegar, 2025).

For the analysis of interest rate outcomes, I construct a separate annual panel from 2002 to 2014, merging branch-level data from RateWatch with bank- and county-level information from the SOD and Call Reports. I estimate fitted farm concentration from the Census of Agriculture, available every five years, and use the most recent fitted value in interim years. This method preserves the structural variation identified at the census years while allowing annual fluctuations in local economic conditions to enter through time-varying controls in the second-stage regressions. The approach reflects the gradual nature of farm consolidation and assumes slow-moving structural change between census years. The data support this assumption. Farm concentration, measured by midpoint acreage, increases by only about 2.5% per year on average. To estimate the specification in first differences, I instead interpolate fitted farm concentration linearly between census years.

Table 1 reports summary statistics. Subsidy payments are widespread and highly skewed. On average, counties receive $1.37 million in subsidy payments per year, with a standard deviation of $1.88 million between counties and $1.15 million within counties over time. These summary statistics show that there is substantial cross-sectional and time-series variation in subsidy payments.

[Insert Table 1 Here]

6 Identification Strategy

I use a continuous difference-in-differences design that combines predetermined county crop shares with changes in national crop-level subsidy allocations. First, I estimate the reduced-form effect of subsidy payments on local banking markets. Second, I evaluate the channels through which payments may affect banks. Third, I instrument farm concentration with subsidy payments to estimate how borrower consolidation affects banking outcomes.

The difference-in-differences design uses county-level predicted subsidy payments under the fixed payment program introduced by the 1996 FAIR Act and extended in 2002 and 2008 (see Section 3 for details). Payments were based on historical planting rather than current production or market conditions. Base acres and yields initially reflected 1981–1985

averages, and farms could update them in 2002 using 1998–2001 plantings.[8] I construct predicted subsidy payments using base acres alone. Excluding historical yields produces a more conservative intent-to-treat measure because yields may be correlated with later banking outcomes through persistent differences in local productivity. The results are similar when yields are included.

Predicted subsidy payments are defined as the inner product of (a) each county's fixed crop-level base acre shares and (b) the national crop-level subsidy allocations in a given year from 1996 to 2014, when the program ended. For example, if a county holds 1% of the nation's corn base acres and the national crop-level subsidy allocation for corn is $2 billion in a given year, that county would be assigned predicted subsidy payments of 1% × $2 billion. The national crop-level subsidy allocations vary over time and apply uniformly across all counties. Identification therefore comes from changes over time in national crop-level subsidy allocations, interacted with county-level crop shares determined by historical base acres. Because both components are determined outside the local banking environment, one by historical planting decisions and the other by national policy, the resulting variation in predicted subsidy payments is plausibly exogenous with respect to local credit conditions.

The timing of the policy and the historical planting decisions underlying predicted payments also help address concerns of reverse causality. For reverse causality to threaten the difference-in-differences design, banking consolidation would need to be systematically correlated with predicted subsidy payments for reasons unrelated to the policy-induced changes in farming. Reverse causality would be a concern if, for instance, banks consolidated in anticipation of the policy or if farmers chose the crops that determine base acres in anticipation of local banking conditions. This is unlikely. The 1996 FAIR Act marked a sharp and largely unexpected break with six decades of price-contingent support, and faced considerable political uncertainty and multiple filibusters prior to passage (Orden, Paarlberg, and Roe, 1999). As a result, banks could not easily anticipate the policy, and planting decisions made more than a decade earlier are unlikely to reflect expectations about future credit conditions.

Using predicted subsidy payments, I estimate the following difference-in-differences specification:

$$Y_{i,t} = \alpha \sum_{c} \left(\frac{\textit{Base Acres}_{c,i,t}}{\textit{Base Acres}_{c,t}} \times \textit{Natl Allocation}_{c,t} \right) + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}, \qquad (1)$$

where c represents the crop type, i represents the county, t represents the year, and d represents state-level banking deregulation. The interpretation of α depends on the outcome. When $Y_{i,t}$ is a local banking outcome, α measures the effect of subsidy payments on the

[8]The results are similar when I construct predicted subsidy payments using only pre-program data (Table A.1). I hold the 1996 base acre shares fixed for the original program crops and use 1992 Census of Agriculture acreage for crops added in 2002.

structure of local banking markets. When $Y_{i,t}$ is a possible channel outcome, α measures the effect of subsidy payments on that channel. For banking outcomes, the subsidy term and the controls enter with a two-year lag. For channel outcomes from the Census of Agriculture, they enter in the same census year. $\textit{Base Acres}_{c,i,t}$ is the number of historical base acres of crop c in county i and year t, $\textit{Base Acres}_{c,t}$ is the total number of historical base acres in the U.S. of crop c in year t, and $\textit{Natl Allocation}_{c,t}$ is the national crop-level subsidy allocation for crop c in year t. $\mathbf{X}_{i,t}$ is a vector of control variables, including county-level economic characteristics and farm characteristics that control for land availability. These controls include the employment rate, log income per capita, and lagged measures of idle cropland, pastureland, and cropland left fallow, lost to crop failure, or otherwise unharvested. I include a fixed effect for state-level branching deregulation using the index from Célérier and Matray (2019), which reflects the staggered implementation of interstate branching reforms between 1994 and 2005. County and year fixed effects absorb time-invariant county characteristics and national shocks.

To understand how subsidy payments affect local banking markets, I study the three main channels through which payments may affect banks: farm concentration, land values, and deposits. Figure 2 shows the geographic distribution of subsidy payments and these three channels. Panel A maps the change in cumulative county-level subsidy payments, and Panels B–D map changes in farm concentration, land values per acre, and deposits per capita.

[Insert Figure 2 Here]

The unconditional correlations with subsidy payments are 51% for farm concentration, −6% for land values per acre, and 12% for deposits per capita. The geographic correlation is strongest for farm concentration, motivating a closer examination of borrower consolidation relative to the alternative channels. Section 7.1 studies these channels formally, testing whether payments change land values, deposits, investment, or production, and shows that farm concentration is the only outcome that increases persistently with subsidy payments. Section 7.2 tests whether payments affect local banking markets in counties where they generate little or no increase in farm concentration.

There are two main challenges with studying the effect of borrower consolidation on the structure of local banking markets. First, borrower consolidation and bank consolidation may be jointly driven by local characteristics. For example, rural areas with limited economic activity may have fewer banks and fewer, but larger, farms because of low population density or substantial land availability. The second challenge is reverse causality. In areas with more concentrated banking markets, limited competition may restrict capital access for

smaller farms. These farms may be forced to exit or sell to larger operators, accelerating consolidation. Alternatively, a more consolidated banking sector may prefer lending to larger farms because it is less suited to lending to informationally opaque borrowers (Berger and Udell, 2002). In both scenarios, the structure of local banking markets could cause borrower consolidation.

To address these concerns, I use predicted subsidy payments as a shift-share-like instrument for borrower consolidation. Because predicted payments depend on national subsidy allocations and historically determined base acres, they are less likely to reflect local factors that jointly drive farm and bank consolidation. Moreover, changes in local banking structure cannot determine the historical plantings or national subsidy allocations underlying predicted payments, mitigating concerns about reverse causality. Last, the exclusion restriction requires that subsidy payments affect local banking markets through farm consolidation and not through the alternative channels discussed above. Sections 7.1 and 7.2 test the alternative channels and whether subsidy payments affect local banking markets in the absence of farm consolidation.

The first stage is given by the difference-in-differences specification above with $Y_{i,t}$ equal to farm concentration, measured as the midpoint farm size described in Section 5. As shown in the following sections, the first-stage F-statistics exceed the Stock and Yogo (2005) thresholds for weak instruments, supporting the relevance condition. The strength of the first stage supports the interpretation that the policy alters the structure of farms. In the baseline specification, I lag farm concentration by two years to allow changes in farm structure to precede changes in local banking structure. The results are similar with three- and four-year lags (Table A.2).

The second-stage estimates measure how exogenous changes in borrower structure affect the organization of local banking markets. The baseline second-stage specification is:

$$Y_{i,t} = \beta_2 \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}, \tag{2}$$

where subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The outcome variable $Y_{i,t}$ represents various measures of the structure of local banking markets at the county level, including the Bank Deposits $\text{HHI}_{i,t}$. All variables are measured in five-year intervals from 1994 to 2014. The vector of control variables $\mathbf{X}_{i,t-2}$ includes the lagged economic variables, log $\text{income per capita}_{i,t-2}$ and $\text{employment rate}_{i,t-2}$, aligned with the timing of the farm data in the first stage. The land availability controls from the first stage also enter every specification that instruments farm concentration or has an outcome from the Census of Agriculture.

Because farm consolidation predicts different effects for small and large banks, I construct

separate county-level measures of local banking outcomes for each group. For these outcomes, I use a Poisson control function approach (Cohn, Liu, and Wardlaw, 2022). About 22% of county-year observations in my sample do not contain a large bank, so this method addresses the high incidence of zeros in the data. Following Wooldridge (2015), I estimate the first stage via OLS, obtain the residual, and include it as a correction term in the Poisson model:

$$E\left[Y_{i,t} \,\middle|\, \textit{Farm Concentration}_{i,t-2},\ \hat{\varepsilon}_{i,t-2},\ \mathbf{X}'_{i,t-2},\ \gamma_i,\ \delta_t,\ \phi_{d(i,t)}\right] = \exp\left(\beta_2\ \textit{Farm Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\boldsymbol{\Gamma} + \beta_3\,\hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\right), \quad (3)$$

where subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The vector of control variables $\mathbf{X}_{i,t-2}$ includes the same control variables as in Equation 2, with the addition of the log number of branches$_{i,t-2}$ to compare counties with similar pre-existing banking infrastructure. For deposits per capita, the first stage also includes log population$_{i,t-2}$ to adjust for the per capita scaling of the outcome. Because the model is estimated in two stages, I bootstrap the full procedure 500 times, clustering standard errors at the county level to preserve the panel structure. All analyses that separate large and small banks follow this estimation strategy, with the exception of linear probability models. For ease of interpretation, coefficients in these tables are reported as decimal percentages.[9]

7 Empirical Results

In this section, I show that subsidy payments increase local banking concentration and consider several channels that could produce this result. Subsidy payments have little effect on local land values, total deposits, or farm production but substantially increase farm concentration. Using subsidy payments as an instrument for farm concentration, I show that farm concentration induces large bank entry and increases large banks' share of branches and deposits, while small banks merge with one another in response to farm consolidation. These results reflect large banks' increase in local agricultural lending and small banks' limited funding capacity, and the changes in presence and deposits are largest among banks that specialize in agricultural lending. The resulting change in the structure of local banking markets spills over to consumer credit markets, where households face higher borrowing costs and earn lower returns on deposits.

[9] All Poisson (PPML) coefficients are reported as implied percentage changes. The relevant tables report $e^{\beta} - 1$ in decimal percentage form rather than the raw Poisson estimate β. For example, a raw estimate of $\beta = 0.53$ is reported as 0.70, corresponding to a 70% increase in the outcome ($e^{0.53} - 1 = 0.70$).

7.1 Subsidy Amount, Banking Concentration, and the Candidate Channels

Subsidy payments increase banking concentration. Figure 3 reports dynamic estimates using the estimator developed by de Chaisemartin and D'Haultfœuille (2026) to accommodate continuous treatment. Panel A shows that the bank deposits HHI increases by about 390 points by 2014. At the sample average level of banking concentration, an increase of this magnitude is nearly twice the 200-point increase that triggers further review of a bank merger for potentially being anticompetitive (U.S. Department of Justice, Antitrust Division, 1995).

[Insert Figure 3 Here]

Subsidy payments may affect banking concentration through several channels, including changes in the composition of borrowers, agricultural land values, and local deposits. First, subsidy payments may enable larger farms to acquire land from smaller farms, increasing farm concentration and the market share of large banks that specialize in serving larger borrowers. Second, payments may increase agricultural land values, providing farmers with additional collateral to borrow against. Higher collateral values may particularly benefit small banks, which specialize in lending to smaller, bank-dependent borrowers who rely more heavily on real estate collateral (Gupta, Sapriza, and Yankov, 2022). Third, subsidy payments may increase local deposits, providing banks with additional funding to expand lending. Because small banks rely more heavily on local deposits, these inflows may help them retain borrowers and compete with large banks. Panels B through D provide evidence on these three potential channels.

Of these candidate channels, only farm concentration increases persistently with subsidy payments. Panel B shows that the midpoint farm size increases by about 650 acres over the program period, or 86% of its average level. Larger farms receive greater payments because they have more base acres (Roberts and Key, 2008). These fixed, predictable payments may support farm expansion by providing a stable cash flow to service additional debt. Larger farms also benefit from economies of scale in commodity crop production (Duffy, 2009; Langemeier, 2013). It follows that lower production costs and more predictable cash flows give larger farms an advantage in acquiring land that would be relatively less profitable for smaller farms to operate. In 1998, for example, large farms received an average annual subsidy payment of about $30,000, which was enough to cover the annual interest cost of financing about 500 acres in Texas, an area equivalent to about 380 football fields.[10]

[10]Based on Texas average cropland values in 1998 of $701 per acre and the average interest rate on farm loans in Texas of 8%. See Agricultural Land Values (https://downloads.usda.library.cornell.edu/usda-esmis/files).

The other two candidate channels, land values and deposits per capita, do not increase during the first decade of the program (Panels C and D). Both increase only during the 2010–2013 farmland boom, when high commodity prices increased farm incomes and agricultural land values while the housing market collapsed (Figure A.3). For additional deposits or collateral to affect banking concentration, the increases generated by subsidy payments would need to be substantial relative to banks' existing funding and farmers' existing collateral.

Even for small banks, subsidy payments are small relative to their deposits. A within-county one-standard-deviation increase in annual payments is $1.15 million. If this entire increase were deposited at a single small bank, it would equal only about 1% of the median small bank's $104 million in total deposits.[11] The implied gains in land values are similarly small. Kirwan (2009) estimates that about a quarter of each subsidy dollar accrues to landowners as higher land rents. If landowners' share of each payment were fully priced into farmland, this estimate implies that payments increase county land values by at most about 1%, an increase that is within the confidence intervals reported in Panel C for the first decade.[12] However, for a representative farm to finance an expansion from 1,000 to 1,500 acres by borrowing against the land it already owns, that land would have to appreciate by about 65%.[13]

Farm consolidation may increase agricultural credit demand without substantially increasing land values when larger farms expand by renting land instead of purchasing it. Subsidy payments reallocate harvested acreage from midsize farms toward the largest farms (Table A.3). Among farms operating both owned and rented land, rented acreage increases more than owned acreage. This result suggests that larger farms can expand their operations and associated financing needs without a proportional increase in land purchases, consistent with the small effect of subsidy payments on land values in Panel C.

[11] Median total deposits across banks with less than $1 billion in inflation-adjusted assets and at least one branch in a sample county, computed from Summary of Deposits data over the sample years.

[12] A within-county one-standard-deviation increase in payments is about $1.15 million per year. Applying Kirwan's 25% incidence estimate, landowners collect about $290,000 per year of this flow in higher rents. Capitalized as a perpetuity at 6%, the long-run average 10-year Treasury rate used by the Economic Research Service (Nickerson et al., 2012), the additional rent adds about $4.8 million to county land values, a 0.5–1.0% increase against county farmland worth $0.5–0.9 billion. This calculation likely overstates the increase in land values for the farms driving the expansion because Kirwan and Roberts (2016) finds that rental incidence decreases by 5–15 cents when total operated acreage doubles.

[13] The representative farm owns 600 of its 1,000 acres (United States Department of Agriculture, National Agricultural Statistics Service, 2004), values land at 2002 Corn Belt cropland values of $2,340 per acre (United States Department of Agriculture, National Agricultural Statistics Service, 2003), borrows against land up to the 85% statutory maximum under the Farm Credit Act (12 U.S.C. §2015(b)), and uses $350 per acre in operating credit (United States Department of Agriculture, Economic Research Service, 2006). Expanding to 1,500 acres under the same tenure share requires about $780,000 in additional credit: a mortgage on the 300 newly owned acres and operating credit on the added acreage. Borrowing $780,000 against appreciation at the 85% cap requires the farm's 600 owned acres to gain about $918,000 in value, a 65% increase.

I also test farm concentration, land values, and deposits per capita jointly, allowing residuals to be correlated across outcomes within a county. The results show that farm concentration increases with subsidy payments, but land values and deposits per capita do not increase over the same period (Table A.4). Last, subsidy payments do not lead to greater agricultural investment or productivity. Crop yields and the share of acreage planted in program crops do not change, and machinery and new tractors decrease (Table A.5). These results indicate that payments primarily increase farm concentration.

7.2 Falsification Tests: Terrain and Never-takers

If payments affect local banking through channels independent of farm consolidation, such as land values or deposits, local banking markets should also change in counties where subsidy payments do not have a large effect on farm concentration. To isolate the borrower consolidation channel, I set up two tests where farm concentration does not respond to subsidy payments.

First, I use variation in terrain ruggedness. Large agricultural machinery operates more efficiently on flat land, so rugged terrain reduces the benefits of farm expansion (MacDonald, Korb, and Hoppe, 2013). Table 2 shows that in rugged counties, where expanding acreage is more difficult, subsidy payments have little effect on farm concentration and do not affect the structure of local banking markets. In flat counties, subsidy payments have a larger effect on farm concentration and increase large banks' number of branches and deposits per capita, with effects about two to three times as large as those for small banks.

[Insert Table 2 Here]

The difference between flat and rugged counties remains after controlling for population, urbanization, and land values (Table A.6).

Second, I compare counties where subsidy payments have different predicted effects on farm concentration. Following Acemoglu et al. (2022), which applies the intuition of D'Haultfœuille, Hoderlein, and Sasaki (2024), I calculate each county's predicted long-run change in farm concentration by multiplying the estimated effect of subsidy payments on farm concentration by the county's change in subsidy payments from 1994 to 2014. I sort counties into terciles based on their predicted responsiveness to subsidy payments and compare the effects of payments across the three groups. Counties in the bottom tercile, where farm consolidation is least responsive to subsidy payments, are interpreted as "never-takers." Counties in the top tercile are those where farm consolidation is most responsive to subsidy payments.

Table 3 shows that in the least responsive counties, a within-tercile one-standard-deviation increase in subsidy payments decreases farm concentration by about 42 acres and does not increase banks' presence, branches, or deposits. In the most responsive counties, the same increase in subsidy payments increases farm concentration by about 141 acres. Large banks' number of branches increases by about 5%, and their deposits per capita by about 11%, about twice the increases for small banks.

[Insert Table 3 Here]

I apply the same tercile analysis to land values and total deposits per capita, restricting the sample to the period before the 2010–2013 farmland boom. Payments do not increase either outcome in any tercile (Table A.7). These results suggest that farm consolidation is the primary channel through which subsidy payments affect local banking markets.

7.3 Farm Concentration and Banking Concentration

Large banks tend to serve larger firms, so they may gain market share when larger farms acquire land from smaller farms. Small banks, which specialize more in relationship lending to smaller borrowers, may instead exit, merge, or be acquired (Stein, 2002; Berger et al., 2005). This prediction is consistent with the broader IO idea that changes in the distribution of buyers may affect the organization of suppliers (Snyder, 1996). I use subsidy payments as an instrument for farm concentration to identify whether borrower consolidation increases local banking concentration.

Table 4 reports the effect of subsidy payments on farm concentration and on banking concentration, followed by IV and OLS estimates of the effect of farm concentration on banking concentration. In Panel A, a within-county one-standard-deviation increase in subsidy payments increases farm concentration by about 105–107 acres. For the average corn farmer in my sample, an additional 107 base acres would generate $2,012 in extra fixed cash payments.[14] The same increase in subsidy payments increases the bank deposits HHI by about 64 points.

[Insert Table 4 Here]

Having established that subsidy payments increase banking concentration, I next examine whether borrower consolidation is the primary channel through which these payments affect the structure of local banking markets. Instrumenting for farm concentration with subsidy

[14]The average historical program yield for corn in my sample is 79 bushels per acre. At the 2002 payment rate of $0.28 per bushel and the program's adjustment factor of 0.85, an additional 107 base acres would generate $79 \times 0.28 \times 107 \times 0.85 = \$2{,}012$.

payments, I find that a within-county one-standard-deviation increase in farm concentration increases the bank deposits HHI by about 252 points, or 8% relative to its unconditional mean (Panel B). An increase of this size is comparable to one out of seven equally sized banks exiting a county.[15] This result suggests that borrower consolidation is an important determinant of local banking concentration. I use these estimates to construct a counterfactual that removes the subsidy-induced component of farm consolidation and measures its contribution to banking concentration. Absent subsidy-induced farm consolidation, local banking concentration would have been about 13% lower by 2014.

Panel B also reports OLS estimates that do not correct for the endogeneity of farm concentration. These coefficients are smaller than the IV estimates, which may be due to downward bias from measurement error in the midpoint. When I instrument the twenty-year change in the midpoint with changes in alternative measures of the farm size distribution, the coefficient is similar to the IV estimate (Table A.11).[16] Following Ishimaru (2024), I decompose the gap into differences in covariate weights, treatment-level weights, and marginal effects to further verify that the gap between the IV and OLS coefficients does not come from differences in how the two estimators weight observations. The marginal effect component, which collects measurement error and unobserved shocks that jointly determine farm structure and local banking markets, constitutes between 95% and 99% of the gap across specifications, including one that controls for land values and deposits (Table A.12).[17]

Last, to test whether banking concentration increases progressively as farms become more concentrated, I follow Saidi and Streitz (2021) and measure changes in the bank deposits HHI across deciles of fitted lagged farm concentration. Figure 4 shows that the bank deposits HHI increases steadily across the deciles, consistent with greater borrower consolidation increasing banking concentration.

[Insert Figure 4 Here]

[15] Assuming deposits are initially evenly distributed across the seven banks and that when one bank exits, its deposits are distributed equally across the six remaining banks. Results are similar when I use the share of harvested acres on farms of at least 1,000 acres or the log number of farms as alternative measures of farm concentration (Table A.8), and in a first-difference specification that addresses concerns about autocorrelated residuals (Table A.9). Price-based measures also indicate greater bank market power. Banks' total interest expense beta (Drechsler, Savov, and Schnabl, 2021) and deposit interest expense beta (Narayanan, Ratnadiwakara, and Strahan, 2025) both decline with farm concentration, indicating that deposits are held at banks that pass less of a change in policy rates through to depositors (Table A.10).

[16] The alternative measures come from census years inside the twenty-year window, so they do not share the midpoint's survey error.

[17] The covariate weight component reflects differences in the weights the two estimators place on counties with different observable characteristics, and the treatment-level weight component reflects differences in the weights placed on ranges of farm concentration.

7.4 Farm Concentration and the Structure of Local Banking Markets

Borrower consolidation also alters the competitive dynamics of local banking. Unlike many buyer-supplier relationships, bank-firm relationships involve information asymmetries and monitoring costs that give incumbent lenders relationship-specific advantages. Small community banks have historically relied on relationship lending to serve smaller or more opaque borrowers (Haynes, Ou, and Berney, 1999; Berger et al., 2005). However, as farms consolidate and expand, large banks may become better positioned to serve the local market because they have greater funding capacity, access to wholesale funding and internal capital markets, and higher regulatory lending limits.[18]

Table 5 shows that large banks are more likely than small banks to enter the consolidating markets. Following a within-county one-standard-deviation increase in farm concentration, the probability of large bank presence increases by about 5 percentage points, compared with about 2 percentage points for small banks. These results suggest that large banks enter as the composition and scale of local credit demand change.

[Insert Table 5 Here]

Large banks also expand their branch networks and deposits more than small banks. Branch networks show whether banks commit physical resources to a market (Cohen and Mazzeo, 2010), and deposits per capita reflect the extent to which customers move deposits across institutions. Large banks expand their local branch presence by about 16%, compared with 10% for small banks, and increase their deposits per capita by about 33%, compared with 22% for small banks. These results remain similar after excluding the 2008 Financial Crisis and after controlling for land values, total county deposits, and production measures (Tables A.13 and A.14). These findings indicate that large banks gain a disproportionate share of market growth in counties with consolidating borrowers.

In contrast, small banks consolidate among themselves. Table 6 shows that mergers and acquisitions among small banks increase by about 23%, whereas mergers between two large banks decline and mergers in which a large bank acquires a small bank do not change. These results suggest that small banks respond to competitive pressure from larger banks and changes in credit demand by merging among themselves. The remaining small banks become fewer but larger.

[18]For literature studying large banks' funding capacity and access to wholesale funding, see LaDue and Duncan (1996); DeYoung, Hunter, and Udell (2004); and Park and Pennacchi (2009). For literature studying internal capital markets, see Stein (1997) and Cortés and Strahan (2017). For regulatory lending limits, see Office of the Comptroller of the Currency (2004).

[Insert Table 6 Here]

These findings align with theoretical predictions that structural changes in downstream markets, such as an increase in borrower size, can trigger consolidation among upstream suppliers (Galbraith, 1952; Snyder, 1996).

7.5 Mechanisms: Agricultural Lending and Small Bank Constraints

This section studies how farm consolidation affects agricultural lending and why large banks gain market share. First, I study which banks provide additional agricultural credit and whether the increase in lending is concentrated among larger farms and larger loans. Second, I study the funding and lending constraints that limit small banks' ability to increase agricultural lending. Last, I test whether farm consolidation has larger effects on banks that specialize in agricultural lending.

7.5.1 Farm Concentration and Agricultural Lending

If farm consolidation drives changes in local banking, the effects should be strongest in agricultural credit markets, where borrower consolidation is most directly relevant. As farms consolidate, their financing needs increase, allowing large banks to increase agricultural lending and gain local deposit market share. I study agricultural lending by bank size, farm revenue, and loan size.

I test whether large banks increase lending to larger farms and, if so, in which loan size categories using CRA loan originations. CRA loan data directly measure county-level bank lending to farms and report borrower revenue and loan size.[19] Table 7 shows that a within-county one-standard-deviation increase in farm concentration increases large banks' lending to farms with more than $1 million in annual revenue by about 30%. The increase is concentrated in loans between $250,000 and $500,000, the largest loan size category reported in the CRA data. In contrast, large banks do not increase lending to farms with $1 million or less in annual revenue or in the smallest loan size category of $100,000 or less. These differences suggest that large banks' additional agricultural lending is concentrated among larger borrowers and larger loans, consistent with their specialization in serving larger firms.

[Insert Table 7 Here]

[19]CRA reporting requirements apply only to banks above an asset threshold. Through 2004, the threshold was $250 million in bank assets or $1 billion in holding company assets. Beginning in 2005, it increased to about $1 billion, and most small banks stopped reporting, so I restrict the main analysis to large banks.

Among the subset of small banks reporting farm loan originations under the CRA through 2004, lending does not increase in any borrower revenue or loan size category (Table A.15).

To study how total agricultural lending changes at both large and small banks, I complement the CRA loan analysis with agricultural lending imputed from bank Call Reports. Because Call Reports provide each bank's agricultural lending only at the bank level, I allocate each bank's lending across counties where it operates branches in proportion to farm interest expenses, which proxy for the geographic distribution of agricultural credit demand for each loan type. Table 8 shows that the increase in agricultural lending is concentrated at large banks. A within-county one-standard-deviation increase in farm concentration increases large banks' total agricultural lending by about 46%. Within total agricultural lending, real estate farm lending increases by about 40% and non-real estate farm lending by about 46%, which suggests that the increase in large banks' agricultural lending is not due to an advantage in valuing real estate collateral. Small banks' total agricultural lending and real estate farm lending do not increase. Their non-real estate farm lending increases by about 6%, about one eighth of the increase at large banks.

[Insert Table 8 Here]

The results are robust to including counties where banks originate CRA farm loans outside their branch networks, with the allocation adjusted for distance (Table A.16). These results suggest that large banks gain local deposit market share in part by providing more agricultural credit as farms consolidate.

7.5.2 Small Bank Funding Constraints

Why does agricultural lending increase at large banks but not at small banks? Small banks fund their lending mainly from local deposits, whereas large banks can finance larger loans by moving funds across branches through their internal capital markets (Gilje, Loutskina, and Strahan, 2016; Cortés and Strahan, 2017).[20] For small banks that rely primarily on local deposits, financing expanding farms may absorb much of the funding available for new lending. Using the same farm example as in Section 7.1, an expansion from 1,000 to 1,500 acres requires about $780,000 in additional credit, about a third of the median small bank's annual change in core deposits.[21] Financing several such expansions could require a substantial share of a small bank's annual deposit growth. When deposit growth is limited,

[20] Only 278, or about 0.2%, of 116,057 syndicated loans from 1994 to 2014 are to U.S. crop farms (DealScan, SIC 01).

[21] The median annual change in core deposits across all small-bank-years from 1997 to 2007, including years in which core deposits declined, is $2.3 million, and core deposits increased by less than $780,000 or declined in 36% of bank-years.

the composition of a bank's existing funding may determine how much it can lend. Core deposits are a stable source of funding, so banks with a larger share of core deposits may be better able to increase lending when farms expand (Berlin and Mester, 1999; Cornett et al., 2011).

To test whether small banks with more stable funding are better able to increase lending as farms consolidate, I split small banks according to their core deposit share before the FAIR Act. Because only a subset of small banks report under the CRA, I use the agricultural lending imputed from bank Call Reports, as in Table 8. Table 9 shows that farm concentration increases agricultural loans above $500,000 at small banks with a larger share of core deposits, but not at small banks with a smaller share. Loans of $100,000 or less do not increase in either group. The additional lending at small banks with a larger share of core deposits is concentrated in larger loans, which suggests that it does not reflect a general increase in agricultural lending. Farm concentration also increases both real estate and non-real estate farm lending at these banks, indicating that their greater lending activity is not due to an advantage in lending against real estate collateral. These results suggest that stable deposit funding can help small banks compete for larger agricultural loans, even though large banks gain market share overall.

[Insert Table 9 Here]

Small banks' ability to increase agricultural lending may also depend on how much of that funding is already committed to existing loans. Small banks with both a larger share of core deposits and unused lending capacity, a pre-program ratio of loans to deposits below the median, show the largest increases in agricultural loans above $500,000 (Table A.17).

Lending capacity constraints may drive banks' differential responses to farm consolidation. In counties where production requires more fixed credit per acre, small banks may face binding funding constraints at smaller farm sizes, allowing large banks to gain more market share in counties with higher production costs. I classify counties as high- or low-cost by the average economic cost per acre of their crops before the program, weighted by harvested acres (Figure A.2). Large banks increase deposits per capita in high-cost counties but not in low-cost counties, whereas small banks increase deposits per capita about three times more in low-cost counties than in high-cost counties (Table A.18). Land values appreciate by similar amounts in the two groups (Figure A.4). Large banks gain more market share in counties where production requires more credit per acre, consistent with small banks' funding constraints limiting their ability to finance larger farms.

A related constraint comes from regulatory limits on lending to a single borrower. The Office of the Comptroller of the Currency (OCC) restricts these loans to 15% of a bank's

capital and surplus. In 2004, the OCC noted that farm consolidation had "resulted in fewer, but larger, farms with expanded credit needs" and that existing limits were too restrictive for community banks (Office of the Comptroller of the Currency, 2004). Over time, higher land values and production costs further increased the credit that expansion requires. For example, a loan that financed about 290 acres of central Illinois farmland in 2001 financed about 110 acres in 2012 (Ellinger, 2012).

7.5.3 The Structure of Agricultural Banking

The prior results are consistent with IO theory and suggest that relationship lending does not limit banks' ability to reorganize, but this finding does not necessarily imply that informational frictions are irrelevant. Agricultural lending becomes more concentrated as farms consolidate. Table 10 shows that a within-county one-standard-deviation increase in farm concentration increases the concentration of agricultural lending by 14%, nearly twice the increase in deposit market concentration.

[Insert Table 10 Here]

For the remaining outcomes, I restrict the sample to agricultural banks, defined as banks whose farm loans exceed 5% of total loans in any of the preceding six years.[22] Large agricultural banks are 15 percentage points more likely to be present, more than three times the increase for large banks overall. Their deposits per capita increase by about 75%, compared with about 24% at small agricultural banks. Large agricultural banks also increase their number of branches by about 55%, but small agricultural banks do not. These results suggest that both bank size and agricultural expertise affect how banks adjust to borrower consolidation.

To quantify the reallocation of market share, I compare the observed split of subsidy-induced deposit growth between large and small agricultural banks with a counterfactual split in proportion to each group's deposit share before the FAIR program was enacted. In counties where large agricultural banks operate, subsidy-induced deposit growth is about 49% at large agricultural banks and about 7% at small agricultural banks. Had the same total growth been divided in proportion to the two groups' initial shares, small agricultural banks' deposits would have grown by about 17 percentage points, or about $47 million, more per county.

[22]The results are monotonic with respect to the 5% threshold. As the share of agricultural lending decreases below 5%, the estimated effects of farm concentration on large agricultural banks' presence and deposits are smaller (Table A.19).

7.6 Farm Concentration and Interest Rates on Broader Consumer Products

IO theory suggests that buyer consolidation can influence prices through two distinct channels. First, consolidated buyers may gain bargaining power over suppliers, negotiating more favorable prices and pushing suppliers to compete on price for all buyers in the market (Snyder, 1996; Chipty and Snyder, 1999). Second, consolidation may trigger restructuring on the supplier side, leading to concentration among the most efficient suppliers, which can result in unchanged or even higher prices (Galbraith, 1952; Tirole, 1988; Farrell and Shapiro, 1990). In agricultural credit markets, Kropp and Whitaker (2011) finds that farms with a greater share of base acres, and thus larger subsidy payments relative to similarly sized farms, obtain lower interest rates on their operating loans.

However, the broader effects of borrower consolidation and the structure of local banking markets on consumer financial products are less clear. New banks entering consolidated markets may lower credit prices to attract customers. Banks may instead increase credit prices if they can segment consumer markets from consolidated borrowers (Inderst and Valletti, 2011). Any pricing effects may also differ between large and small banks. Large banks tend to offer lower deposit rates because they have access to wholesale funding, whereas small banks may offer higher deposit rates to attract local retail deposits and fund their lending activities (Park and Pennacchi, 2009).

To assess these spillover effects, I examine how borrower consolidation affects the pricing of consumer deposit and loan products. Among deposit products, I focus on three commonly offered types: time, savings, and checking deposits. These deposit types differ in liquidity, interest elasticity, and their role in bank funding strategies. Liquid deposits, such as savings and checking deposits, provide a stable and relatively inelastic source of funds. Because depositors rarely withdraw these funds in response to rate changes, banks can maintain low-cost funding even as they reduce offered interest rates. In contrast, time deposits are more sensitive to interest rates and are often used to fund commercial lending, including agricultural loans (Drechsler, Savov, and Schnabl, 2017; Supera, 2021). I use real, inflation-adjusted interest rates in all analyses. In addition to the baseline farmland and local economic condition controls, I control for changes in local household and farm credit demand by including county-level measures of total deposits and total farm interest expenses.

Panel A of Table 11 shows that a within-county one-standard-deviation increase in farm concentration decreases the county-level average savings deposit interest rate by 32 basis points and the checking deposit interest rate by about 22 basis points, or about 26% and 14% relative to their unconditional means. The highest savings deposit interest rate offered

by any bank branch in the county also decreases, by more than the lowest rate offered (Table A.20). County-level time deposit interest rates increase by about 13 basis points, consistent with banks offering higher rates on deposits that are more sensitive to interest rates to preserve their funding base.

[Insert Table 11 Here]

I also analyze offered interest rates on auto loans, which are standardized, broadly held products unrelated to farm lending, and subject primarily to local competition (Argyle, Nadauld, and Palmer, 2023). The county-level average auto loan interest rate increases by about 103 basis points.[23] The lowest and highest auto loan interest rates offered by any bank branch in the county also increase, so households that compare rates across the county's branches still face higher auto loan interest rates (Table A.20). The result is similar in first differences (Table A.21). Mortgage interest rates should be less sensitive to local banking conditions because mortgages are commonly sold into national secondary markets and priced using national benchmarks (Fuster, Lo, and Willen, 2024). Consistent with this prediction, the county-level average mortgage interest rate does not increase.

Small and large banks price consumer products differently as farms consolidate. Panels B and C report the effects separately for small and large banks, comparing interest rates offered by the same bank in the same county over time. Auto loan interest rates increase by about 97 basis points at large banks but decrease by about 37 basis points at small banks. Small banks also increase time deposit interest rates by about 30 basis points, compared with a decrease of about 18 basis points at large banks. Interest rates on savings and checking deposits decrease at both large and small banks. Mortgage interest rates increase by only about 11 basis points at large banks, about 4% relative to the unconditional mean, with no corresponding increase at small banks. The small magnitude is consistent with mortgage rates being less sensitive to local banking conditions. These results suggest that small banks compete by increasing time deposit interest rates to secure local funding and by lowering auto loan interest rates to attract consumer borrowers, while offering lower rates on deposits that are less sensitive to interest rate changes. Large banks, on the other hand, reduce deposit interest rates and increase auto loan interest rates, consistent with greater pricing power in more concentrated local banking markets.

[23]This magnitude is within the range of price differences that search frictions sustain in auto lending, where borrowers on opposite sides of a credit score threshold at the same lender face rate differences of about 130 basis points (Argyle, Nadauld, and Palmer, 2023) and dealer-arranged loans carry discretionary markups averaging about 113 basis points (Grunewald et al., 2023).

7.7 Robustness

This section tests whether the subsidy program changed government lending, controls for greater crop insurance take-up and other government payments, and considers counties where farming is not central to the local economy. Last, I also control for the number of Farm Credit System branches, which provide agricultural credit outside the commercial banking system and compete with banks for farm borrowers.

7.7.1 Entry and Credit Demand

Although my main results show that fixed subsidy payments increase farm consolidation, one alternative mechanism is that these payments decrease income risk and encourage new entry into farming. Subsidy payments may also simply increase credit demand more broadly rather than specifically among large, consolidating operations. If either of these mechanisms were at work, federal direct loans, which are targeted to small, new, or socially disadvantaged farmers, should increase. However, I find no evidence that subsidy-induced consolidation affects the number or volume of direct loans at the county-year level (Table A.22).

These results support the interpretation that subsidy payments primarily finance the expansion of existing farms and not new entry by smaller farms, which would decrease the midpoint farm size. The absence of an increase in direct loans is consistent with consolidation driven by incumbent farms and not by broader access to credit.

7.7.2 Crop Insurance

Crop insurance decreases downside risk, improves credit access, and supports farm expansion (Cornaggia, 2013; Azzam, Walters, and Kaus, 2021; Ifft et al., 2024), so greater take-up may affect these results. Controlling for the log number of county-level crop insurance policies leaves the effects of payments on farm concentration and of farm concentration on banks largely unchanged (Table A.23).

7.7.3 Countercyclical Payments

The 2002 FSRI Act included countercyclical payments, which were paid on base acres when crop prices were below thresholds set by the FSRI Act (Monke, 2006). To address the possibility that these payments also affected local banking markets, I control for predicted countercyclical payments, constructed using the same approach as predicted subsidy payments. The main results remain similar (Table A.24).

7.7.4 Urban Counties

If farm consolidation drives changes in the structure of agricultural credit markets, the effect should be muted in urban counties where farming is not a major part of the local

economy. For this test, I add back the large central metropolitan counties excluded from the main sample. I find no relationship between farm concentration and banking concentration in these urban counties. Although the number of urban observations is small, the instrument strengthens and the effect increases in progressively more rural counties. These results suggest that farm consolidation affects banking concentration most where the consolidating sector is central to the local economy, and that the effect is not driven by broader macroeconomic trends in banking (Table A.25).

7.7.5 The Farm Credit System

Nonbank lenders may also affect the structure of local commercial banking. In agricultural credit markets, the FCS is the main alternative to commercial banks. Established in 1916 as a government-sponsored enterprise, the FCS is a borrower-owned cooperative system designed to provide credit to farmers and ranchers who historically faced limited access to commercial bank financing. FCS institutions may lend only to persons owning agricultural land or engaged in the production of agricultural commodities, and they are prohibited from offering nonfarm loans. In 2016, the FCS held nearly 41% of the farm sector's total debt, compared with the 42% share held by commercial banks.[24]

Because the FCS competes with commercial banks for agricultural borrowers, counties with more FCS branches may experience different levels of commercial banking concentration, deposit holdings, or branching activity. To address this possibility, I control for the log number of FCS branches in each county. The effects of farm consolidation on commercial bank outcomes remain similar. These findings suggest that the observed reorganization of commercial banks in consolidating agricultural markets is not driven by variation in FCS presence across counties (Table A.26).

8 Conclusion

In this paper, I introduce a new driver of bank consolidation: the consolidation of borrowers. Using U.S. agriculture as a setting, I show that farm consolidation, driven by fixed federal subsidies, increases local banking concentration. As farms consolidate, large banks increase their presence, branch networks, deposits, and agricultural lending, while small banks consolidate through mergers. Large banks gain deposit market share particularly where farms require more capital, and small banks increase their agricultural lending above $500,000 only when they have a larger share of core deposits. Changes in bank composition also affect the local community. Following an increase in borrower concentration, large banks

[24] See Farm Credit System (https://www.congress.gov/crs_external_products/RS/HTML/RS21278.html).

reduce deposit interest rates and increase auto loan interest rates. In contrast, small banks increase time deposit rates and lower auto loan rates, potentially to attract retail customers and deposits to fund lending.

Large banks may also withdraw from these markets when farm incomes decrease. When farm incomes decreased in the decade following the subsidy program, the earlier expansion reversed, and several of the largest banks decreased their farm lending.[25] While borrower consolidation may attract large banks in the short run, it may also decrease long-run financial stability.

The effect on banking should be strongest where firms depend heavily on bank credit, so changes in firm size alter the distribution of credit demand across banks. For example, firms in construction, trucking, and manufacturing rely on bank debt to finance investment. Where firms rely less on bank credit or credit is supplied through national or securitized markets, the same mechanism should be weaker, consistent with the absence of an effect of borrower consolidation on county-level average mortgage interest rates.

[25]See Rabobank takes on US agriculture's lending gap as large banks exit, Big banks exiting farm loan business, and Wall Street banks bailing on troubled U.S. farm sector.

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Figure 1. Timeline of U.S. Farm Policy Changes from 1996 to 2014

This figure illustrates the timeline of base acre reference periods and key policy events under successive U.S. farm bills. The 1996 Federal Agriculture Improvement and Reform (FAIR) Act introduced fixed direct payments based on average planted acres from 1981 to 1985. The 2002 Farm Security and Rural Investment (FSRI) Act expanded the program to include oilseeds and peanuts and allowed base acres to be updated to reflect average plantings from 1998 to 2001. The program was renewed in the 2008 Food, Conservation, and Energy Act and ended in the 2014 Agricultural Act.

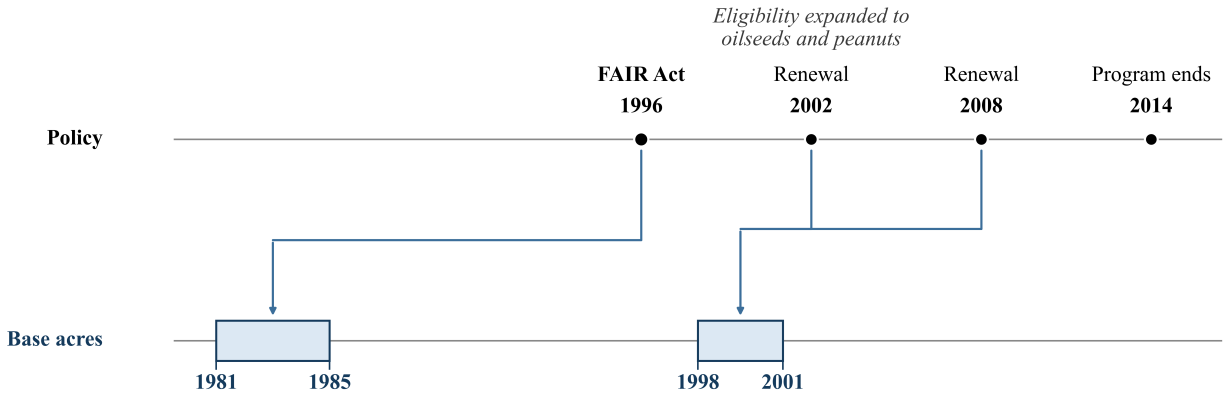
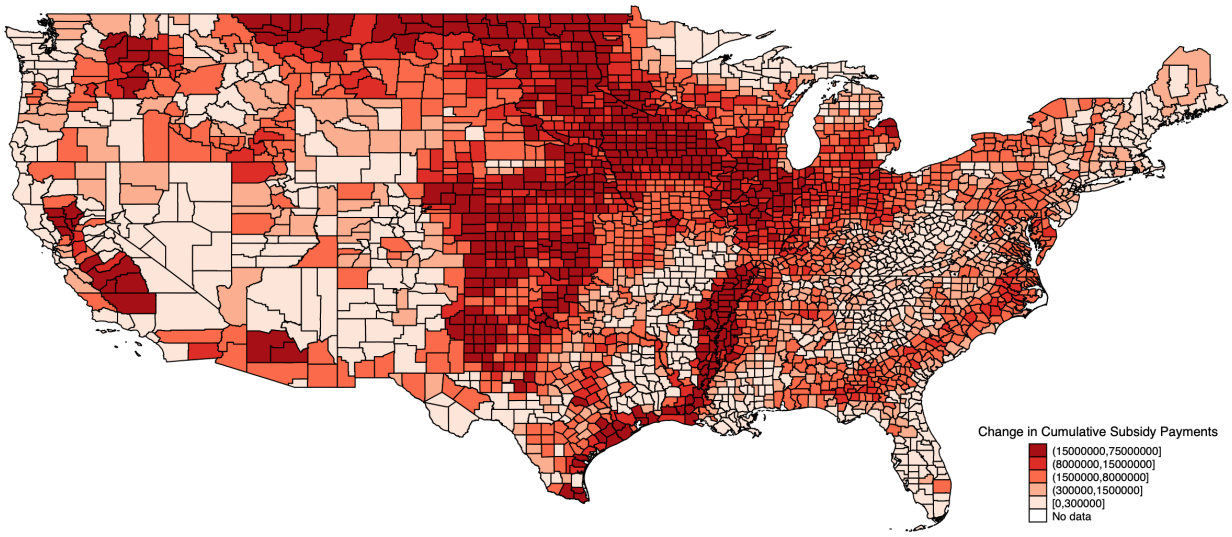


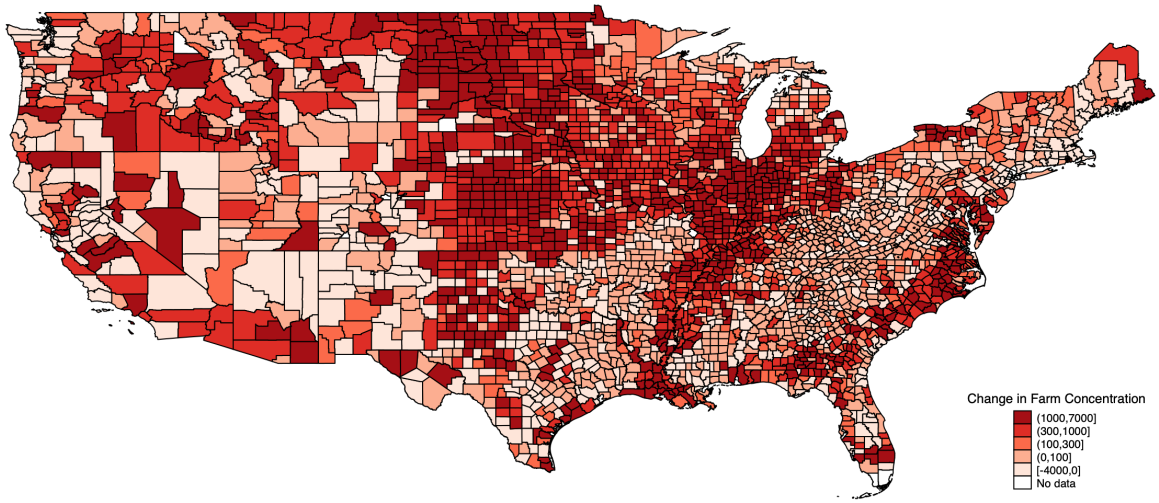
Figure 2. Changes in County-Level Subsidy Payments, Farm Concentration, Land Values, and Deposits

This figure maps the change in subsidy payments and in the three candidate channels across counties. Panel A shows the change in cumulative subsidy payments per county between 1992 and 2012. Panel B shows the change in farm concentration between 1992 and 2012. Panel C shows the change in land values per acre between 1992 and 2012. Panel D shows the change in total deposits per capita between 1994 and 2014.

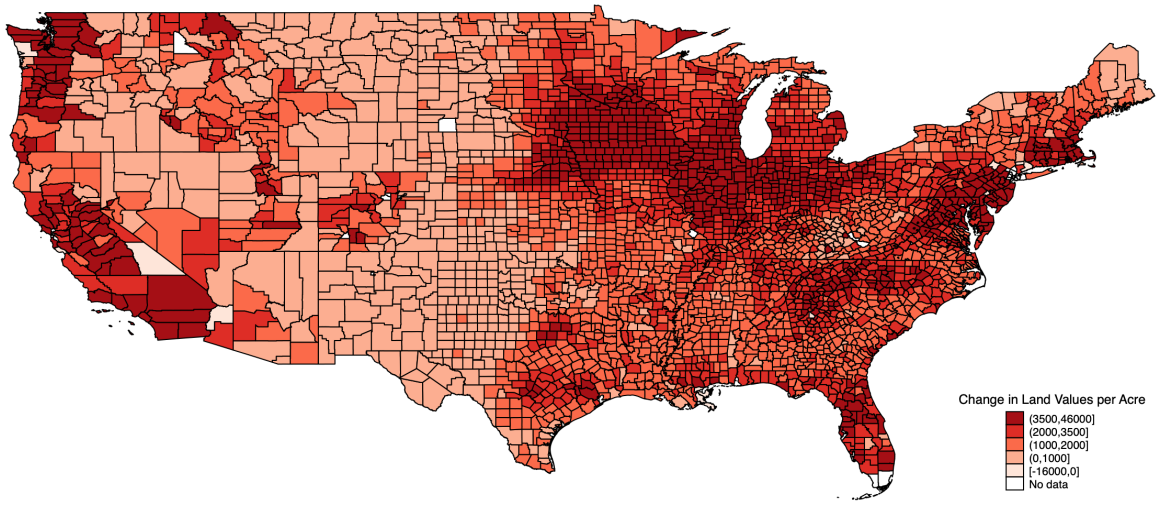
Panel A. Change in Cumulative Subsidy Payments between 1992 and 2012



Panel B. Change in Farm Concentration between 1992 and 2012



Panel C. Change in Land Values per Acre between 1992 and 2012



Panel D. Change in Total Deposits per Capita between 1994 and 2014

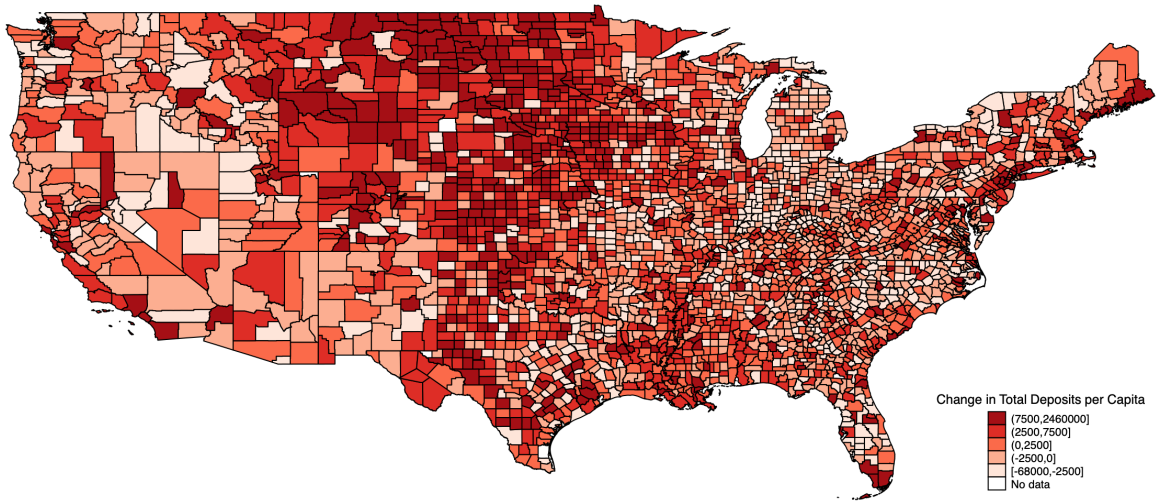
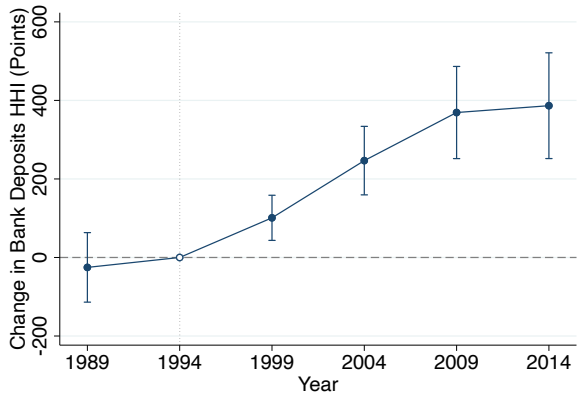


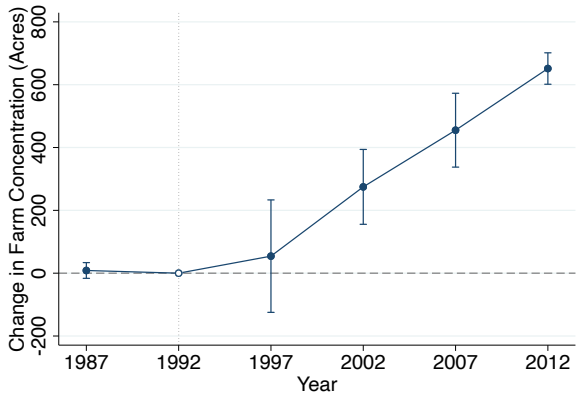
Figure 3. Effect of Subsidy Amount on Banking Concentration and Candidate Channels

This figure plots the dynamic effect of Subsidy Amount on the bank deposits HHI, farm concentration, land values, and deposits per capita, with 95% confidence intervals. Estimates are obtained using the difference-in-differences estimator developed by de Chaisemartin and D'Haultfœuille (2026), which accommodates continuous treatments. The specification supplied to the estimator is: $Y_{i,t} = \alpha\, \textit{Subsidy Amount}_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$, and each panel plots the dynamic coefficients for the periods relative to the change in $\textit{Subsidy Amount}_{i,t}$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. Panels A–D report the bank deposits HHI, farm concentration, the log market value of agricultural land and buildings per acre, and total county deposits per capita, respectively. The bank deposits HHI is a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. The sample excludes large central metropolitan counties and is observed in five-year snapshots. Panels A and D use banking years 1994–2014, and Panels B and C use Census of Agriculture years 1992–2012. Subsidy Amount is a within-county standardized measure of cumulative total fixed subsidy payments. In Panels A and D, the controls are log income per capita, the employment rate, and indicators for state-level banking deregulation, and Panel D additionally includes the log number of bank branches in the county. In Panels B and C, the controls are log income per capita, the employment rate, and lagged measures of idle cropland, pastureland, and cropland left fallow, lost to crop failure, or otherwise unharvested from the prior Census of Agriculture. All panels allow nonparametric trends across state-level banking deregulation regimes.

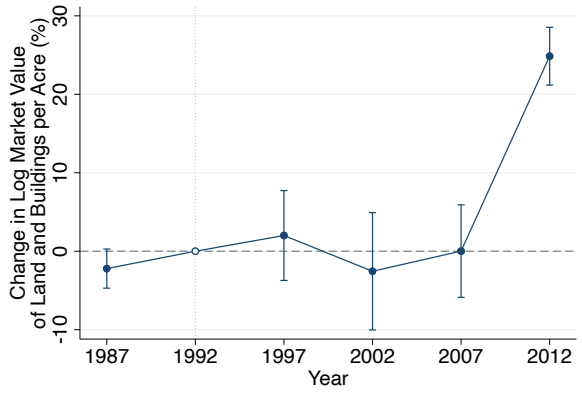
Panel A. Bank Deposits HHI



Panel B. Farm Concentration



Panel C. Log Land Value Per Acre



Panel D. Total Deposits per Capita

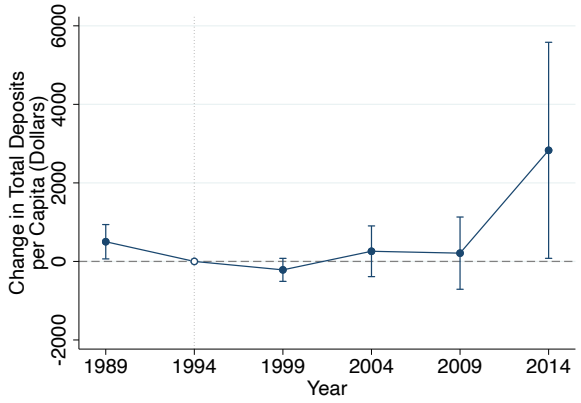


Figure 4. Effect of Farm Concentration on Banking Concentration

This figure illustrates the effect of fitted Farm Concentration$_{i,t-2}$ on the Bank Deposits Herfindahl-Hirschman Index$_{i,t}$ (HHI), following Saidi and Streitz (2021). The plotted coefficients are second-stage coefficients on deciles of fitted Farm Concentration$_{i,t-2}$, with 95% confidence intervals. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \sum_{k=2}^{10} \beta_k\, \mathbf{1}\left(Farm\ Concentration_{i,t-2} \in D_k\right) + \beta_{11}\hat{\varepsilon}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$, where D_k denotes the k-th decile of fitted farm concentration and the first decile is the omitted category. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. The outcome variable, Bank Deposits HHI$_{i,t}$, is a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. In both stages, the controls are log income per capita, the employment rate, indicators for state-level banking deregulation, and lagged measures of idle cropland, pastureland, and cropland left fallow, lost to crop failure, or otherwise unharvested from the prior Census of Agriculture.

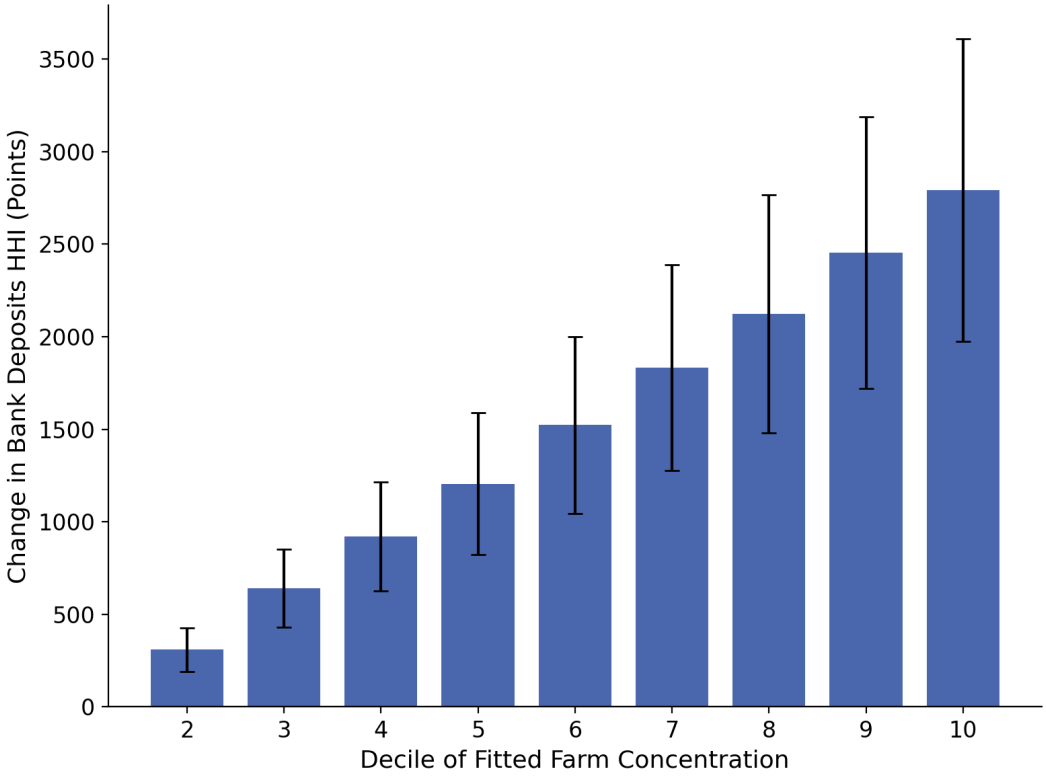


Table 1. Descriptive Statistics

The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Subscript i represents the county and t represents the year. $Farm\ Concentration_{i,t-2}$ is the midpoint farm size at the county level. $Bank\ Deposits\ HHI_{i,t}$ is the Herfindahl-Hirschman Index, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. $Agricultural\ Loan\ HHI_{i,t}$ is calculated using the same method based on agricultural loan shares held by each bank within a county. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. $Probability\ of\ Large\ Bank\ Entry_{i,t}$ and $Probability\ of\ Small\ Bank\ Entry_{i,t}$ are indicator variables for whether a large or small bank, respectively, is present in a county in a given year. $Number\ of\ Large\ Bank\ Branches_{i,t}$ and $Number\ of\ Small\ Bank\ Branches_{i,t}$ refer to the number of branches operated by large and small banks at the county level, respectively. $Large\ Banks'\ Deposits\ Per\ Capita_{i,t}$ and $Small\ Banks'\ Deposits\ Per\ Capita_{i,t}$ measure total deposits per capita held at large and small banks, respectively. $Large\ Acquires\ Small_{i,t}$ measures the total number of mergers or acquisitions in which a large bank acquires a small bank, aggregated over five-year intervals. $Small\ Acquires\ Small_{i,t}$ and $Large\ Acquires\ Large_{i,t}$ measure the number of mergers between two small or two large banks, respectively, also aggregated over five-year intervals. $Total\ Agricultural\ Loans_{i,t}$ is the total imputed agricultural loan volume at the county level, disaggregated by bank size. $Real\ Estate\ Farm\ Loans_{i,t}$ and $Non\text{-}Real\ Estate\ Farm\ Loans_{i,t}$ represent county-level imputed loan volumes secured by farmland and used to finance agricultural production, respectively. $Total\ Agricultural\ Loan\ Growth_{i,t}$, $Real\ Estate\ Loan\ Growth_{i,t}$, and $Non\text{-}Real\ Estate\ Loan\ Growth_{i,t}$ are large or small banks' lending in each loan type relative to the prior period's total volume of that loan type. $Probability\ of\ Large\ Agricultural\ Bank\ Entry_{i,t}$ and $Probability\ of\ Small\ Agricultural\ Bank\ Entry_{i,t}$ are indicator variables for whether a large or small agricultural bank, respectively, is present in a county in a given year, where agricultural banks are those whose farm loans exceed 5% of total loans in any of the preceding six years. $Urban_{i,t}$ is an indicator variable for whether the county is classified as a large central metropolitan area by the National Center for Health Statistics (NCHS). $Metro_{i,t}$ is an indicator variable for whether the county is a broader metropolitan area, and $Rural_{i,t}$ is an indicator variable for whether the county is a noncore area. $Savings\ Deposit\ Rate_{i,t}$ is the real interest rate offered on $10,000 money market accounts, and $Time\ Deposit\ Rate_{i,t}$ is the real interest rate offered on 12-month $10,000 certificates of deposit. $Checking\ Deposit\ Rate_{i,t}$ is the real interest rate offered on interest-bearing checking accounts. $Auto\ Loan\ Interest\ Rate_{i,t}$ is the real interest rate offered on auto loans, and $Mortgage\ Interest\ Rate_{i,t}$ is the real interest rate offered on residential mortgages. The instrument, $Subsidy\ Amount_{i,t}$, is a continuous measure of the predicted value of total fixed subsidy payments in a given county-year. The control variables for county-level land availability are idle cropland, pastureland, and fallow or unharvested cropland from the prior Census of Agriculture. The control variables for local economic conditions and pre-existing banking infrastructure are the county-level employment rate, log income per capita, and total number of branches. $Population_{i,t-2}$ is the county population from the Bureau of Economic Analysis, and $State\text{-}Level\ Deregulation\ Index_{i,t}$ is the interstate branching deregulation index of Célérier and Matray (2019).

	Mean	SD	P10	P50	P90	N
Main Dependent Variables						
Farm Concentration$_{i,t-2}$ (Acres)	752	833	65	327	2,072	13,808
Bank Deposits HHI$_{i,t}$	3,123	2,011	1,308	2,540	5,474	13,808
Agricultural Loan HHI$_{i,t}$	4,472	2,661	1,720	3,785	10,000	13,808
Probability of Large Bank Entry$_{i,t}$	0.78	0.42	0.00	1.00	1.00	13,808
Probability of Small Bank Entry$_{i,t}$	0.96	0.20	1.00	1.00	1.00	13,808
Number of Large Bank Branches$_{i,t}$	13	29	0	3	33	13,808
Large Banks' Deposits Per Capita$_{i,t}$ ($)	7,498	24,887	0	5,684	15,437	13,808
Number of Small Bank Branches$_{i,t}$	9	10	1	7	19	13,808
Small Banks' Deposits Per Capita$_{i,t}$ ($)	11,814	9,679	1,552	9,836	24,462	13,808
Large Acquires Small$_{i,t}$	0.35	0.78	0.00	0.00	1.00	13,808
Small Acquires Small$_{i,t}$	0.43	0.84	0.00	0.00	1.00	13,808
Large Acquires Large$_{i,t}$	0.67	1.21	0.00	0.00	2.00	13,808
Large Banks' Total Agricultural Loans$_{i,t}$ ($1,000s)	12,900	32,656	0	1,534	35,074	13,808
Small Banks' Total Agricultural Loans$_{i,t}$ ($1,000s)	28,027	40,403	1	11,944	77,869	13,808
Large Banks' Real Estate Farm Loans$_{i,t}$ ($1,000s)	6,089	15,127	0	730	16,667	13,808
Small Banks' Real Estate Farm Loans$_{i,t}$ ($1,000s)	13,491	19,336	0	5,973	36,340	13,808
Large Banks' Non-Real Estate Farm Loans$_{i,t}$ ($1,000s)	6,632	18,965	0	400	17,198	13,808
Small Banks' Non-Real Estate Farm Loans$_{i,t}$ ($1,000s)	14,321	23,329	0	4,265	43,186	13,808
Large Banks' Total Agricultural Loan Growth$_{i,t}$	0.39	0.56	0.00	0.16	1.04	13,808
Small Banks' Total Agricultural Loan Growth$_{i,t}$	0.70	0.57	0.00	0.66	1.33	13,808
Large Banks' Real Estate Loan Growth$_{i,t}$	0.41	0.62	0.00	0.14	1.12	13,808
Small Banks' Real Estate Loan Growth$_{i,t}$	0.84	0.80	0.00	0.73	1.62	13,808
Large Banks' Non-Real Estate Loan Growth$_{i,t}$	0.39	0.62	0.00	0.14	1.04	13,808
Small Banks' Non-Real Estate Loan Growth$_{i,t}$	0.62	0.60	0.00	0.55	1.25	13,808
Direct Loan Amount$_{i,t}$ ($1,000s)	431	890	0	89	1,258	13,808
Number of Direct Loans$_{i,t}$	12	24	0	3	35	13,808
Urban$_{i,t}$	0.01	0.12	0.00	0.00	0.00	14,072
Metro$_{i,t}$	0.53	0.50	0.00	1.00	1.00	14,072
Rural$_{i,t}$	0.45	0.50	0.00	0.00	1.00	14,072
Agricultural Banks' Dependent Variables						
Probability of Large Agricultural Bank Entry$_{i,t}$	0.32	0.47	0.00	0.00	1.00	13,808
Probability of Small Agricultural Bank Entry$_{i,t}$	0.77	0.42	0.00	1.00	1.00	13,808
Number of Large Agricultural Bank Branches$_{i,t}$	1.32	3.94	0.00	0.00	4.00	13,808
Large Agricultural Banks' Deposits Per Capita$_{i,t}$ ($)	1,431	3,519	0	0	4,951	13,808
Number of Small Agricultural Bank Branches$_{i,t}$	4.56	4.96	0.00	3.00	11.00	13,808
Small Agricultural Banks' Deposits Per Capita$_{i,t}$ ($)	8,803	10,271	0	5,258	22,882	13,808
Interest Rate Dependent Variables						
Savings Deposit Rate$_{i,t}$ (%)	-1.21	0.45	-1.76	-1.19	-0.65	36,539
Time Deposit Rate$_{i,t}$ (%)	-0.21	1.07	-1.58	-0.18	1.40	36,544
Checking Deposit Rate$_{i,t}$ (%)	-1.63	0.41	-2.09	-1.66	-0.93	36,489
Auto Loan Interest Rate$_{i,t}$ (%)	4.64	1.56	2.37	4.91	6.42	27,141
Mortgage Interest Rate$_{i,t}$ (%)	3.19	1.10	1.72	3.43	4.41	24,336
Instrument						
Subsidy Amount$_{i,t}$ ($1,000s)	1,369	2,193	0	277	4,265	13,808
Control Variables						
Income Per Capita$_{i,t-2}$ ($)	36,523	9,217	27,227	34,930	47,124	13,808
Employment Rate$_{i,t-2}$	0.50	0.13	0.34	0.49	0.66	13,808
Population$_{i,t-2}$	63,234	115,890	5,909	25,012	149,639	13,808
State-Level Deregulation Index$_{i,t}$	1	2	0	1	4	13,808
Number of Branches$_{i,t}$	22	36	3	11	49	13,808
Idle Land$_{i,t-7}$ (Acres)	9,959	16,403	751	4,851	23,274	13,808
Unharvested Cropland$_{i,t-7}$ (Acres)	21,009	42,398	1,047	7,868	48,167	13,808
Pastureland$_{i,t-7}$ (Acres)	148,400	328,613	6,864	40,478	387,947	13,808

Table 2. Flat vs. Rugged Counties: Effect of Subsidy Amount on Farm Concentration and the Presence, Branches, and Deposits of Large and Small Banks

This table reports reduced-form estimates of the effect of subsidy payments on farm concentration and on the presence, branches, and deposits of large and small banks in flat counties (Panel A) and rugged counties (Panel B). The specifications are: $\textit{Farm Concentration}_{i,t} = \beta_1 \textit{Subsidy Amount}_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ in Column 1, $Y_{i,t} = \alpha\, \textit{Subsidy Amount}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ in Columns 2 and 5, and $E\big[Y_{i,t} \,\big|\, \textit{Subsidy Amount}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\alpha\, \textit{Subsidy Amount}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in Columns 3, 4, 6, and 7. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Counties are classified as rugged if their average Terrain Ruggedness Index (TRI), calculated by the Economic Research Service (ERS), is in the top quartile of the national distribution, and flat otherwise. In Column 1, the outcome variable is $\textit{Farm Concentration}_{i,t}$, the midpoint farm size at the county level. In Columns 2 and 5, the outcome variable is an indicator that a large or small bank, respectively, is present in the county in a given year. In Columns 3 and 6, the outcome variable is $\textit{Branches}_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 4 and 7, the outcome variable is $\textit{Deposits Per Capita}_{i,t}$, which is the deposits per capita in large and small banks, respectively. $\textit{Subsidy Amount}_{i,t}$ is a continuous measure of the predicted value of total fixed subsidy payments in a given county-year, standardized by its within-county standard deviation within each subsample. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The control variables for local economic conditions are the county-level employment rate and log income per capita, and the control variables for county-level land availability are lagged measures of idle cropland, pastureland, and cropland left fallow, lost to crop failure, or otherwise unharvested from the prior Census of Agriculture. Column 1 includes both sets of control variables. Columns 2 through 7 include the control variables for local economic conditions and additionally control for pre-existing banking infrastructure by including the log number of bank branches$_{i,t-2}$ in the county. F-statistics are from the first stage. Coefficients are reported as decimal percentages in Columns 2 through 7. Means of the dependent variable are calculated within the relevant subsample. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Large Banks			Small Banks		
	Farm Concentration	Probability of Entry	Branches	Deposits Per Capita	Probability of Entry	Branches	Deposits Per Capita
Panel A: Flat Counties							
Subsidy Amount$_{i,t-2}$	85.11***	0.014***	0.031***	0.065*	0.003**	0.020***	0.021***
	(5.68)	(0.005)	(0.010)	(0.038)	(0.001)	(0.005)	(0.004)
Observations	10,376	10,368	9,080	9,080	10,368	10,313	10,313
R-squared	0.84	0.72	0.89	0.80	0.58	0.62	0.87
F-statistic	224.52	224.52	224.52	224.52	224.52	224.52	224.52
Mean of Dep. Variable	892	0.77	15	8,279	0.97	10	12,897
Panel B: Rugged Counties							
Subsidy Amount$_{i,t-2}$	16.23***	-0.002	0.008	-0.012	-0.002	0.008	0.016*
	(4.31)	(0.006)	(0.007)	(0.010)	(0.002)	(0.005)	(0.008)
Observations	3,442	3,440	3,151	3,146	3,440	3,354	3,354
R-squared	0.86	0.71	0.84	0.76	0.65	0.57	0.80
F-statistic	14.18	14.18	14.18	14.18	14.18	14.18	14.18
County Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	328	0.81	12	9,015	0.93	7	8,981

Table 3. Falsification Test: Effect of Subsidy Amount on Farm Concentration and on Large and Small Banks in the Least and Most Responsive Counties

This table reports reduced-form estimates of the effect of subsidy payments on farm concentration and on the presence, branches, and deposits of large and small banks in the counties where farm concentration is least and most responsive to subsidy payments. The specifications are: $Farm\ Concentration_{i,t} = \alpha\, Subsidy\ Amount_{i,t} + \sum_{g\in\{1,3\}} \alpha_g\, Subsidy\ Amount_{i,t} \times \mathbf{1}\{g(i)=g\} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ in Column 1, $Y_{i,t} = \alpha\, Subsidy\ Amount_{i,t-2} + \sum_{g\in\{1,3\}} \alpha_g\, Subsidy\ Amount_{i,t-2} \times \mathbf{1}\{g(i)=g\} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ in Columns 2 and 5, and $E\big[Y_{i,t} \,\big|\, Subsidy\ Amount_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\alpha\, Subsidy\ Amount_{i,t-2} + \sum_{g\in\{1,3\}} \alpha_g\, Subsidy\ Amount_{i,t-2} \times \mathbf{1}\{g(i)=g\} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in Columns 3, 4, 6, and 7, where $g(i)$ denotes the responsiveness tercile of county i and the middle tercile is the omitted group. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Counties are classified as least responsive if their predicted long-run change in farm concentration attributable to subsidy payments, the first-stage coefficient on subsidy payments multiplied by the county's change in subsidy payments from 1994 to 2014, is in the bottom tercile of the national distribution, and as most responsive if it is in the top tercile. The first two rows report the effect of a within-tercile one-standard-deviation increase in subsidy payments in the least and most responsive terciles, $\alpha + \alpha_1$ and $\alpha + \alpha_3$, and the third row reports their difference, $\alpha_3 - \alpha_1$. In Column 1, the outcome variable is $Farm\ Concentration_{i,t}$, the midpoint farm size at the county level. In Columns 2 and 5, the outcome variable is an indicator that a large or small bank, respectively, is present in the county in a given year. In Columns 3 and 6, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 4 and 7, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small banks, respectively. $Subsidy\ Amount_{i,t}$ is a continuous measure of the predicted value of total fixed subsidy payments in a given county-year, standardized by its within-county standard deviation within each tercile. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of controls as in Table 2. Coefficients are reported as decimal percentages in Columns 2 through 7. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Large Banks			Small Banks		
	Farm Concentration	Probability of Entry	Branches	Deposits Per Capita	Probability of Entry	Branches	Deposits Per Capita
Least Responsive	-41.66***	-0.006	0.000	-0.003	-0.001	-0.003	0.008
	(4.92)	(0.004)	(0.001)	(0.006)	(0.002)	(0.004)	(0.008)
Most Responsive	141.17***	0.009	0.049***	0.107*	0.005**	0.028***	0.043***
	(8.94)	(0.007)	(0.015)	(0.062)	(0.002)	(0.007)	(0.007)
Most Minus Least	182.83***	0.015*	0.049***	0.109*	0.005**	0.031***	0.035***
	(8.85)	(0.007)	(0.015)	(0.059)	(0.002)	(0.008)	(0.008)
Observations	12,088	12,088	10,976	10,971	12,088	12,045	12,045
R-squared	0.84	0.70	0.88	0.79	0.53	0.60	0.86
County Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	715	0.80	15	8,361	0.97	10	11,699

Table 4. Effect of Subsidy Amount on Farm Concentration and Banking Concentration and of Farm Concentration on Banking Concentration

This table reports reduced-form estimates of the effect of subsidy payments on farm concentration and banking concentration (Panel A) and instrumental variable (IV) and OLS estimates of the effect of a within-county one-standard-deviation increase in farm concentration on banking concentration (Panel B). The specifications are: $Y_{i,t} = \alpha\, Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ in Panel A and $Y_{i,t} = \beta_2\, \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Panel B. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Columns 1 and 2, the outcome variable is $Farm\ Concentration_{i,t}$, the midpoint farm size at the county level. In the remaining columns, the outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. $Subsidy\ Amount_{i,t}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. The control variables for local economic conditions are the county-level employment rate and log income per capita, and the control variables for county-level land availability are lagged measures of idle cropland, pastureland, and cropland left fallow, lost to crop failure, or otherwise unharvested from the prior Census of Agriculture. F-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
	Farm Concentration		Bank Deposits HHI	
Panel A: Effect of Subsidy Amount				
Subsidy Amount$_{i,t-2}$	105.24***	107.50***	57.39***	63.55***
	(5.49)	(5.88)	(8.45)	(8.84)
Log Income Per Capita$_{i,t-2}$		186.49***		21.48
		(69.30)		(114.74)
Employment Rate$_{i,t-2}$		-146.08		242.76
		(157.97)		(312.70)
Idle Land$_{i,t-7}$		20.22***		
		(5.00)		
Unharvested Cropland$_{i,t-7}$		-11.16***		
		(3.59)		
Pastureland$_{i,t-7}$		4.24		
		(4.30)		
Observations	13,818	13,818	13,818	13,818
F-statistic	367.14	334.06	367.14	357.27
Deregulation Index FE	No	Yes	No	Yes
County & Year FE	Yes	Yes	Yes	Yes
Mean of Dep. Variable	752	752	3,121	3,121

	(1)	(2)	(3)	(4)
	Bank Deposits HHI			
Panel B: Effect of Farm Concentration				
$\widehat{\text{Farm Concentration}}_{i,t-2}$	252.18***	275.28***		
	(38.58)	(41.18)		
Farm Concentration$_{i,t-2}$			20.80**	17.73**
			(9.04)	(8.98)
Log Income Per Capita$_{i,t-2}$		-90.17		24.31
		(124.19)		(116.19)
Employment Rate$_{i,t-2}$		330.96		320.72
		(327.48)		(313.94)
Idle Land$_{i,t-7}$		-12.36		-4.34
		(7.87)		(7.34)
Unharvested Cropland$_{i,t-7}$		7.80		-8.89
		(7.10)		(7.19)
Pastureland$_{i,t-7}$		-1.90		0.42
		(5.89)		(4.91)
Observations	13,818	13,818	13,818	13,818
F-statistic	367.14	334.06	5.29	4.96
Deregulation Index FE	No	Yes	No	Yes
County & Year FE	Yes	Yes	Yes	Yes
Mean of Dep. Variable	3,121	3,121	3,121	3,121

Table 5. Effect of Farm Concentration on the Presence, Branches, and Deposits of Large and Small Banks

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the presence, branches, and deposits of large and small banks. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ Farm\ \widehat{Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Columns 1 and 4 and $E\big[Y_{i,t} \,\big|\, Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Columns 1 and 4, the outcome variable is an indicator that a large or small bank, respectively, is present in the county in a given year. In Columns 2 and 5, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 3 and 6, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small banks, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 4. I additionally control for pre-existing banking infrastructure by including the log number of bank branches$_{i,t-2}$ in the county. Log population$_{i,t-2}$ is included in the first-stage regression for Columns 3 and 6 to adjust for the per capita scaling of the outcome variable. F-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Large Banks			Small Banks		
	Probability of Entry	Branches	Deposits Per Capita	Probability of Entry	Branches	Deposits Per Capita
Farm $\widehat{\text{Concentration}}_{i,t-2}$	0.046**	0.160***	0.327**	0.021***	0.099***	0.225***
	(0.022)	(0.050)	(0.128)	(0.006)	(0.023)	(0.031)
Observations	13,808	12,231	12,226	13,808	13,667	13,667
F-statistic	322.68	322.68	174.91	322.68	322.68	174.91
County Controls	Yes	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	0.78	14	8,468	0.96	9	11,936

Table 6. Effect of Farm Concentration on Bank Acquisitions, by Bank Size

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on bank acquisitions by bank size. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t} \,\big|\, Farm\ \ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Column 1, the outcome variable is $Large\ Acquires\ Small_{i,t}$, which is the number of mergers or acquisitions in which a large bank acquires a small bank, aggregated over five-year intervals. In Columns 2 and 3, the outcome variables are $Small\ Acquires\ Small_{i,t}$ and $Large\ Acquires\ Large_{i,t}$, which are the number of mergers or acquisitions between two small banks and between two large banks, respectively, also aggregated over five-year intervals. Each merger or acquisition is counted in every county where the target bank operated branches, with bank sizes and branch counties measured at the last Summary of Deposits before the transaction date. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 5. F-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1) Large Acquires Small	(2) Small Acquires Small	(3) Large Acquires Large
Farm $\widehat{\text{Concentration}}_{i,t-2}$	0.153	0.234**	-0.325***
	(0.138)	(0.103)	(0.076)
Observations	8,902	10,236	9,848
F-statistic	322.68	322.68	322.68
County Controls	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes
Mean of Dep. Variable	0.55	0.58	0.94

Table 7. Effect of Farm Concentration on Large Banks' Farm Lending by Farm Revenue and Loan Size

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on large banks' farm lending by farm revenue and loan size. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t}\,\big|\,Farm\ \ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ \ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data at the annual level (see Section 5 for variable and sample construction details). The sample is restricted to 1996 to 2014, the years with CRA reporting. The outcome variable is the total volume of farm loans of $500,000 or less that large banks originate in a given county-year, for loans to farms with gross annual revenue above $1 million in Column 1 and $1 million or less in Column 2. In Columns 3 through 5, the outcome variable is the total volume of farm loans at or below $100,000, between $100,000 and $250,000, and between $250,000 and $500,000, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size, carried forward annually until the next census observation. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 5. F-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
	Farm Revenue		Loan Size		
	>$1M	≤$1M	≤$100K	$100K–$250K	$250K–$500K
Farm $\widehat{\text{Concentration}}_{i,t-2}$	0.299***	0.017	-0.046	0.031	0.177**
	(0.082)	(0.059)	(0.051)	(0.076)	(0.077)
Observations	52,533	52,665	52,675	51,212	48,919
F-statistic	326.06	326.38	326.37	318.33	301.52
County Controls	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	1,039	3,393	1,512	1,516	1,552

Table 8. Effect of Farm Concentration on Agricultural Loan Growth

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on agricultural loan growth by bank size. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t}\,\big|\,Farm\ \ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Columns 1 and 2, the outcome variable is $Total\ Agricultural\ Loan\ Growth_{i,t}$, in Columns 3 and 4 it is $Real\ Estate\ Loan\ Growth_{i,t}$, and in Columns 5 and 6 it is $Non\text{-}Real\ Estate\ Loan\ Growth_{i,t}$, each measuring large or small banks' growth in the loan type relative to the prior period's total volume of that loan type. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 5. The first stage also includes the prior period's log total loan volume by loan type. F-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Total Agricultural Loan Growth		Real Estate Loan Growth		Non-Real Estate Loan Growth	
	Large Banks	Small Banks	Large Banks	Small Banks	Large Banks	Small Banks
Farm $\widehat{\text{Concentration}}_{i,t-2}$	0.455***	0.028	0.398**	0.015	0.460***	0.060*
	(0.124)	(0.032)	(0.164)	(0.037)	(0.120)	(0.036)
Observations	11,719	13,465	11,709	13,418	11,683	13,306
F-statistic	323.27	323.27	322.45	322.45	321.62	321.62
County Controls	Yes	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	0.46	0.73	0.49	0.87	0.47	0.65

Table 9. Effect of Farm Concentration on Small Banks' Agricultural Lending by Loan Size and Core Funding

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on small banks' agricultural lending by loan size and core deposit share. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t} \,\big|\, Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Panels A and B, the outcome variable is high- and low-core small banks' growth in real estate farm lending, respectively, and in Panels C and D it is high- and low-core small banks' growth in non-real estate farm lending, each relative to the prior period's total volume of that loan type. In Columns 1 through 4, the outcome variable is the growth in loans at or below \$100,000, between \$100,000 and \$250,000, between \$250,000 and \$500,000, and above \$500,000, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than \$1 billion in inflation-adjusted assets, and large otherwise. High-core small banks are those with a pre-program (1990–1995) core deposit share above the within-small-bank median, and low-core small banks otherwise. The controls are the same set of first- and second-stage controls as in Table 5. F-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
	Agricultural Loan Growth			
	≤$100K	$100K–$250K	$250K–$500K	>$500K
Panel A: Real Estate Loans, High-Core Funding				
Farm $\widehat{\text{Concentration}}_{i,t-2}$	-0.199***	-0.090	0.133	0.941***
	(0.049)	(0.080)	(0.118)	(0.273)
Observations	10,848	10,069	9,592	10,406
F-statistic	322.65	322.65	322.65	322.65
Mean of Dep. Variable	0.28	0.13	0.12	0.14
Panel B: Real Estate Loans, Low-Core Funding				
Farm $\widehat{\text{Concentration}}_{i,t-2}$	-0.176***	-0.158**	-0.079	-0.002
	(0.055)	(0.075)	(0.095)	(0.151)
Observations	11,617	11,187	10,931	11,416
F-statistic	322.65	322.65	322.65	322.65
Mean of Dep. Variable	0.27	0.12	0.11	0.17
Panel C: Non-Real Estate Loans, High-Core Funding				
Farm $\widehat{\text{Concentration}}_{i,t-2}$	-0.063	0.193*	0.446**	0.805**
	(0.051)	(0.113)	(0.175)	(0.362)
Observations	10,724	9,315	8,684	10,205
F-statistic	322.65	322.65	322.65	322.65
Mean of Dep. Variable	0.28	0.07	0.07	0.10
Panel D: Non-Real Estate Loans, Low-Core Funding				
Farm $\widehat{\text{Concentration}}_{i,t-2}$	-0.170***	-0.353***	-0.249***	-0.033
	(0.052)	(0.069)	(0.085)	(0.145)
Observations	11,536	10,380	9,876	11,024
F-statistic	322.65	322.65	322.65	322.65
Mean of Dep. Variable	0.29	0.07	0.07	0.09
County Controls	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes

Table 10. Effect of Farm Concentration on Agricultural Loan Concentration and on Large and Small Agricultural Banks

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the concentration of agricultural lending and on the presence, branches, and deposits of large and small agricultural (ag) banks. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Columns 1, 2, and 5 and $E\big[Y_{i,t}\,\big|\,Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Column 1, the sample includes all banks, and the outcome variable is $Agricultural\ Loan\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared agricultural loan shares held by each bank within a county. In the remaining columns, the sample is restricted to ag banks, those with farm loans exceeding 5% of total loans in any of the preceding six years. In Columns 2 and 5, the outcome variable is an indicator that a large or small ag bank, respectively, is present in the county in a given year. In Columns 3 and 6, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small ag banks, respectively. In Columns 4 and 7, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small ag banks, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. In Column 1, the controls are the same set of first- and second-stage controls as in Table 4, and in the remaining columns as in Table 5. F-statistics are from the first stage. Coefficients are reported as decimal percentages in Columns 2 through 7. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Large Ag Banks			Small Ag Banks		
	Agricultural Loan HHI	Probability of Entry	Branches	Deposits Per Capita	Probability of Entry	Branches	Deposits Per Capita
$\widehat{\text{Farm Concentration}}_{i,t-2}$	627.78***	0.154***	0.546***	0.752***	0.016	-0.003	0.237***
	(92.66)	(0.024)	(0.170)	(0.228)	(0.010)	(0.021)	(0.034)
Observations	13,818	13,808	8,559	8,554	13,808	12,218	12,218
F-statistic	334.06	333.74	333.74	187.52	333.74	333.74	187.52
County Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	4,468	0.32	2	2,310	0.77	5	9,948

Table 11. Effect of Farm Concentration on Offered Interest Rates, 2002 to 2014

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the real interest rates offered by banks, for all banks (Panel A) and separately for small banks (Panel B) and large banks (Panel C). The specifications for the first and second stages, respectively, are: $\textit{Farm Concentration}_{i,t} = \beta_1 \textit{Subsidy Amount}_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2 \widehat{\textit{Farm Concentration}}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data at the annual level (see Section 5 for variable and sample construction details). In Panel A, each observation is a county-level average of the June interest rates offered by bank branches within the county. In Panels B and C, the second stage is estimated at the bank-county-year level within each size subsample, with bank-by-county and year fixed effects in place of the county fixed effects γ_i, and each observation is a bank's average offered rate across its branches within a county in June of a given year. In Columns 1 through 5, the outcome variables are the offered interest rates on savings deposit accounts ($10,000 money market accounts), time deposit accounts (12-month $10,000 certificates of deposit), checking deposit accounts (interest-bearing checking accounts), auto loans, and residential mortgage loans, all expressed in real, inflation-adjusted terms. $\textit{Farm Concentration}_{i,t-2}$ is the lagged within-county standardized midpoint farm size, carried forward annually until the next census observation. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 4. In both stages, the control variables for local credit demand are county-level annual log total deposits and log total farm interest expenses. F-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
	Deposit Rates			Loan Rates	
	Savings	Time	Checking	Auto	Mortgage
Panel A: All Banks					
Farm $\widehat{\text{Concentration}}_{i,t-2}$	-0.316***	0.125***	-0.223***	1.028***	0.009
	(0.055)	(0.041)	(0.035)	(0.208)	(0.065)
Observations	34,828	34,834	34,786	25,770	23,178
F-statistic	323.16	323.16	323.16	323.16	323.16
County Controls	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	-1.20	-0.17	-1.64	4.69	3.23
Panel B: Small Banks					
Farm $\widehat{\text{Concentration}}_{i,t-2}$	-0.118**	0.302***	-0.132***	-0.369***	-0.104
	(0.054)	(0.051)	(0.029)	(0.130)	(0.125)
Observations	110,968	109,764	107,648	30,366	18,209
F-statistic	323.16	323.16	323.16	323.16	323.16
County Controls	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes
Bank × County & Year FE	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	-1.07	-0.06	-1.55	4.83	3.58
Panel C: Large Banks					
Farm $\widehat{\text{Concentration}}_{i,t-2}$	-0.221***	-0.181***	-0.041*	0.971***	0.107**
	(0.070)	(0.063)	(0.023)	(0.284)	(0.053)
Observations	106,470	103,140	101,628	70,313	59,102
F-statistic	323.16	323.16	323.16	323.16	323.16
County Controls	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes
Bank × County & Year FE	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	-1.36	-0.49	-1.74	4.35	2.96

Appendix A. Additional Analyses

Figure A.1. Distribution of Counties by Average Terrain Ruggedness

This figure shows the distribution of average terrain ruggedness across U.S. counties, measured using the 2010 Terrain Ruggedness Index (TRI) from the USDA Economic Research Service (ERS). Counties are divided into quartiles based on the national TRI distribution, with the top quartile corresponding to an average ruggedness index greater than or equal to 28.

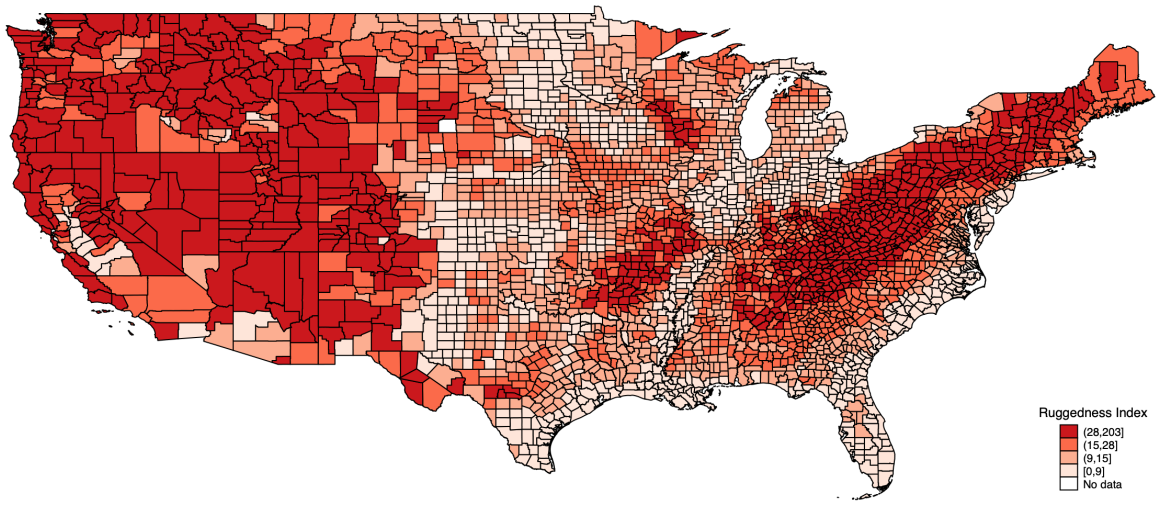


Figure A.2. Distribution of High- and Low-Cost Counties

This figure shows the distribution of U.S. counties classified as high- or low-cost based on per acre economic costs. The measure is constructed as a time-invariant, county-level weighted average of per acre economic costs, where weights are each crop's share of harvested acreage from the 1992 Census of Agriculture. Economic costs per acre for each major commodity crop are calculated as the six-year average (1990–1995) from the Farm Costs and Returns Survey (FCRS) compiled by the U.S. Department of Agriculture's Economic Research Service (ERS). Counties with weighted average costs above the national median are classified as high-cost, and those below the median as low-cost.

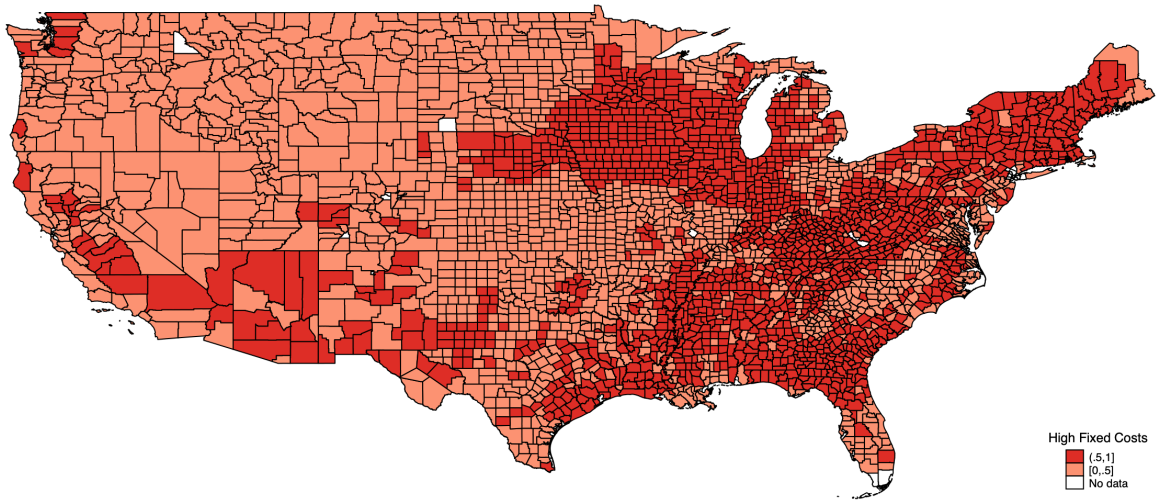


Figure A.3. Agricultural Land Values in Above-Median Subsidy Payment and Housing Bust Counties

This figure plots the market value of agricultural land and buildings per acre from the Census of Agriculture, in real terms, for two groups of counties between 1992 and 2012, indexed to 100 in 1992. Each series is computed as the exponential of the within-group mean of the log market value per acre in each census year, relative to the group's 1992 mean. Above-Median Subsidy Payment counties have total predicted fixed subsidy payments per acre of cropland, summed over the sample period, above the national median. Housing Bust counties have below-median payments per cropland acre and a decline in the annual county-level all-transactions House Price Index (HPI) from the Federal Housing Finance Agency (FHFA) between 2007 and 2012. The sample is restricted to counties with land value data in all five census years.

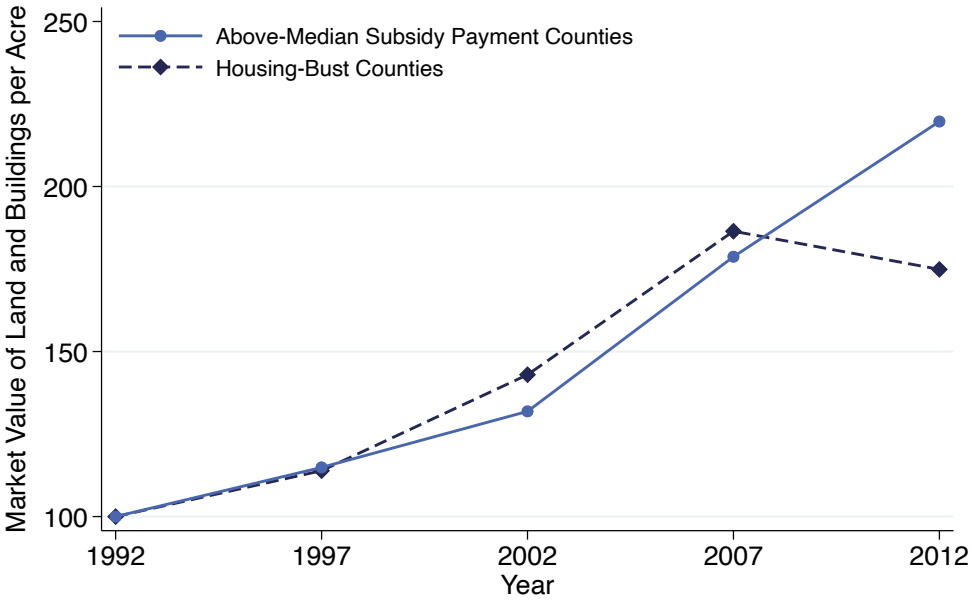


Figure A.4. Distribution of Changes in Land Values by Farming Cost

This figure shows the distribution of percent changes in market value of agricultural land and buildings per acre between 1992 and 2012 for high- and low-cost counties. Counties are classified as high-cost if their weighted average per acre economic cost exceeds the nationwide median, and low-cost otherwise. Economic costs per acre for each major commodity crop are calculated as the six-year average (1990–1995) from the Farm Costs and Returns Survey (FCRS) compiled by the U.S. Department of Agriculture's Economic Research Service (ERS). The measure is constructed as a time-invariant, county-level weighted average of per acre economic costs, where weights are each crop's share of harvested acreage from the 1992 Census of Agriculture.

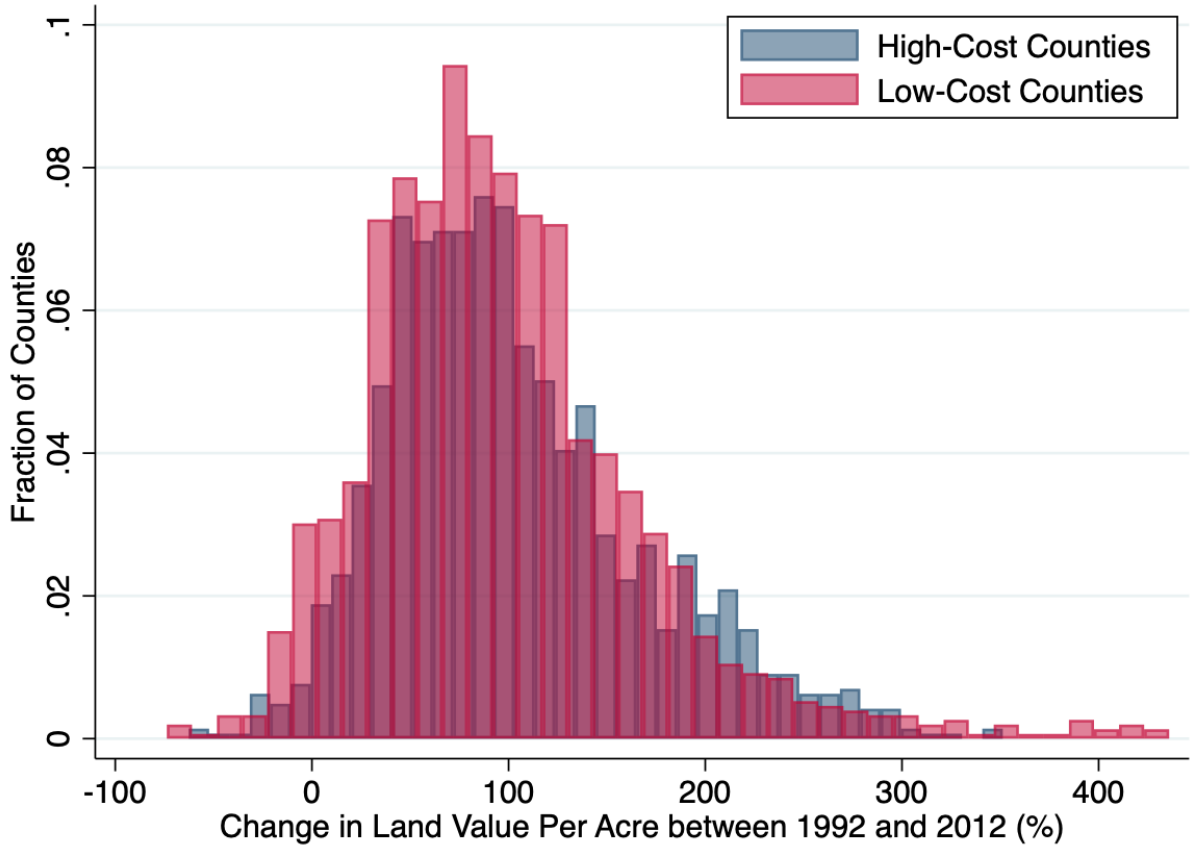


Table A.1. Pre-Program Shares: Effect of Subsidy Amount and Farm Concentration on Banking Concentration and the Presence, Branches, and Deposits of Large and Small Banks

This table reports reduced-form estimates of the effect of subsidy payments (Panel A) and second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration (Panel B) on banking concentration and on the presence, branches, and deposits of large and small banks. In Panel A, the specification is $Y_{i,t} = \alpha\, Subsidy\ Amount_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$. In Panel B, the specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Columns 1, 2, and 5 and $E\big[Y_{i,t} \,\big|\, Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Column 1, the outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. In Columns 2 and 5, the outcome variable is an indicator that a large or small bank, respectively, is present in the county in a given year. In Columns 3 and 6, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 4 and 7, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small banks, respectively. $Subsidy\ Amount_{i,t-2}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year, constructed from pre-program shares, using the 1996 base acre shares for the seven original program crops in all years and county shares of national 1992 Census of Agriculture harvested acres for the crops added in 2002. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than \$1 billion in inflation-adjusted assets, and large otherwise. In Column 1, the controls are the same set of controls as in Table 4, and in Columns 2 through 7, the controls are the same set of controls as in Table 5. F-statistics are from the first stage. Coefficients are reported as decimal percentages in Columns 2 through 7. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Large Banks			Small Banks		
	Bank Deposits HHI	Probability of Entry	Branches	Deposits Per Capita	Probability of Entry	Branches	Deposits Per Capita
Panel A: Effect of Subsidy Amount							
Subsidy Amount$_{i,t-2}$	49.61***	0.007*	0.025***	0.066*	0.003***	0.017***	0.020***
	(7.18)	(0.004)	(0.008)	(0.037)	(0.001)	(0.004)	(0.003)
Observations	13,818	13,808	12,231	12,226	13,808	13,667	13,667
R-squared	0.92	0.72	0.88	0.79	0.61	0.61	0.86
Mean of Dep. Variable	3,121	0.78	14	8,468	0.96	9	11,936
Panel B: Effect of Farm Concentration							
Farm $\widehat{\text{Concentration}}_{i,t-2}$	268.59***	0.047**	0.155***	0.369**	0.021***	0.095***	0.219***
	(41.86)	(0.022)	(0.052)	(0.149)	(0.006)	(0.023)	(0.032)
Observations	13,818	13,808	12,231	12,226	13,808	13,667	13,667
F-statistic	337.25	325.95	325.95	177.39	325.95	325.95	177.39
County Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	3,121	0.78	14	8,468	0.96	9	11,936

Table A.2. Alternative Lag Structures: Effect of Farm Concentration on Large and Small Banks

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on banking concentration and on the branches and deposits of large and small banks with alternative time lags for the second stage. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-k} + \mathbf{X}'_{i,t-k}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Column 1 and $E\big[Y_{i,t}\,\big|\,Farm\ Concentration_{i,t-k}, \hat{\varepsilon}_{i,t-k}, \mathbf{X}'_{i,t-k}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-k} + \mathbf{X}'_{i,t-k}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-k} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in Columns 2 through 5. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-k}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Panels A and B, the first stage is lagged by three and four years, respectively. In Column 1, the outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. In Columns 2 and 4, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 3 and 5, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small banks, respectively. $Farm\ Concentration_{i,t-k}$ is the within-county standardized midpoint farm size, lagged by either three or four years. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. In Column 1, the controls are the same set of first- and second-stage controls as in Table 4. In Columns 2 through 5, the controls are the same set of first- and second-stage controls as in Table 5. All controls carry the same lag as $Farm\ Concentration_{i,t-k}$. F-statistics are from the first stage. Coefficients are reported as decimal percentages in Columns 2 through 5. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
		Large Banks		Small Banks	
	Bank Deposits HHI	Branches	Deposits Per Capita	Branches	Deposits Per Capita
Panel A: Three-Year Lag					
Farm $\widehat{\text{Concentration}}_{i,t-3}$	224.66***	0.169***	0.414***	0.102***	0.236***
	(38.32)	(0.065)	(0.154)	(0.022)	(0.031)
Observations	13,808	12,276	12,271	13,653	13,653
F-statistic	94.49	92.15	79.15	92.15	79.15
Mean of Dep. Variable	3,120	15	8,787	9	11,759
Panel B: Four-Year Lag					
Farm $\widehat{\text{Concentration}}_{i,t-4}$	227.53***	0.117**	0.469***	0.108***	0.244***
	(38.02)	(0.050)	(0.144)	(0.019)	(0.032)
Observations	13,808	12,357	12,347	13,654	13,651
F-statistic	93.58	91.21	77.71	91.21	77.71
County Controls	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	3,114	15	9,110	9	11,629

Table A.3. Effect of Subsidy Amount on Farm Size and Land Tenure

This table reports reduced-form estimates of the effect of subsidy payments on farm size and land tenure. The specification is $Y_{i,t} = \alpha\, \textit{Subsidy Amount}_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Column 1, the outcome variable is the log of the number of farms with harvested cropland. In Column 2, it is the log of total harvested acres. In Columns 3 and 4, it is rented acres and owned acres per part owner farm, where part owners operate both owned and rented land. In Columns 5 through 8, it is the share of the county's harvested acres on farms harvesting 1 to 49, 50 to 499, 500 to 999, and 1,000 or more acres. Rented and owned acres are from the Census of Agriculture tenure tables. $\textit{Subsidy Amount}_{i,t}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year. The controls are the same set of first-stage controls as in Table 4. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
					Share of Harvested Acres			
	Number of Farms	Harvested Acres	Rented Acres	Owned Acres	1–49	50–499	500–999	1,000+
Subsidy Amount$_{i,t}$	-0.033***	0.003	32.66***	20.15***	0.004***	-0.018***	-0.016***	0.030***
	(0.002)	(0.002)	(2.90)	(2.51)	(0.001)	(0.001)	(0.001)	(0.002)
Observations	13,817	13,772	13,595	13,646	13,772	13,772	13,772	13,772
R-squared	0.98	0.98	0.93	0.91	0.92	0.88	0.76	0.93
County Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	5.8	10.8	553	488	0.14	0.43	0.16	0.28

Table A.4. Effect of Subsidy Amount on Farm Concentration, Land Values, and Deposits per Capita

This table reports reduced-form estimates of the effect of subsidy payments on farm concentration, land values, and deposits per capita, estimated jointly in a stacked system. The specification is $Y_{i,t}^{j} = \alpha_j \, \textit{Subsidy Amount}_{i,t} + \mathbf{X}_{i,t}'\mathbf{\Gamma}_j + \gamma_i^j + \delta_t^j + \phi_{d(i,t)}^j + \varepsilon_{i,t}^j$. Column j reports equation j, where $j = 1, 2, 3$ denotes the farm concentration, land value, and deposits per capita equations, respectively. Subscript i represents the county, t represents the year, d represents state-level banking deregulation, and j represents the outcome equation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). The common sample includes county-years with nonmissing outcomes and controls in all three equations and excludes the final five-year snapshot. The outcomes are county-level midpoint farm size in Column 1, log market value of agricultural land and buildings per acre in Column 2, and total county deposits per capita in Column 3. In the deposits per capita equation, the subsidy payments and the controls are measured two years before the outcome, as in Table 2. Each outcome and $\textit{Subsidy Amount}_{i,t}$, a continuous measure of predicted total fixed subsidy payments, is divided by its pooled within-county standard deviation on this sample. The controls are the same set of first-stage controls as in Table 4. Control coefficients are equation-specific, and the system covariance matrix allows residuals to be correlated across outcomes within counties. Farm Concentration Exceeds Alternatives is the p-value from a Wald test that the coefficient on Subsidy Amount in the farm concentration equation is larger in absolute value than the coefficients in the land value and deposits per capita equations. Coefficients report the change in outcome standard deviations associated with a one-standard-deviation increase in Subsidy Amount. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1) Farm Concentration	(2) Land Values	(3) Deposits Per Capita
Subsidy Amount$_{i,t}$	0.243***	-0.031***	0.015
	(0.014)	(0.007)	(0.010)
Farm Concentration Exceeds Alternatives		< 0.001	
Observations	11,146	11,146	11,146
R-squared	0.86	0.97	0.71
County Controls	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes
Mean of Dep. Variable	697	7.67	18,868

Table A.5. Effect of Subsidy Amount on Investment, Crop Mix, and Production Outcomes

This table reports reduced-form estimates of the effect of subsidy payments on county-level agricultural production outcomes from the Census of Agriculture. The specification is $Y_{i,t} = \alpha\, \textit{Subsidy Amount}_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Column 1, the outcome variable is the Törnqvist crop-yield index, which aggregates per acre yield growth across the nine census crops with quantity data using revenue-share weights. In Column 2, the outcome variable is the inverse hyperbolic sine of the real value of total county machinery in thousands of dollars, and in Column 3 the inverse hyperbolic sine of the county count of recently manufactured tractors. In Column 4, the outcome variable is the program crop share, the share of county crop acres planted in the seven program crops of the 1996 FAIR Act (wheat, corn, sorghum, barley, oats, upland cotton, and rice). $\textit{Subsidy Amount}_{i,t}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year. The controls are the same set of first-stage controls as in Table 4. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1) Törnqvist Yield Index	(2) Machinery Stock	(3) Tractor Investment	(4) Program Crop Share
Subsidy Amount$_{i,t}$	-0.0030	-0.0145***	-0.0324***	0.0007
	(0.0052)	(0.0023)	(0.0051)	(0.0009)
Observations	12,140	13,818	13,818	13,818
R-squared	0.97	0.96	0.86	0.95
County Controls	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes
Mean of Dep. Variable	-10	11	5	0.35

Table A.6. Effect of Subsidy Amount on Farm Concentration and on Large and Small Banks, by Terrain, Population, Urbanization, and Land Value

This table reports reduced-form estimates of the effect of subsidy payments on farm concentration and on the presence, branches, and deposits of large and small banks in a pooled sample of flat and rugged counties, in which the effect of payments varies with terrain, population, urbanization, and land value. The specification is $Y_{i,t} = \alpha\, \textit{Subsidy Amount}_{i,t-2} + \sum_m \alpha_m\, \textit{Subsidy Amount}_{i,t-2} \times Z_{m,i} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ in Columns 1, 2, and 5, and $E\big[Y_{i,t} \,\big|\, \textit{Subsidy Amount}_{i,t-2}, Z_{m,i}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\alpha\, \textit{Subsidy Amount}_{i,t-2} + \sum_m \alpha_m\, \textit{Subsidy Amount}_{i,t-2} \times Z_{m,i} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Counties are classified as rugged if their average Terrain Ruggedness Index (TRI), calculated by the Economic Research Service (ERS), is in the top quartile of the national distribution, and flat otherwise. In Column 1, the outcome variable is $\textit{Farm Concentration}_{i,t}$, the midpoint farm size at the county level. In Columns 2 and 5, the outcome variable is an indicator that a large or small bank, respectively, is present in the county in a given year. In Columns 3 and 6, the outcome variable is $\textit{Branches}_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 4 and 7, the outcome variable is $\textit{Deposits Per Capita}_{i,t}$, which is the deposits per capita in large and small banks, respectively. $\textit{Subsidy Amount}_{i,t-2}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year, standardized on the full sample. $Z_{m,i}$ collects the rugged indicator and pre-program population, urbanization, and land value, each standardized across counties. Pre-program characteristics are county means over banking years 1994 to 1997 of log population, the 1990 National Center for Health Statistics (NCHS) urbanization code, which runs from 2 for large fringe metropolitan counties to 6 for noncore counties, and the CPI-adjusted market value of agricultural land and buildings per acre in dollars. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of controls as in Table 2. Coefficients are reported as decimal percentages in Columns 2 through 7. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Large Banks			Small Banks		
	Farm Concentration	Probability of Entry	Branches	Deposits Per Capita	Probability of Entry	Branches	Deposits Per Capita
Subsidy Amount$_{i,t-2}$	87.70***	0.010**	0.083***	0.051***	0.003***	0.026***	0.012***
	(5.46)	(0.004)	(0.011)	(0.016)	(0.001)	(0.003)	(0.004)
Subsidy Amount$_{i,t-2}$ × Rugged$_i$	-76.03***	0.011	-0.006	-0.110***	-0.010	0.004	0.029
	(13.89)	(0.022)	(0.032)	(0.039)	(0.007)	(0.019)	(0.028)
Subsidy Amount$_{i,t-2}$ × Population$_i$	-28.13***	-0.010**	-0.032***	0.046	0.001	-0.016***	-0.033***
	(6.99)	(0.005)	(0.008)	(0.047)	(0.001)	(0.005)	(0.005)
Subsidy Amount$_{i,t-2}$ × Urbanization$_i$	-23.26***	0.017**	0.009	0.057***	-0.000	0.001	0.006
	(8.13)	(0.007)	(0.009)	(0.020)	(0.001)	(0.004)	(0.005)
Subsidy Amount$_{i,t-2}$ × Land Value$_i$	-7.68	0.028***	0.002	-0.022	-0.001	-0.005	0.005
	(10.27)	(0.007)	(0.009)	(0.054)	(0.002)	(0.006)	(0.007)
Observations	12,693	12,693	11,430	11,425	12,693	12,625	12,625
R-squared	0.84	0.70	0.88	0.79	0.56	0.61	0.86
County Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	729	0.79	15	8,380	0.97	10	11,849

Table A.7. Effect of Subsidy Amount on Farm Concentration, Land Values, and Deposits per Capita, by Predicted Responsiveness

This table reports reduced-form estimates of the effect of subsidy payments on farm concentration, land values, and total county deposits per capita by the predicted responsiveness of farm consolidation to subsidy payments. The specifications are: $Y_{i,t} = \alpha\, \textit{Subsidy Amount}_{i,t} + \sum_{g\in\{1,3\}} \alpha_g\, \textit{Subsidy Amount}_{i,t} \times \mathbf{1}\{g(i) = g\} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ in Columns 1 and 2, and $E\left[Y_{i,t} \,\middle|\, \textit{Subsidy Amount}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\right] = \exp\left(\alpha\, \textit{Subsidy Amount}_{i,t-2} + \sum_{g\in\{1,3\}} \alpha_g\, \textit{Subsidy Amount}_{i,t-2} \times \mathbf{1}\{g(i) = g\} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)}\right)$ in Column 3, where $g(i)$ denotes the responsiveness tercile of county i and the middle tercile is the omitted group. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Counties are classified as least responsive if their predicted long-run change in farm concentration attributable to subsidy payments, the first-stage coefficient on subsidy payments multiplied by the county's change in subsidy payments from 1994 to 2014, is in the bottom tercile of the national distribution, and as most responsive if it is in the top tercile. The first two rows report the effect of a within-tercile one-standard-deviation increase in subsidy payments in the least and most responsive terciles, $\alpha + \alpha_1$ and $\alpha + \alpha_3$, and the third row reports their difference, $\alpha_3 - \alpha_1$. The sample ends in banking year 2009, excluding the final five-year snapshot. In Column 1, the outcome variable is $\textit{Farm Concentration}_{i,t}$, the midpoint farm size at the county level. In Column 2, it is the log market value of agricultural land and buildings per acre. In Column 3, it is total county deposits per capita, total county deposits divided by county population. $\textit{Subsidy Amount}_{i,t}$ is a continuous measure of the predicted value of total fixed subsidy payments in a given county-year, standardized by its within-county standard deviation within each tercile. The controls are the same set of controls as in Table 4. In Column 3, I additionally control for pre-existing banking infrastructure by including the log number of bank $\text{branches}_{i,t-2}$ in the county. Coefficients are reported as decimal percentages in Column 3. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1) Farm Concentration	(2) Land Value Per Acre	(3) Total Deposits Per Capita
Least Responsive	-28.51***	-0.003	0.001
	(9.12)	(0.004)	(0.003)
Most Responsive	121.11***	-0.010***	0.005
	(8.82)	(0.004)	(0.006)
Most Minus Least	149.62***	-0.007*	0.003
	(9.79)	(0.004)	(0.006)
Observations	9,651	9,651	9,651
R-squared	0.85	0.97	0.89
County Controls	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes
Mean of Dep. Variable	653	7.74	18,776

Table A.8. Effect of Alternative Measures of Farm Concentration on Banking Concentration

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on banking concentration using two alternative measures of farm concentration. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). The outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. In Column 1, farm concentration is measured by $Share\ of\ Acres\ on$ 1,000+ $Acre\ Farms_{i,t-2}$, the share of harvested acres on farms of 1,000 acres or more. In Column 2, it is measured by $Log\ Number\ of\ Farms_{i,t-2}$, the log number of farms, for which a negative coefficient corresponds to greater consolidation. Both measures are lagged, constructed from the Census of Agriculture acreage bins, and standardized within county. The controls are the same set of first- and second-stage controls as in Table 4. F-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)
	Bank Deposits HHI	
$\widehat{\text{Share of Acres on 1,000+ Acre Farms}}_{i,t-2}$	190.35***	
	(27.69)	
$\widehat{\text{Log Number of Farms}}_{i,t-2}$		-335.98***
		(52.01)
Observations	13,772	13,818
F-statistic	369.99	167.15
County Controls	Yes	Yes
Deregulation Index FE	Yes	Yes
County & Year FE	Yes	Yes
Mean of Dep. Variable	3,111	3,121

Table A.9. First Differences: Effect of Farm Concentration on Banking Concentration

This table reports first- and second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation change in farm concentration on the change in banking concentration. This specification uses first differences, where the first and second stages, respectively, are: $\Delta Farm\ Concentration_{i,t} = \beta_1\ \Delta Subsidy\ Amount_{i,t} + \Delta \mathbf{X}'_{i,t}\mathbf{\Gamma} + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $\Delta Y_{i,t} = \beta_2\ \Delta Farm\ \widehat{Concentration}_{i,t-2} + \Delta \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Columns 1 and 2, the outcome variable is $\Delta Farm\ Concentration_{i,t}$, the change in the midpoint farm size at the county level. In Columns 3 and 4, the outcome variable is $\Delta Bank\ Deposits\ HHI_{i,t}$, the change in $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. $\Delta Subsidy\ Amount_{i,t}$ is a continuous within-county standardized measure of the predicted change in total fixed subsidy payments in a given county-year. $\Delta Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized change in the midpoint farm size. The controls are the same set of first- and second-stage controls as in Table 4, but in first differences. F-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
	First Stage		Second Stage	
	Δ Farm Concentration		Δ Bank Deposits HHI	
Δ Farm $\widehat{\text{Concentration}}_{i,t-2}$			123.28*	157.77**
			(74.10)	(65.55)
Δ Subsidy Amount$_{i,t-2}$	36.56***	44.85***		
	(3.92)	(4.22)		
Δ Log Income Per Capita$_{i,t-2}$		90.05		-55.57
		(65.85)		(89.83)
Δ Employment Rate$_{i,t-2}$		-123.39		71.99
		(133.26)		(274.07)
Δ Idle Land$_{i,t-7}$		20.54***		4.51
		(4.34)		(6.36)
Δ Unharvested Cropland$_{i,t-7}$		15.70***		-7.17
		(4.19)		(7.00)
Δ Pastureland$_{i,t-7}$		5.98*		-6.41
		(3.37)		(5.39)
Observations	13,173	13,173	13,173	13,173
F-statistic	86.97	113.07	86.97	113.07
Deregulation Index FE	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes
Mean of Dep. Variable	119	119	13	13

Table A.10. Effect of Subsidy Amount and Farm Concentration on Banks' Interest Expense Betas

This table reports reduced-form estimates of the effect of subsidy payments and second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on county-level bank interest expense betas. Columns 1 and 3 report the reduced form: $Y_{i,t} = \alpha\, Subsidy\ Amount_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$. Columns 2 and 4 report second-stage IV estimates, where the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\, Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\, \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Columns 1 and 2, the outcome variable is the total interest expense beta of Drechsler, Savov, and Schnabl (2021), the sensitivity of banks' total interest expense per dollar of assets to the federal funds rate. In Columns 3 and 4, the outcome variable is the deposit interest expense beta, the sensitivity of banks' interest expense on deposits per dollar of deposits to the federal funds rate. A lower beta indicates greater deposit market power. The outcome variables are standardized by their within-county standard deviations. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. The controls are the same set of first- and second-stage controls as in Columns 1, 2, 4, and 5 of Table 5. F-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
	Total Interest Expense Beta		Deposit Interest Expense Beta	
Subsidy Amount$_{i,t-2}$	-0.058***		-0.041***	
	(0.016)		(0.014)	
Farm $\widehat{\text{Concentration}}_{i,t-2}$		-0.234***		-0.183***
		(0.068)		(0.063)
Observations	13,621	13,621	13,621	13,621
F-statistic	4.63	89.57	3.56	89.57
County Controls	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes
Mean of Dep. Variable	17	17	16	16

Table A.11. Effect of Farm Concentration on Banking Concentration, Correcting for Attenuation Bias

This table reports ordinary least squares (OLS) and instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on banking concentration in twenty-year differences, where Δ denotes the change between 1994 and 2014. Column 1 reports the OLS specification: $\Delta Y_{i,t} = \beta\, \Delta \textit{Farm Concentration}_{i,t-2} + \Delta \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \delta_t + \varepsilon_{i,t}$. Columns 2 and 3 report second-stage IV estimates, where the second stage is $\Delta Y_{i,t} = \beta_2\, \Delta \widehat{\textit{Farm Concentration}}_{i,t-2} + \Delta \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \delta_t + \epsilon_{i,t}$. In Column 2, the first stage is $\Delta \textit{Farm Concentration}_{i,t} = \pi_1\, \Delta^{1999,2009} \textit{Farm Share}_i + \pi_2\, \Delta^{1999,2009} \textit{Land Share}_i + \Delta \mathbf{X}'_{i,t}\mathbf{\Gamma} + \delta_t + \nu_{i,t}$, where $\Delta^{1999,2009}$ denotes the change between 1999 and 2009. In Column 3, the first stage is $\Delta \textit{Farm Concentration}_{i,t} = \beta_1\, \Delta \textit{Subsidy Amount}_{i,t} + \Delta \mathbf{X}'_{i,t}\mathbf{\Gamma} + \delta_t + \nu_{i,t}$. Subscript i represents the county and t represents the year. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). The sample is restricted to counties observed in both 1994 and 2014 whose farm share and land share are also observed in 1999 and 2009. The outcome variable, $\Delta Y_{i,t}$, is the change in *Bank Deposits HHI*$_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. *Farm Concentration*$_{i,t-2}$ is the lagged within-county standardized midpoint farm size. *Farm Share*$_{i,t}$ is the within-county standardized share of farms with harvested cropland that harvest at least 1,000 acres, and *Land Share*$_{i,t}$ is the within-county standardized share of land in farms on farms of at least 1,000 acres. *Subsidy Amount*$_{i,t}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year. The controls are the same set of first- and second-stage controls as in Column 2 of Panel B of Table 4, in twenty-year differences and without the deregulation index. Columns 1 and 2 include only the second-stage controls. F-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)
	Δ Bank Deposits HHI		
Δ Farm Concentration$_{i,t-2}$	35.16**		
	(17.09)		
Δ Farm $\widehat{\text{Concentration}}_{i,t-2}$		221.59***	317.29***
		(48.92)	(71.44)
Observations	2,340	2,340	2,340
F-statistic	2.63	142.53	76.47
County Controls	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	No	No	No
Mean of Dep. Variable	-33	-33	-33

Table A.12. Decomposition of the IV–OLS Gap in the Effect of Farm Concentration on Banking Concentration

This table decomposes the gap between the instrumental variable (IV) and ordinary least squares (OLS) estimates of the effect of farm concentration on banking concentration following Ishimaru (2024). The baseline specification is: $Y_{i,t} = \beta \, \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). The outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. The decomposition separates the IV–OLS gap, $\hat{\beta}_{IV} - \hat{\beta}_{OLS}$, into three components: differences in covariate weights, treatment-level weights, and marginal effects. Columns 1 and 3 use a cubic polynomial in farm concentration as the basis function. Columns 2 and 4 use a fifth-order polynomial. The baseline controls are the same set of first- and second-stage controls as in Column 2 of Panel B of Table 4. Columns 3 and 4 augment the baseline controls with the log market value of agricultural land and buildings per acre from the Census of Agriculture and the log of total county deposits. F-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
	Bank Deposits HHI			
Δ Covariate Weight	-1.97	-1.97	4.66	4.66
	(3.91)	(3.91)	(4.61)	(4.61)
Δ Treatment-Level Weight	4.49	10.80*	4.86	11.27*
	(3.86)	(6.09)	(3.52)	(6.03)
Δ Marginal Effect	255.03***	248.72***	306.34***	299.93***
	(39.42)	(39.10)	(40.61)	(40.22)
Observations	13,818	13,818	13,790	13,790
F-statistic	334.04	334.04	334.14	334.14
Marginal Effect / Gap (%)	99.0	96.6	97.0	95.0
Cubic Polynomial Basis	Yes	No	Yes	No
Quintic Polynomial Basis	No	Yes	No	Yes
Collateral & Deposit Controls	No	No	Yes	Yes
County Controls	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes
Mean of Dep. Variable	3,121	3,121	3,121	3,121

Table A.13. Excluding 2009: Effect of Farm Concentration on Large and Small Banks

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the presence, branches, and deposits of large and small banks, excluding the 2008 Financial Crisis. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Columns 1 and 4 and $E\big[Y_{i,t}\,\big|\,Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). The sample is further restricted to exclude 2009. In Columns 1 and 4, the outcome variable is an indicator that a large or small bank, respectively, is present in the county in a given year. In Columns 2 and 5, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 3 and 6, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small banks, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 5. F-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Large Banks			Small Banks		
	Probability of Entry	Branches	Deposits Per Capita	Probability of Entry	Branches	Deposits Per Capita
$\widehat{\text{Farm Concentration}}_{i,t-2}$	0.049*	0.178***	0.361**	0.024***	0.100***	0.273***
	(0.027)	(0.058)	(0.170)	(0.008)	(0.029)	(0.043)
Observations	11,154	9,759	9,755	11,154	11,021	11,021
F-statistic	246.65	246.65	131.41	246.65	246.65	131.41
County Controls	Yes	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	0.76	14	8,128	0.96	9	12,111

Table A.14. Controlling for Land Values, Deposits, and Production Outcomes: Effect of Subsidy Amount and Farm Concentration on Large and Small Banks

This table reports reduced-form estimates of the effect of subsidy payments (Panel A) and second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration (Panel B) on the presence, branches, and deposits of large and small banks, controlling directly, in both stages, for land values, total county deposits, and the production outcomes reported in Table A.5. In Panel A, the specifications are $Y_{i,t} = \alpha\, \textit{Subsidy Amount}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ in Columns 1 and 4, and $E\big[Y_{i,t} \,\big|\, \textit{Subsidy Amount}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\alpha\, \textit{Subsidy Amount}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. In Panel B, the specification for the first stage is the specification of Columns 1 and 4 in Panel A with $\textit{Farm Concentration}_{i,t-2}$ as the outcome variable. The specifications for the second stages are $Y_{i,t} = \beta_2\, \widehat{\textit{Farm Concentration}}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Columns 1 and 4 and $E\big[Y_{i,t} \,\big|\, \textit{Farm Concentration}_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ \textit{Farm Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns using a control function approach. In the table, I include a hat on $\textit{Farm Concentration}_{i,t-2}$ to indicate this is an IV regression. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). The sample ends in banking year 2009 and excludes the final five-year snapshot. In Columns 1 and 4, the outcome variable is an indicator that a large or small bank, respectively, is present in the county in a given year. In Columns 2 and 5, the outcome variable is $\textit{Branches}_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 3 and 6, the outcome variable is $\textit{Deposits Per Capita}_{i,t}$, which is the deposits per capita in large and small banks, respectively. $\textit{Subsidy Amount}_{i,t-2}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year. $\textit{Farm Concentration}_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The baseline controls are the same set of controls as in Columns 2 through 7 of Table 2 in Panel A and Table 5 in Panel B, augmented by the following set of controls. The first is $\textit{Log Land Value Per Acre}_{i,t-2}$, the log market value per acre of agricultural land and buildings from the Census of Agriculture. The second is $\textit{Log Total County Deposits}_{i,t-2}$. The third is the production controls, the Törnqvist crop-yield index, machinery stock, tractor investment, and program crop share. F-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
		Large Banks			Small Banks	
	Probability of Entry	Branches	Deposits Per Capita	Probability of Entry	Branches	Deposits Per Capita
Panel A: Effect of Subsidy Amount						
Subsidy Amount$_{i,t-2}$	0.008*	0.026***	0.029***	0.002	0.015***	0.026***
	(0.005)	(0.009)	(0.011)	(0.001)	(0.004)	(0.004)
Observations	11,146	9,723	9,723	11,146	11,027	11,027
R-squared	0.73	0.88	0.78	0.64	0.62	0.88
Mean of Dep. Variable	0.76	14	7,883	0.97	9	12,127
Panel B: Effect of Farm Concentration						
Farm Concentration$_{i,t-2}$	0.059**	0.157**	0.453***	0.014**	0.079***	0.283***
	(0.029)	(0.067)	(0.140)	(0.007)	(0.026)	(0.045)
Observations	11,146	9,723	9,723	11,146	11,027	11,027
F-statistic	262.68	262.68	146.20	262.68	262.68	146.20
County Controls	Yes	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	0.76	14	7,883	0.97	9	12,127

Table A.15. Small Banks: Effect of Farm Concentration on Farm Lending by Farm Revenue and Loan Size

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on small banks' farm lending by farm revenue and loan size. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t}\,\big|\,Farm\ \ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ \ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data at the annual level (see Section 5 for variable and sample construction details). The sample is restricted to 1996 to 2004, the years in which small banks report under the CRA. The outcome variable is the total volume of farm loans of $500,000 or less that small banks originate in a given county-year, for loans to farms with gross annual revenue above $1 million in Column 1 and $1 million or less in Column 2. In Columns 3 through 5, the outcome variable is the total volume of farm loans at or below $100,000, between $100,000 and $250,000, and between $250,000 and $500,000, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size, carried forward annually until the next census observation. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 7. F-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
	Farm Revenue		Loan Size		
	>$1M	≤$1M	≤$100K	$100K–$250K	$250K–$500K
Farm $\widehat{\text{Concentration}}_{i,t-2}$	0.332	-0.258	-0.242	-0.289	-0.209
	(0.515)	(0.201)	(0.200)	(0.208)	(0.234)
Observations	20,301	24,351	24,351	20,898	17,934
F-statistic	38.34	48.34	48.11	46.88	46.72
County Controls	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	412	2,563	1,250	1,111	951

Table A.16. Alternative County Allocation: Effect of Farm Concentration on Agricultural Loan Growth

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on agricultural loan growth by bank size. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t}\,\big|\,Farm\ \ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Each bank's agricultural lending is allocated across counties under the alternative allocation described in Section B.1. In Columns 1 and 2, the outcome variable is $Total\ Agricultural\ Loan\ Growth_{i,t}$, in Columns 3 and 4 it is $Real\ Estate\ Loan\ Growth_{i,t}$, and in Columns 5 and 6 it is $Non\text{-}Real\ Estate\ Loan\ Growth_{i,t}$, each measuring large or small banks' growth in the loan type relative to the prior period's total volume of that loan type. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 8. F-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Total Agricultural Loan Growth		Real Estate Loan Growth		Non-Real Estate Loan Growth	
	Large Banks	Small Banks	Large Banks	Small Banks	Large Banks	Small Banks
$\widehat{\text{Farm Concentration}}_{i,t-2}$	0.529***	0.025	0.438**	-0.004	0.403***	0.051
	(0.144)	(0.040)	(0.179)	(0.053)	(0.122)	(0.039)
Observations	13,780	13,735	13,769	13,711	13,766	13,702
F-statistic	323.26	323.26	322.48	322.48	321.98	321.98
County Controls	Yes	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	0.49	0.73	0.54	0.89	0.56	0.66

Table A.17. Effect of Farm Concentration on Small Banks' Agricultural Lending by Loan Size, Core Funding, and Funding Utilization

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on small banks' agricultural lending, splitting small banks on two predetermined funding measures at once. The specifications for the first and second stages, respectively, are: $\textit{Farm Concentration}_{i,t} = \beta_1\ \textit{Subsidy Amount}_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t} \,\big|\, \textit{Farm Concentration}_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ \textit{Farm Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $\textit{Farm Concentration}_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. High-core small banks are those with a pre-program (1990–1995) core deposit share above the within-small-bank median, and low-core small banks otherwise. Slack small banks are those with a pre-program (1990–1995) ratio of loans to deposits below the within-small-bank median, and at limit otherwise. In Columns 1 and 3, the outcome variable is small banks' growth in real estate farm lending, and in Columns 2 and 4 it is small banks' growth in non-real estate farm lending, each relative to the prior period's total volume of that loan type. In Columns 1 and 2, the outcome variable is the growth in loans above $500,000, and in Columns 3 and 4 it is the growth in loans at or below $100,000. The controls are the same set of first- and second-stage controls as in Table 9. F-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
	Agricultural Loan Growth			
	>$500K		≤$100K	
	Real Estate	Non-Real Estate	Real Estate	Non-Real Estate
High-Core, Slack	1.391**	3.008***	-0.184***	-0.017
	(0.560)	(0.799)	(0.065)	(0.059)
High-Core, At Limit	0.823**	0.236	-0.288***	-0.162**
	(0.349)	(0.374)	(0.071)	(0.071)
Low-Core, Slack	0.098	0.388	-0.303***	-0.183**
	(0.230)	(0.245)	(0.065)	(0.077)
Low-Core, At Limit	-0.012	-0.207	-0.070	-0.168**
	(0.217)	(0.184)	(0.085)	(0.071)
High-Core, Slack − Low-Core, At Limit	1.404**	3.215***	-0.114	0.151*
	(0.703)	(0.885)	(0.099)	(0.083)
High-Core, Slack − High-Core, At Limit	0.568	2.772**	0.104	0.145*
	(0.967)	(1.188)	(0.083)	(0.083)
High-Core, Slack − Low-Core, Slack	1.293*	2.621**	0.119	0.166*
	(0.663)	(1.030)	(0.076)	(0.093)
Observations	13,821	13,821	13,821	13,821
F-statistic	322.65	322.65	322.65	322.65
County Controls	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes
Mean of Dep. Variable	0.11	0.06	0.18	0.19

Table A.18. High- vs. Low-Cost Counties: Effect of Farm Concentration on the Presence, Branches, and Deposits of Large and Small Banks

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the presence, branches, and deposits of large and small banks, in high-cost counties (Panel A) and low-cost counties (Panel B). The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ Farm\ \widehat{Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Columns 1 and 4 and $E\big[Y_{i,t}\,\big|\,Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Counties are classified as high-cost if their average per acre economic costs are above the nationwide median, and low-cost otherwise. Economic costs per acre for each major commodity crop are calculated as the six-year average (1990–1995) from the Farm Costs and Returns Survey (FCRS) compiled by the U.S. Department of Agriculture's Economic Research Service (ERS). The measure is constructed as a time-invariant, county-level weighted average of per acre economic costs, where weights are each crop's share of harvested acreage from the 1992 Census of Agriculture. In Columns 1 and 4, the outcome variable is an indicator that a large or small bank, respectively, is present in the county in a given year. In Columns 2 and 5, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 3 and 6, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small banks, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size, standardized within the relevant subsample. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 5. F-statistics are from the first stage. Coefficients are reported as decimal percentages. Means of the dependent variable are calculated within the relevant subsample. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Large Banks			Small Banks		
	Probability of Entry	Branches	Deposits Per Capita	Probability of Entry	Branches	Deposits Per Capita
Panel A: High-Cost Counties						
$\widehat{\text{Farm Concentration}}_{i,t-2}$	0.041*	0.120**	0.411**	0.010*	0.062***	0.060**
	(0.022)	(0.048)	(0.191)	(0.006)	(0.021)	(0.025)
Observations	6,910	6,274	6,274	6,910	6,882	6,882
F-statistic	56.84	56.84	45.47	56.84	56.84	45.47
Mean of Dep. Variable	0.80	14	8,181	0.97	10	11,250
Panel B: Low-Cost Counties						
$\widehat{\text{Farm Concentration}}_{i,t-2}$	0.005	0.077**	0.070	0.013**	0.122***	0.197***
	(0.022)	(0.037)	(0.057)	(0.006)	(0.026)	(0.028)
Observations	6,898	5,957	5,952	6,898	6,785	6,785
F-statistic	53.76	53.76	50.83	53.76	53.76	50.83
County Controls	Yes	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	0.76	15	8,772	0.95	8	12,632

Table A.19. Alternative Definition of Agricultural Banks: Effect of Farm Concentration on Large and Small Banks

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the presence, branches, and deposits per capita of large and small agricultural (ag) banks. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Columns 1 and 4 and $E\big[Y_{i,t} \,\big|\, Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in the remaining columns. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). The sample is restricted to ag banks, those with farm loans exceeding 1% of total loans in any of the preceding six years. In Columns 1 and 4, the outcome variable is an indicator that a large or small ag bank, respectively, is present in the county in a given year. In Columns 2 and 5, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small ag banks, respectively. In Columns 3 and 6, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small ag banks, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Columns 2 through 7 of Table 10. F-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Large Ag Banks			Small Ag Banks		
	Probability of Entry	Branches	Deposits Per Capita	Probability of Entry	Branches	Deposits Per Capita
$\widehat{\text{Farm Concentration}}_{i,t-2}$	0.069***	0.231***	0.607***	0.023***	0.037*	0.228***
	(0.023)	(0.064)	(0.120)	(0.008)	(0.022)	(0.032)
Observations	13,808	11,912	11,897	13,808	13,432	13,432
F-statistic	331.31	331.31	185.11	331.31	331.31	185.11
County Controls	Yes	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	0.67	7	5,032	0.90	7	10,502

Table A.20. Effect of Farm Concentration on the Best and Worst Offered Interest Rates, 2002 to 2014

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the best and worst real interest rates offered by banks in the county. The specifications for the first and second stages, respectively, are: $\textit{Farm Concentration}_{i,t} = \beta_1 \textit{Subsidy Amount}_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2 \widehat{\textit{Farm Concentration}}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data at the annual level (see Section 5 for variable and sample construction details). In Panel A, the outcome variable is the best interest rate offered by any bank branch in the county, which is the highest rate for deposit accounts and the lowest rate for auto loans and residential mortgage loans. In Panel B, the outcome variable is the worst interest rate offered by any bank branch in the county, which is the lowest rate for deposit accounts and the highest rate for auto loans and residential mortgage loans. Each branch's rate is its June rate. In Columns 1 through 5, the outcome variables are the offered interest rates on savings deposit accounts ($10,000 money market accounts), time deposit accounts (12-month $10,000 certificates of deposit), checking deposit accounts (interest-bearing checking accounts), auto loans, and residential mortgage loans, all expressed in real, inflation-adjusted terms. $\textit{Farm Concentration}_{i,t-2}$ is the lagged within-county standardized midpoint farm size, carried forward annually until the next census observation. The controls are the same set of first- and second-stage controls as in Table 11. F-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
	Deposit Rates			Loan Rates	
	Savings	Time	Checking	Auto	Mortgage
Panel A: Best Offered Rate in the County					
Farm $\widehat{\text{Concentration}}_{i,t-2}$	-0.298***	0.106**	-0.273***	1.084***	0.050
	(0.079)	(0.044)	(0.051)	(0.228)	(0.065)
Observations	34,828	34,834	34,786	25,770	23,178
F-statistic	323.16	323.16	323.16	323.16	323.16
Mean of Dep. Variable	-0.76	0.17	-1.40	3.85	2.94
Panel B: Worst Offered Rate in the County					
Farm $\widehat{\text{Concentration}}_{i,t-2}$	-0.181***	0.048	-0.080***	0.577***	-0.093
	(0.048)	(0.059)	(0.028)	(0.186)	(0.104)
Observations	34,828	34,834	34,786	25,770	23,178
F-statistic	323.16	323.16	323.16	323.16	323.16
County Controls	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	-1.54	-0.53	-1.78	5.62	3.59

Table A.21. First Differences: Effect of Farm Concentration on Offered Interest Rates, 2003 to 2014

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation change in farm concentration on the change in the real interest rates offered by banks. This specification uses first differences, where the first and second stages, respectively, are: $\Delta Farm\ Concentration_{i,t} = \beta_1\ \Delta Subsidy\ Amount_{i,t} + \Delta \mathbf{X}'_{i,t}\mathbf{\Gamma} + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $\Delta Y_{i,t} = \beta_2\ \Delta Farm\ \widehat{Concentration}_{i,t-2} + \Delta \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data at the annual level (see Section 5 for variable and sample construction details). In Columns 1 through 5, the outcome variables are the changes in the county-level average offered interest rates on savings deposit accounts ($10,000 money market accounts), time deposit accounts (12-month $10,000 certificates of deposit), checking deposit accounts (interest-bearing checking accounts), auto loans, and residential mortgage loans, all expressed in real, inflation-adjusted terms. $\Delta Farm\ Concentration_{i,t-2}$ is the change in the lagged within-county standardized midpoint farm size, interpolated annually until the next census observation. The controls are the same set of first- and second-stage controls as in Table 11, but in first differences. F-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
	Δ Deposit Rates			Δ Loan Rates	
	Savings	Time	Checking	Auto	Mortgage
Δ Farm $\widehat{\text{Concentration}}_{i,t-2}$	-0.191***	0.093**	-0.123***	0.391***	0.047
	(0.048)	(0.041)	(0.032)	(0.134)	(0.074)
Observations	29,718	29,720	29,687	20,985	18,805
F-statistic	86.68	86.68	86.68	86.68	86.68
County Controls	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	-0.07	-0.13	-0.00	-0.31	-0.17

Table A.22. Falsification Test: Effect of Farm Concentration on Federal Government Loans

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on direct loans from the Farm Service Agency. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1\ Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t}\,\big|\,Farm\ \ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Columns 1 and 2, the outcome variables are $Direct\ Loan\ Amount_{i,t}$, in \$1,000s, and $Number\ of\ Direct\ Loans_{i,t}$, respectively, for direct loans issued at the county level by the federal government. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. The controls are the same set of first- and second-stage controls as in Table 5. F-statistics are from the first stage. Coefficients are reported as decimal percentages. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1) Direct Loan Amount	(2) Number of Direct Loans
$\widehat{\text{Farm Concentration}}_{i,t-2}$	0.170	-0.142
	(0.211)	(0.135)
Observations	12,556	12,556
F-statistic	322.68	322.68
County Controls	Yes	Yes
Deregulation Index FE	Yes	Yes
County & Year FE	Yes	Yes
Mean of Dep. Variable	474	13

Table A.23. Effect of Farm Concentration on Banking Concentration, Controlling for Crop Insurance Coverage

This table reports first- and second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on banking concentration, controlling for crop insurance coverage. The specifications for the first and second stages, respectively, are: $\textit{Farm Concentration}_{i,t} = \beta_1 \textit{Subsidy Amount}_{i,t} + \beta_3 \textit{Log Crop Insurance Policies}_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2 \widehat{\textit{Farm Concentration}}_{i,t-2} + \beta_4 \textit{Log Crop Insurance Policies}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Columns 1 and 2, the outcome variable is $\textit{Farm Concentration}_{i,t}$, the midpoint farm size at the county level. In Columns 3 and 4, the outcome variable is $\textit{Bank Deposits HHI}_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. $\textit{Subsidy Amount}_{i,t}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year. $\textit{Farm Concentration}_{i,t-2}$ is the lagged within-county standardized midpoint farm size. $\textit{Log Crop Insurance Policies}_{i,t}$ is the log total number of crop insurance policies in the county-year. The controls are the same set of first- and second-stage controls as in Table 4. F-statistics are from the first stage. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
	First Stage		Second Stage	
	Farm Concentration		Bank Deposits HHI	
Farm $\widehat{\text{Concentration}}_{i,t-2}$			251.79***	274.36***
			(38.66)	(41.49)
Log Crop Insurance Policies$_{i,t-2}$	23.43***	22.87***	10.62	5.82
	(4.29)	(4.33)	(9.16)	(9.47)
Subsidy Amount$_{i,t-2}$	105.05***	106.66***		
	(5.52)	(5.90)		
Log Income Per Capita$_{i,t-2}$		195.47***		-87.51
		(69.11)		(124.42)
Employment Rate$_{i,t-2}$		-132.33		334.17
		(157.30)		(327.48)
Idle Land$_{i,t-7}$		20.81***		-12.17
		(4.98)		(7.90)
Unharvested Cropland$_{i,t-7}$		-11.97***		7.57
		(3.61)		(7.14)
Pastureland$_{i,t-7}$		4.02		-1.95
		(4.30)		(5.87)
Observations	13,818	13,818	13,818	13,818
F-statistic	362.31	326.49	362.31	326.49
Deregulation Index FE	No	Yes	No	Yes
County & Year FE	Yes	Yes	Yes	Yes
Mean of Dep. Variable	752	752	3,121	3,121

Table A.24. Effect of Subsidy Amount and Farm Concentration on Banking Concentration and Large and Small Banks, Controlling for Countercyclical Payments (CCP)

This table reports reduced-form estimates of the effect of subsidy payments on farm concentration and banking concentration (Columns 1 and 2) and second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on the branches and deposits of large and small banks (Columns 3 through 6), controlling for countercyclical payments (CCP). In Columns 1 and 2, the specification is $Y_{i,t} = \alpha\, Subsidy\ Amount_{i,t-2} + \theta\, CCP_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$. In Columns 3 through 6, the specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \theta_1\, CCP_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $E\big[Y_{i,t}\,\big|\, Farm\ \ Concentration_{i,t-2}, CCP_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \theta_2\, CCP_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_3\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$, estimated using a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Column 1, the outcome variable is $Farm\ Concentration_{i,t}$, the midpoint farm size at the county level. In Column 2, the outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. In Columns 3 and 5, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 4 and 6, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small banks, respectively. $Subsidy\ Amount_{i,t-2}$ is a continuous within-county standardized measure of the predicted value of total fixed subsidy payments in a given county-year. $CCP_{i,t-2}$ is a continuous within-county standardized measure of the predicted value of countercyclical payments in a given county-year. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. Small banks are those with less than \$1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first-stage controls as in Table 4 in Column 1, the same set of controls as in Column 4 of Panel A of Table 4 in Column 2, and the same set of first- and second-stage controls as in Table 5 in Columns 3 through 6. F-statistics are from the first stage. Coefficients are reported as decimal percentages in Columns 3 through 6. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
			Large Banks		Small Banks	
	Farm Concentration	Bank Deposits HHI	Branches	Deposits Per Capita	Branches	Deposits Per Capita
Subsidy Amount$_{i,t-2}$	107.97***	63.57***				
	(5.91)	(8.85)				
$\widehat{\text{Farm Concentration}}_{i,t-2}$			0.164***	0.333**	0.097***	0.222***
			(0.051)	(0.132)	(0.023)	(0.031)
CCP$_{i,t-2}$	11.20***	3.44	0.001	-0.000	-0.009***	-0.009***
	(2.55)	(3.57)	(0.002)	(0.005)	(0.002)	(0.002)
Observations	13,818	13,818	12,231	12,226	13,667	13,667
F-statistic	333.84	333.84	322.44	174.79	322.44	174.79
County Controls	Yes	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	752	3,121	14	8,468	9	11,936

Table A.25. Rural vs. Urban: Effect of Farm Concentration on Banking Concentration

This table reports first- and second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on banking concentration by county type. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-2} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). Columns 1 and 4 restrict the sample to $Urban_{i,t}$ counties, classified as large central metropolitan areas by the National Center for Health Statistics (NCHS). Columns 2 and 5, and Columns 3 and 6, restrict the sample to $Metro_{i,t}$, or broader metropolitan, and $Rural_{i,t}$, or noncore, counties, respectively. The outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. The controls are the same set of first- and second-stage controls as in Table 4. F-statistics are from the first stage. Means of the dependent variable are calculated within the relevant subsample. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	First Stage			Second Stage		
	Farm Concentration			Bank Deposits HHI		
	Urban	Metro	Rural	Urban	Metro	Rural
Subsidy Amount$_{i,t-2}$	65.48	71.06***	87.17***			
	(54.53)	(6.55)	(6.70)			
$\widehat{\text{Farm Concentration}}_{i,t-2}$				267.76	152.48***	364.99***
				(529.23)	(52.14)	(70.26)
Observations	193	7,398	6,021	193	7,398	6,021
F-statistic	1.45	117.88	169.20	1.45	117.88	169.20
County Controls	Yes	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	597	715	812	1,825	2,412	3,944

Table A.26. Effect of Farm Concentration on Large and Small Banks, Controlling for Farm Credit System (FCS) Branches

This table reports second-stage instrumental variable (IV) estimates of the effect of a within-county one-standard-deviation increase in farm concentration on banking concentration and on the branches and deposits of large and small banks, controlling for Farm Credit System (FCS) branches. The specifications for the first and second stages, respectively, are: $Farm\ Concentration_{i,t} = \beta_1 Subsidy\ Amount_{i,t} + \beta_3 Log\ FCS\ Branches_{i,t} + \mathbf{X}'_{i,t}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \varepsilon_{i,t}$ and $Y_{i,t} = \beta_2\ \widehat{Farm\ Concentration}_{i,t-2} + \beta_4 Log\ FCS\ Branches_{i,t} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \gamma_i + \delta_t + \phi_{d(i,t)} + \epsilon_{i,t}$ in Column 1 and $E\big[Y_{i,t}\,\big|\,Farm\ Concentration_{i,t-2}, \hat{\varepsilon}_{i,t-2}, \mathbf{X}'_{i,t-2}, Log\ FCS\ Branches_{i,t}, \gamma_i, \delta_t, \phi_{d(i,t)}\big] = \exp\big(\beta_2\ Farm\ Concentration_{i,t-2} + \beta_4 Log\ FCS\ Branches_{i,t} + \mathbf{X}'_{i,t-2}\mathbf{\Gamma} + \beta_5\ \hat{\varepsilon}_{i,t-2} + \gamma_i + \delta_t + \phi_{d(i,t)}\big)$ in Columns 2 through 5. Subscript i represents the county, t represents the year, and d represents state-level banking deregulation. For this specification, I use a control function approach. In the table, I include a hat on $Farm\ Concentration_{i,t-2}$ to indicate this is an IV regression. The county-year data are from merged bank and agriculture data (see Section 5 for variable and sample construction details). In Column 1, the outcome variable is $Bank\ Deposits\ HHI_{i,t}$, a continuous variable ranging from 0 to 10,000 calculated as the sum of squared deposit shares held by each bank within a county. In Columns 2 and 4, the outcome variable is $Branches_{i,t}$, which is the county-level number of branches owned by large and small banks, respectively. In Columns 3 and 5, the outcome variable is $Deposits\ Per\ Capita_{i,t}$, which is the deposits per capita in large and small banks, respectively. $Farm\ Concentration_{i,t-2}$ is the lagged within-county standardized midpoint farm size. $Log\ FCS\ Branches_{i,t}$ is the log number of Farm Credit System (FCS) branches in the county in year t. Small banks are those with less than $1 billion in inflation-adjusted assets, and large otherwise. The controls are the same set of first- and second-stage controls as in Table 5. F-statistics are from the first stage. Coefficients are reported as decimal percentages in Columns 2 through 5. Standard errors clustered at the county level are reported in parentheses. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
		Large Banks		Small Banks	
	Bank Deposits HHI	Branches	Deposits Per Capita	Branches	Deposits Per Capita
$\widehat{\text{Farm Concentration}}_{i,t-2}$	272.84***	0.159***	0.313***	0.101***	0.235***
	(40.71)	(0.050)	(0.115)	(0.023)	(0.032)
Log FCS Branches$_{i,t}$		0.007	0.082	-0.015	-0.069***
		(0.024)	(0.138)	(0.023)	(0.023)
Observations	13,818	12,231	12,226	13,667	13,667
F-statistic	329.44	316.79	172.33	316.79	172.33
County Controls	Yes	Yes	Yes	Yes	Yes
Deregulation Index FE	Yes	Yes	Yes	Yes	Yes
County & Year FE	Yes	Yes	Yes	Yes	Yes
Mean of Dep. Variable	3,121	14	8,468	9	11,936

Appendix B. Data Methods and Technical Details

This appendix provides additional detail on the procedures used to construct key variables and allocate bank- and loan-level data to the county level for empirical analysis.

B.1 Estimating County-Level Agricultural Lending

Because agricultural loan volumes are reported at the bank level in the FFIEC Call Reports, I allocate each bank's lending across counties based on its branches at the start of each five-year period. For the period ending in year t, I use the counties in which the bank operated a branch in year $t-5$, identified from the FDIC's Summary of Deposits. The five-year period matches the frequency with which the Census of Agriculture updates the interest expense weights.

I allocate each bank's agricultural lending in year t across these counties using farm interest expenses from the most recent Census of Agriculture as of year $t-5$. I use interest expenses on real estate loans for real estate farm lending, interest expenses on non-real estate loans for non-real estate farm lending, and their sum for total agricultural lending. For example, if a bank operated branches in three counties with equal farm interest expenses in year $t-5$, each county receives one third of the bank's agricultural lending in year t. I allocate the prior period's lending across the same counties using the same weights. I repeat this procedure for every bank and period, winsorize the allocated lending of each bank-county pair at the 1st and 99th percentiles, and aggregate it by county separately for small and large banks.

Table A.16 reports results under an alternative allocation that also assigns lending to counties in which a bank lends without operating a branch. The alternative footprint follows Key, Burns, and Lyons (2019) and Ifft et al. (2024), who use the CRA data to identify counties in which a bank lends outside its branch network. The alternative footprint adds to the bank's branch counties in year t the counties in which the bank originated farm loans during the five years ending in year t, identified using the CRA disclosure files. I use originations with original amounts of \$500,000 or less from table D2-1 and match CRA respondents to Call Report identifiers using the CRA transmittal files. Because CRA reporting begins in 1996 and applies only to institutions above an asset threshold, this footprint contains branch counties alone for all years before 1996 and whenever CRA data are unavailable. The alternative allocation uses farm interest expenses from the most recent Census of Agriculture as of year t and decreases the weight assigned to counties in which the bank does not operate a branch. I multiply the county's interest expenses by $(1 + r_{bi})^{\theta}$, where r_{bi} is the distance in kilometers between county i's banking centroid and the nearest branch of bank b. I define

the banking centroid as the average coordinates of all bank branches in the county. Counties in which the bank operates a branch receive no distance adjustment. I estimate θ from a Poisson regression of the bank's farm loan originations in each county on log distance, log farm interest expenses, and an indicator for whether the bank operates a branch in the county, with fixed effects for each bank and year combination. The estimate of θ is -0.23, consistent with the distance deterrence in commercial lending documented by Brevoort and Hannan (2006). I apply the same distance adjustment to real estate and non-real estate farm lending.

To assess the distance adjustment, I estimate the decay parameter using a random half of the CRA reporting banks and evaluate the resulting county shares using the remaining banks. Among the remaining banks, the correlation between the allocated county shares and observed origination shares is 0.63, compared with 0.56 when I use the same interest expense weights without the distance adjustment. For example, suppose a bank's lending set contains three counties with equal farm interest expenses. The bank operates branches in two counties and recently originated farm loans in a third county 100 kilometers from its nearest branch. Each branch county receives an unadjusted weight of one, and the third county receives a weight of $(1 + 100)^{-0.23} = 0.35$, after rounding. Normalizing these weights assigns 42.6% of the bank's agricultural lending to each branch county and 14.7% to the county outside its branch network. I repeat this procedure for every bank and year and aggregate the allocated lending by county, separately for small and large banks.

Two limitations apply to both allocations. First, the main allocation assumes that a bank lends only in its branch counties, and the CRA data that relax this assumption cover only institutions above the reporting threshold and only farm loans of $500,000 or less. Second, both allocations assume that past interest expenses measure current county loan demand, and that lending remains proportional to the resulting weights.

B.2 Loan Size Categories and Funding Measures

Call Report Schedule RC-C Part II reports each bank's outstanding farm loans of $500,000 or less by original loan amount in three categories, $100,000 or less, between $100,000 and $250,000, and between $250,000 and $500,000. The Call Reports provide these categories separately for real estate farm loans and non-real estate farm loans. I construct the category above $500,000 by subtracting the three reported categories from the bank's total farm lending in Schedule RC-C. The reporting thresholds are fixed in nominal terms. I express loan volumes in constant dollars using the CPI but retain the reported origination categories. I allocate each bank's loans in each category across the set of counties described above using the same interest expense weights, then aggregate the allocated lending by

county, separately for small and large banks.

The funding measures are fixed before the subsidy program begins. The core deposit share equals total deposits less time deposits of $100,000 or more and brokered deposits, divided by total deposits, averaged over 1990–1995. Core deposits are the stable component of bank funding, and institutions that rely more heavily on them sustain lending when funding conditions tighten (Berlin and Mester, 1999; Cornett et al., 2011). The loans-to-deposits ratio, averaged over the same period, measures how much of the deposit base is already deployed in lending. A lower ratio indicates that more deposit funding remains available for additional loans (Acharya and Mora, 2015). I calculate each median among small banks, classify small banks as above or below it, and hold these classifications fixed throughout the analysis.

B.3 Community Reinvestment Act Loans

Table 7 uses farm loan originations that banks report under the CRA. Each reporting bank lists the dollar amount of its farm loans with original amounts of $500,000 or less in each year by the county of the borrowing farm. I use originations from 1996 to 2014, and I match each CRA lender to its Call Report identifier using the lender information that each reporting bank files with its CRA data.

Only banks above an asset threshold report these originations. Through 2004, a bank reports if it has $250 million or more in total assets or belongs to a holding company with $1 billion or more in total assets. From 2005, the threshold is about $1 billion in total assets, adjusted each year for inflation. Most small banks stop reporting after 2004, so the small bank sample ends in 2004.

The CRA data report loans to farms with revenue of $1 million or less. I measure loans to farms with revenue above $1 million as the remaining loans, which also include loans to farms whose revenue the bank does not report. In 113 of the 353,207 bank-county-year observations, all in 1996, the reported amount to farms with revenue of $1 million or less exceeds the bank's total originations. I set the remaining loans in these observations to zero. Each bank also divides its originations into three categories of original amount, $100,000 or less, above $100,000 and up to $250,000, and above $250,000 and up to $500,000. These thresholds are fixed in nominal terms. I express amounts in thousands of 2020 dollars using the CPI and aggregate them by county and year, separately for small and large banks, with bank size measured in the year of the origination.

B.4 Instrument Construction and Interpolation

I construct predicted subsidy payments from county base acres and national crop-level subsidy allocations. Base acres initially reflect planting between 1981 and 1985. Beginning in 2002, I use updated base acres based on planting between 1998 and 2001 and include soybeans and peanuts, which became eligible under the Farm Security and Rural Investment Act. Actual payments also depended on historical program yields, but I exclude yields because they may be correlated with local productivity. Including historical yields produces similar estimates.

For each county and year, I multiply its share of national base acres in each crop by the corresponding national subsidy allocation and sum across crops. Base acres remain fixed within each policy period, and national subsidy allocations vary across crops and over time. Counties therefore experience different changes in predicted payments based on their historical crop shares. County fixed effects absorb permanent differences across counties, and year fixed effects absorb national changes common to all counties. The resulting measure is the continuous treatment used in the reduced-form and first-stage specifications.

In the five-year panel, I calculate predicted payments in each Census of Agriculture year and use them as the instrument for farm concentration. The annual RateWatch panel requires annual values of farm concentration, but the Census measures farm structure every five years. I estimate fitted farm concentration in each census year and carry the most recent fitted value forward until the next Census. Annual local economic controls continue to vary in the second-stage regressions. Observed farm concentration changes slowly, increasing by 2.5% per year on average over the sample. To estimate the specification in first differences (Table A.21), I instead interpolate fitted farm concentration linearly between census years.

B.5 Lag Structure and Timing Alignment

In the first stage, I relate farm concentration to predicted payments in the same Census of Agriculture year. In the banking specifications, predicted payments, fitted farm concentration, and the agricultural controls enter with a two-year lag. The lag allows changes in farm structure to precede the banking outcomes. The SOD records the location and deposits of each branch as of June 30 each year. I use each branch's June RateWatch rate, the average of its weekly June quotes, so the interest rate and SOD outcomes refer to the same month.

B.6 USDA Direct Loan Program

To study whether borrower consolidation affects access to federal agricultural credit, I use administrative data on direct loans from the USDA's Farm Service Agency (FSA). These

loans primarily serve family farms and are intended for small, new, or socially disadvantaged farmers who cannot obtain commercial credit at reasonable rates. The FSA defines a family farm as an operation in which members of one family own and operate most of the business. Direct loans support farm ownership, operation, or expansion.

Eligibility does not depend on a minimum credit score. Although applicants must demonstrate reasonable ability to repay and cannot be delinquent on other federal debt, individuals with no formal credit history may still qualify. The program includes both ownership and operating credit. Farm ownership loans can be used to purchase farmland or expand physical infrastructure and carry repayment terms of up to 40 years. Operating loans are designed for inputs such as seed, livestock, or equipment and carry repayment terms of up to seven years. Over the sample period, the loan limit for both programs was $200,000 through 2007 and $300,000 from 2008 onward, following the loan limit increase in the 2008 Farm Bill.

I aggregate annual FSA direct loan records to the county level from 1994 to 2014. The two outcomes are the number of loans issued and their total dollar amount, combining farm ownership and operating loans. These outcomes measure whether borrower consolidation changes the use of federally supported agricultural credit.

B.7 Total and Deposit Interest Expense Betas

I estimate the interest expense betas in Table A.10 using the quarterly Call Report panel distributed by Drechsler, Savov, and Schnabl (2021). The sample includes commercial banks from 1984 to 2017 with at least 60 quarterly observations and excludes merger quarters and quarters with extreme asset growth. For each bank, I regress the quarterly change in total interest expense per dollar of assets on the contemporaneous change in the federal funds rate and three lags. The total interest expense beta equals the sum of these four coefficients. I construct the deposit interest expense beta in the same manner, using interest expense on deposits divided by total deposits (Narayanan, Ratnadiwakara, and Strahan, 2025). Following Drechsler, Savov, and Schnabl (2021), I winsorize both betas at the 5th and 95th percentiles. County-level betas are deposit-weighted averages across banks operating in the county, using each bank's branch deposits from the SOD in the corresponding census year.

Internet Appendix

IA.1 A Model of Farm Consolidation and Bank Lending

This section develops a one-period model of farm borrowing and bank lending. The model first shows how fixed cash payments tied to land increase farms' pledgeable resources and the loan a lender is willing to extend. The payments reduce lenders' break-even interest rates and allow some previously rationed farms to borrow. The model also shows why larger farms face lower borrowing costs and why merging farms may obtain cheaper financing when consolidation increases mean returns.

The model then introduces small and large banks that differ in their information technology, underwriting costs, recovery rates, and sources of funding. When farms expand or merge, credit demand becomes concentrated in fewer, larger loans. Small banks' informational advantage declines with farm size, while fixed underwriting costs fall per dollar as loan size increases. Larger loans also become more costly for small banks to fund relative to their core deposit base. Borrower consolidation increases large banks' share of lending, although small banks with greater core funding remain better able to serve larger farms.

The model develops separate predictions for the scale and funding, collateral, deposit, and production channels. Changes in collateral or production reduce both banks' break-even interest rates across loan sizes, and they reduce a small bank's rate by more when its core deposit base is smaller. Deposit inflows instead benefit small banks, particularly those closest to their funding constraints. Among these channels, only the scale and funding channel matches the empirical evidence. A within-county one-standard-deviation increase in subsidy amount increases farm concentration by 105–107 acres. Within small banks, a one-standard-deviation increase in farm concentration increases lending above $500,000 only at banks with high-core funding, and mergers among small banks increase by about 23%. Subsidy payments do not increase land values or deposits per capita during the first decade of the program, and they do not increase measures of farm production. These predictions provide the basis for the empirical tests in the main paper.

IA.1 Setup

A farm is characterized by its scale S, measured in acres, and the market price of farmland q per acre. The farm seeks a loan L to finance a project with payoff $X \sim N(\mu L, \sigma_j^2 L^2)$, where μ is the expected return per dollar invested and σ_j measures payoff uncertainty. Both the expected payoff and its risk scale with the size of the project. The subscript j on the payoff uncertainty indexes the lender and is defined below. The farm also holds fixed cash payments A, which reflect the subsidies tied to its land. Investment needs and fixed cash payments

scale with size, $L = \pi_2 S$ and $A = \pi_1 S$, which implies that $A = \pi L$ for a constant π. At the end of the period the farm owes $L(1 + r)$, where r is the interest rate on a loan, if one is received. The farm repays when the project payoff X plus its cash A and the value of its land qS cover this obligation, and defaults when $X + A + qS < L(1 + r)$. In default the lender liquidates the farm's resources $A + qS + X$. Lending is competitive, and the farm borrows from the lender offering the lowest interest rate.

Two bank types, small (s) and large (ℓ), lend to the farm and differ along three dimensions. The first is their information technology. The small bank holds a soft information advantage in lending to opaque borrowers, so that payoff uncertainty when the small bank lends is $\sigma_s(S) = \sigma - \delta(S)$, with $\delta(S) > 0$, $\delta'(S) < 0$, and $\delta(S) \to 0$ as S increases, while uncertainty when the large bank lends is $\sigma_\ell(S) = \sigma$. The advantage $\delta(S)$ reflects relationship lending. It is valuable for small, opaque farms and erodes as borrowers become larger and more transparent (Petersen and Rajan, 1994; Berger et al., 2005).

The second is underwriting and liquidation technology. Underwriting a loan costs bank j a fixed amount F_j, with $F_\ell > F_s$, so that the large bank operates a standardized, fixed-cost-intensive technology whose cost per dollar falls quickly with loan size, while the small bank's relationship technology has a low fixed cost. In default, bank j recovers a fraction ℓ_j of the farm's resources. The recovery rates may differ across bank types, and no result depends on which bank recovers more. The propositions below involve ℓ_s and ℓ_ℓ only through the difference $\ell_\ell - \ell_s$. However, it is natural to assume that the large bank is the weakly better liquidator of standardized assets, $\ell_\ell \geq \ell_s$. Note that the value of the recovery difference increases with the resources recovered in default, and with collateral qS.

The third is funding technology. Bank j pays a cost of funds w_j per dollar lent. The small bank raises its funds from a local core deposit base D, measured in dollars of core deposits, and its cost of funds is

$$w_s = w\left(\frac{L}{D}\right), \qquad w' > 0,$$

with $w(L/D)$ increasing without bound as L/D increases. While lending is small relative to the deposit base, the bank funds loans with cheap core deposits. As lending outgrows the deposit base, it must substitute toward rate-sensitive wholesale and brokered funds, whose cost increases with the amount of funding raised. This matches evidence that small banks' lending tracks their local deposits and that non-core funds carry a rising marginal cost (Jayaratne and Morgan, 2000; Kashyap and Stein, 2000; Stein, 1998). The large bank operates in many markets and allocates funds through an internal capital market, so the cost of funds it applies in any one county reflects its system-wide funding position rather than its local lending. I therefore take w_ℓ to be constant in L (Stein, 1997; Houston, James, and

Marcus, 1997; Campello, 2002). Consistent with this assumption, large banks move funding across markets in the data (Gilje, Loutskina, and Strahan, 2016; Cortés and Strahan, 2017). Treating w_ℓ as constant is a normalization rather than a substantive restriction. Suppose instead that the large bank draws on a system-wide funding base D_ℓ, so that $w_\ell = w(L/D_\ell)$. Its funding cost then increases with local lending, but less steeply than the small bank's because a county's loans are small relative to D_ℓ. All results continue to hold when D_ℓ is sufficiently large.

IA.2 Equilibrium

For an interest rate r, bank j's expected profit from lending L to a farm of type (S, q) is

$$\Pi_j(r; S, q) \;=\; E\Big[\mathbb{1}_{\{X+A+qS<L(1+r)\}}\,\ell_j \max\big(0,\; A + qS + X\big) \\ +\; \mathbb{1}_{\{X+A+qS\geq L(1+r)\}}\,L(1+r)\Big] \;-\; L\,(1+w_j) \;-\; F_j,$$

where $w_s = w(L/D)$ and w_ℓ is constant. The deposit base D enters the small bank's profit through its cost of funds. Two observations reduce the expected profit to a formula in per-dollar quantities. First, scaling by the loan, the return per dollar X/L is normal with mean μ and standard deviation σ_j. Second, the farm's cash and land are worth $a \equiv \pi + qS/L$ per dollar of loan. I call a the farm's pledgeable resources per dollar. The farm defaults when the return per dollar falls below $(1 + r) - a$, and its liquidated resources are positive only when the return exceeds $-a$. Using the mean of a truncated normal distribution for the region in between, expected profit per dollar of loan is

$$\frac{\Pi_j(r; S, q)}{L} \;=\; (1+r)\Big(1 - \Phi(\bar{z})\Big) \;+\; \ell_j\Big[(\mu + a)\big(\Phi(\bar{z}) - \Phi(z_0)\big) - \sigma_j\big(\phi(\bar{z}) - \phi(z_0)\big)\Big] \\ -\; (1+w_j) \;-\; \frac{F_j}{L}, \qquad \bar{z} = \frac{(1+r) - a - \mu}{\sigma_j}, \quad z_0 = \frac{-a - \mu}{\sigma_j},$$

where ϕ and Φ denote the standard normal p.d.f. and c.d.f., and $\Phi(\bar{z})$ is the probability of default. The first term is the repayment when the farm stays solvent. The second term is the expected recovery in default, earned over the states where the farm's resources are positive. The last two terms are the funds advanced, including the cost of funds, and the fixed underwriting cost. Denote the first two terms, the expected payoff per dollar, by $g(r, a, \sigma_j)$. One property of g does much of the work below: the mean return and the farm's resources enter only through the sum $\mu + a$, so a dollar of expected project return, a dollar of subsidy cash, and a dollar of collateral value raise the lender's expected payoff in exactly the same way.

The lending market is competitive, so each bank quotes the minimum rate $r_j^*(S, q)$ at

which its expected profit is nonnegative, the lowest solution of

$$g\big(r_j^*,\, a,\, \sigma_j\big) \;=\; 1 + w_j + \frac{F_j}{L},$$

whenever a solution exists. The right-hand side of this condition, $1 + w_j + F_j/L$, is bank j's cost per dollar of lending, and the phrase "cost per dollar" refers to it throughout. If none exists, the bank declines to lend. The farm borrows from the bank with the lower break-even rate, so the equilibrium lender is $j^*(S, q) = \arg\min_j\, r_j^*(S, q)$. Two forces determine j^*. The small bank's advantage is informational, since $\delta(S)$ lowers payoff uncertainty for opaque borrowers, and its fixed cost is lower. The large bank's advantage is financial, through superior recovery ℓ_ℓ, a fixed cost per dollar that falls quickly with loan size, and a cost of funds that does not increase with its lending in a single market.

Throughout, I maintain two regularity conditions. First, the probability of default at the break-even rate is below one half, which matches the single-digit delinquency rates on agricultural loans in the data. Second, expected profit crosses zero strictly from below at the break-even rate, $\partial g/\partial r > 0$ at r_j^*. Because r_j^* is the lowest solution and profit is negative at lower rates, the weak inequality holds automatically. The assumption that the crossing is strict rules out only the case in which profit touches zero without crossing it.[26] The next lemma signs the derivatives of the break-even rate with respect to pledgeable resources, payoff uncertainty, and the cost of lending, and is the basis for every proposition that follows.

LEMMA 1: *Under the regularity conditions, at the break-even rate r_j^*: (i) $\dfrac{\partial g}{\partial a} = (1-\ell_j)(1+r)\dfrac{\phi(\bar{z})}{\sigma_j} + \ell_j\big(\Phi(\bar{z}) - \Phi(z_0)\big) > 0$, so r_j^* is strictly decreasing in the farm's pledgeable resources per dollar a and, because g depends on μ and a only through $\mu + a$, r_j^* is strictly decreasing in the mean return μ with the same derivative. (ii) $\dfrac{\partial g}{\partial \sigma_j} = (1-\ell_j)(1+r)\,\bar{z}\,\dfrac{\phi(\bar{z})}{\sigma_j} + \ell_j\big(\phi(z_0) - \phi(\bar{z})\big) < 0$, so r_j^* is strictly increasing in payoff uncertainty σ_j. (iii) r_j^* is strictly increasing in the cost per dollar $1 + w_j + F_j/L$. For the small bank, r_s^* is strictly decreasing in the deposit base D. (iv) The bank is willing to lend at some rate if and only if $\max_r g(r, a, \sigma_j) \geq 1 + w_j + F_j/L$, and the left side is strictly increasing in a, so farms whose pledgeable resources fall below a threshold cannot borrow at any rate, and the threshold falls as the cost per dollar falls.*

Proof. (i) Raising a by da has two effects. It moves the default boundary down, transferring probability mass $\phi(\bar{z})\, da/\sigma_j$ from default to repayment. At the default boundary the farm's resources equal the repayment $(1 + r)$ per dollar, so each transferred state gains $(1 + r) - \ell_j(1 + r) = (1 - \ell_j)(1 + r)$. And it raises the resources recovered by $\ell_j\, da$ in every default

[26] The probability of default is below one half at every break-even rate shown in the figures. The largest default probability at any plotted break-even rate is about 9%, which is at the low end of the mean return range in Figure IA.9.

state with positive resources, an event of probability $\Phi(\bar{z}) - \Phi(z_0)$. At the boundary z_0 the farm's resources are exactly zero, so states crossing z_0 contribute nothing. Summing these effects gives the stated derivative, which is strictly positive. Because g depends on μ and a only through $\mu + a$, the derivative in μ is identical. By the second regularity condition and the implicit function theorem, $dr_j^*/da = -(\partial g/\partial a)/(\partial g/\partial r) < 0$, and similarly with μ. (ii) Differentiating g in σ_j and simplifying with the identities $\mu + a + \sigma_j \bar{z} = 1 + r$ and $\mu + a + \sigma_j z_0 = 0$ gives the stated expression. Under the first regularity condition $\bar{z} < 0$, so the first term is strictly negative, and $z_0 < \bar{z} < 0$ implies $\phi(z_0) < \phi(\bar{z})$, so the second term is strictly negative as well. (iii) and (iv) By the second regularity condition $\partial g/\partial r > 0$ at r_j^*, so the implicit function theorem gives $dr_j^*/d\theta = \big(\partial k/\partial\theta - \partial g/\partial\theta\big)/\big(\partial g/\partial r\big)$ for any parameter θ, where $k = 1 + w_j + F_j/L$ is the cost per dollar. k is increasing in w_j and F_j/L and, for the small bank, decreasing in D since $w' > 0$. The participation statement follows because $\max_r g$ inherits the strict monotonicity of g in a by the envelope theorem. Moreover, $\max_r g$ exists because g is continuous and tends to $\ell\, E[\max(0, a + X/L)]$ as $r \to \infty$. □

IA.3 How Subsidies and Scale Increase Borrowing

I first study how fixed payments increase borrowing capacity, allow farms that could not borrow at any interest rate to obtain credit, and increase the gains from consolidation. To isolate the borrower side of the model, I set aside bank heterogeneity and consider a single competitive lender with recovery rate ℓ, payoff uncertainty σ, fixed underwriting cost F, a constant cost of funds normalized to zero, and no collateral value ($q = 0$). The farm's pledgeable resources per dollar are its cash, $a = \pi$. Proposition 1 states the demand-side results.

PROPOSITION 1 (Payments, size, and the terms of credit): *Assume the regularity conditions stated above hold. (i) A farm with more base acres per dollar loaned (higher π) borrows at a strictly lower equilibrium rate, and the highest lending cost F at which it can borrow at all is strictly larger, so farms with few base acres are rationed first. (ii) Holding π fixed, a larger farm (higher S) borrows at a strictly lower rate, and among farms facing the same lending cost, small farms are rationed first. (iii) A merged farm that doubles the loan and the lending cost and raises the mean return to $\mu' > \mu$ borrows at a strictly lower rate than the two standalone farms, and it can borrow at lending costs that would ration them.*

Proof. (i) With $q = 0$, $a = \pi$, so Lemma 1(i) gives the rate result and Lemma 1(iv) the participation result. (ii) Raising S at fixed π leaves a and σ unchanged and lowers the cost per dollar F/L. Lemma 1(iii) gives the rate result, and participation expands because the right side of the condition in Lemma 1(iv) falls while the left side is unchanged. (iii)

Doubling L and F leaves the cost per dollar F/L unchanged, and $\mu' > \mu$ raises g pointwise by Lemma 1(i), so the break-even rate falls and participation expands. □

Let us study the bank's decision problem first, then provide intuition for each of the three parts of the proposition. Figure IA.1 illustrates the bank's decision problem. It plots, against the interest rate, the lender's expected payoff net of the loan amount, together with the two components that generate it. The repayment part is the payment $L(1 + r)$ the lender receives per solvent state, net of the loan. It increases one for one with the interest rate but is earned in fewer states, because a higher rate makes default more likely, as the default probability line shows. The recovery part is the liquidation value $\ell(A+X)$ the lender expects per default state, net of the loan. It is a loss, because a defaulting farm's resources fall short of the amount owed and the lender recovers only a fraction of them. The loss narrows as the rate increases, because farms with larger payoffs enter the default region, but a recovery always remains worth less than a full repayment. The expected payoff averages the two parts, weighted by the probability of default. Raising the rate enlarges the repayment while shifting probability mass from repayment states to loss-making default states. Beyond a point the shift dominates and the expected payoff falls. This hump is why participation can fail. If the peak of the expected payoff falls short of the lending cost, no interest rate makes the loan profitable.

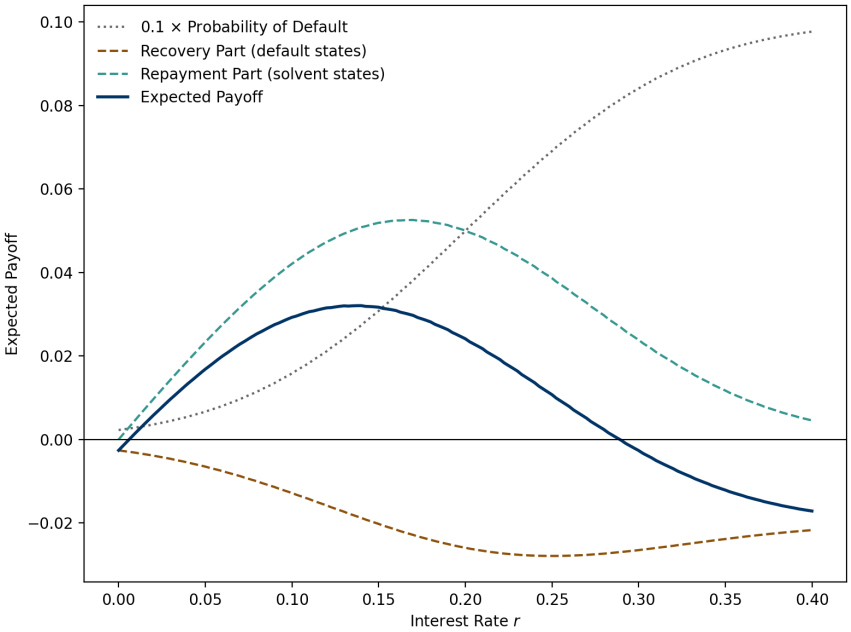


Figure IA.1. Lender's Expected Payoff by Interest Rate

This figure plots the lender's expected payoff, net of the loan amount, as a function of the interest rate, together with its two components. These are the repayment part, the payoff per solvent state, and the recovery part, the payoff per default state, each net of the loan amount. The expected payoff is the average of the two parts weighted by the probability of default. The dotted line plots one tenth of that probability. The parameter values are $L = 50{,}000$, $\pi = 0.15$, $\mu = 1.05$, $\sigma = 0.1$, and $\ell = 0.8$.

Figure IA.2 illustrates part (i) of Proposition 1. Because fixed payments are tied to base acres, a farm's cash holdings A correspond to its base acreage. The bank's payoff before lending costs is increasing in A, since a farm with more base acres has more states in which it does not default, and yields a larger recovery in the default states that remain, because the liquidation value increases with the farm's cash. These are the two effects that appear in Lemma 1(i). The dotted horizontal lines indicate the level of lending costs, and the x values of the solid dots denote equilibrium interest rates. When the cost of lending is high, there is no interest rate at which the lender would lend to the farm with low base acreage, because profit would be negative at any rate. Such farms are credit rationed. When the cost is sufficiently low, the bank is willing to lend to farms with low base acreage, but only at a higher interest rate than for farms with greater base acreage.

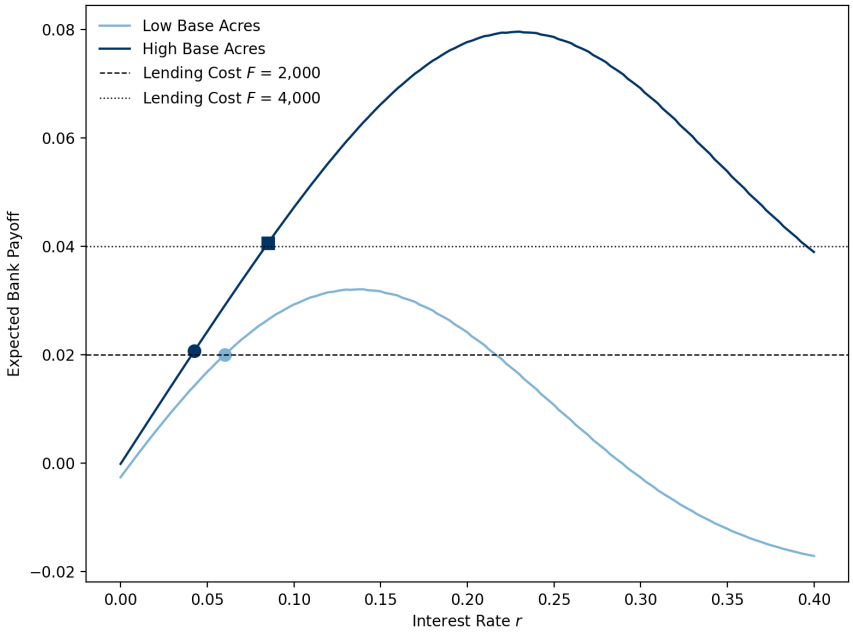


Figure IA.2. Bank Payoffs by Base Acreage

This figure shows bank payoffs as a function of rates for farms with a large and a small number of base acres. π is set to 0.15 (light blue) and 0.25 (dark blue). The other parameter values are L = 50,000, $\mu = 1.05$, $\sigma = 0.1$, and $\ell = 0.8$. Two lending cost levels, F = 2,000 and 4,000, are considered. The solid dots mark equilibrium rates.

Figure IA.3 illustrates part (ii) of Proposition 1 with the analogous comparison for farms of different sizes. Expected lender payoff per dollar increases with farm size because larger farms undertake larger projects, over which the fixed cost of lending is spread. Because subsidies are tied to land, larger farms also mechanically hold more base acreage and more cash, so pledgeable resources per dollar of loan remain constant as the loan increases. As a result, sufficiently small farms are credit rationed when the cost of lending is high, holding lending costs fixed across farms. When the cost of lending is sufficiently low, the bank is willing to lend to both small and large farms, but charges the large farm a lower equilibrium rate. This result suggests that larger farms are better positioned to secure cheaper financing after the introduction of fixed subsidies, allowing them to fund larger upfront investments such as farmland acquisitions.

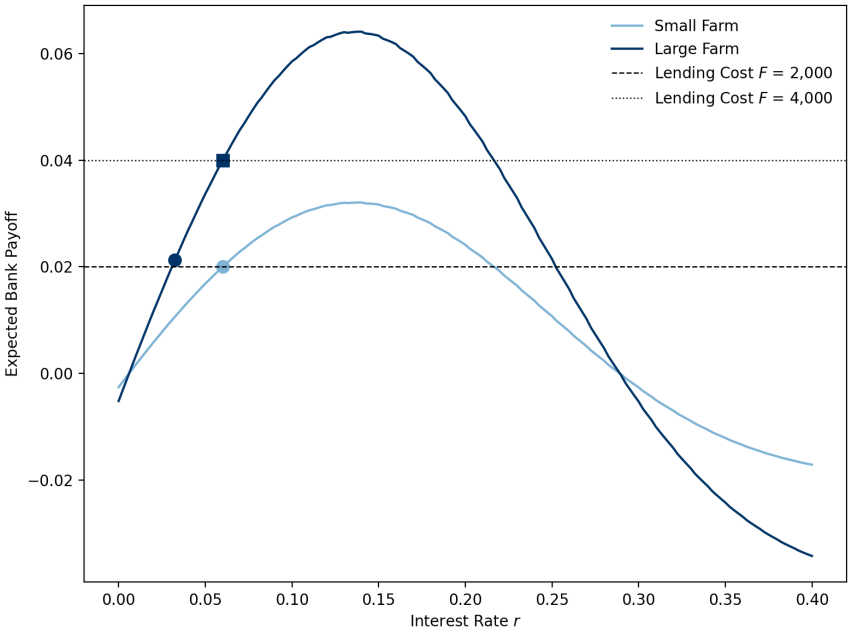


Figure IA.3. Bank Payoffs by Farm Size

This figure shows bank payoffs as a function of rates for a large and a small farm. L is varied from 50,000 to 100,000. The other parameter values are $\mu = 1.05$, $\sigma = 0.1$, $\pi = 0.15$, and $\ell = 0.8$. Two lending cost levels, $F = 2{,}000$ and 4,000, are considered. The solid dots mark equilibrium rates.

Figure IA.4 illustrates part (iii) of Proposition 1. I compare the bank's decision to lend to two identical standalone farms with its decision to lend to a merged farm that operates with increasing returns to scale, modeled as a higher mean return μ. The merged farm's project requires double the upfront investment and incurs double the lending cost, $2F$. Holding the total lending cost at $2F$ is conservative, since any saving in the cost of lending to one merged farm rather than two separate farms would only strengthen the result. The bank's expected payoff from lending to the merged farm exceeds the combined expected payoff from lending to the two farms separately, so for a given lending cost the merged farm is less likely to be credit rationed and borrows at a lower equilibrium rate.

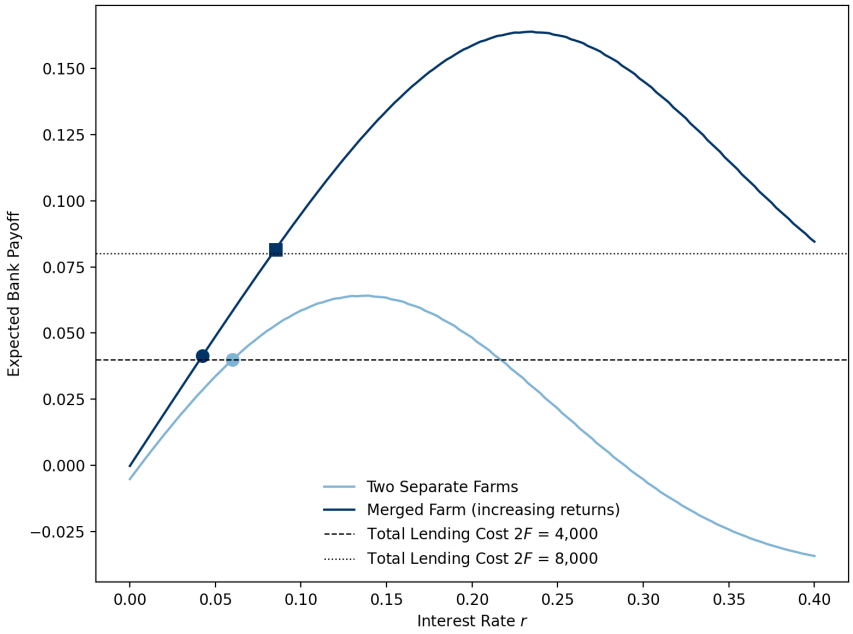


Figure IA.4. Bank Payoffs under a Merger

This figure compares expected bank payoffs under a merger with increasing returns to scale to payoffs when the bank lends to two farms separately. $\mu = 1.05$ when the farms are separate. Under a merger, $\mu = 1.1 \cdot 1.05$. Each farm has $L = 50{,}000$. The other parameter values are $\sigma = 0.1$, $\pi = 0.15$, and $\ell = 0.8$. Two total lending cost levels, $2F = 4{,}000$ and $8{,}000$, are considered, and the total cost of lending to either the two separate farms or the merged farm is the same.

IA.4 Empirical Implications for the Demand Side

Proposition 1 generates the demand-side predictions tested by two results in the main paper. The first-stage estimates show that farm concentration increases more in counties receiving larger fixed payments, as part (i) predicts when payments increase pledgeable resources per dollar, $a = \pi$, and ease the terms of credit. A within-county one-standard-deviation increase in subsidy amount increases farm concentration by 105–107 acres (Table 4). The event-study estimates show that farm concentration increases gradually after payments begin, accumulating to about 650 acres over 15 years (Figure 3). Part (iii) predicts that merging farms obtain cheaper financing when consolidation increases mean returns. Increasing returns in farming make this force economically plausible (MacDonald, Korb, and Hoppe, 2013; Key, 2019), but it is not required for the main prediction. Evidence that farms with more base acres obtain lower interest rates on operating loans also supports part (i)'s rate prediction (Kropp and Whitaker, 2011).

IA.5 Examining Four Channels

I next study which lenders serve borrowers as farms consolidate. Subsidy payments may affect local banking markets through four parameters in the model. Farm size increases S. The capitalization of payments into farmland increases the collateral price q. Payments deposited in local bank accounts increase the deposit base D. Increasing returns increase the mean return μ. I study each parameter in turn. The scale channel benefits large banks because larger borrowers demand loans that small banks are less able to fund from their core deposits. The deposit channel benefits small banks because deposit inflows increase the loans they fund cheaply. The collateral and production channels increase the resources recovered in default by both banks, so their effect on bank choice depends on differences in liquidation technology.

The following proposition characterizes the effect of S on the equilibrium.

PROPOSITION 2 (Scale and funding channel): *Hold q, D, and μ fixed and let farm scale S vary, with $L = \pi_2 S$. Suppose that wherever the two banks quote the same rate, the small bank's cost of funds increases with the loan at least as fast as its average fixed cost falls, $w'(L/D)/D \geq F_s/L^2$. Then (i) the large bank's break-even rate is strictly decreasing in S, and if the small bank offers the lower rate for some farm scale, there is a unique cutoff $\bar{S}(q, D)$ such that the small bank lends to all farms with $S < \bar{S}$ and the large bank to all farms with $S > \bar{S}$. (ii) $\bar{S}(q, D)$ is strictly increasing in D, so better funded small banks retain larger borrowers. (iii) Across markets with the same pledgeable resources per dollar of loan, $a = \pi + q/\pi_2$, $\bar{S}$ is strictly decreasing in the loan each acre requires, π_2. Where farming is more capital intensive, the large bank's break-even rate falls below the small bank's at smaller farm scales S.*

Proof. Fix q. Then $a = \pi + qS/L = \pi + q/\pi_2$ does not vary with S, so scale moves the break-even rates only through the costs per dollar F_j/L and w_j and through $\sigma_s = \sigma - \delta(S)$. For the large bank only F_ℓ/L moves, and it falls, so r_ℓ^* is strictly decreasing in S by Lemma 1(iii). Let $H(S) = r_s^*(S) - r_\ell^*(S)$. By the implicit function theorem, at any scale where the two banks tie,

$$H'(S) \;=\; \frac{\pi_2\left(\dfrac{w'(L/D)}{D} - \dfrac{F_s}{L^2}\right) + \left(-\dfrac{\partial g}{\partial \sigma}\right)\left(-\delta'(S)\right)}{\partial g_s/\partial r} \;+\; \frac{\pi_2 F_\ell/L^2}{\partial g_\ell/\partial r}.$$

The first numerator is nonnegative by the funding-slope condition and since $\delta' < 0$ and $\partial g/\partial \sigma < 0$ by Lemma 1(ii), and the second numerator is strictly positive. Both denominators are positive by the second regularity condition. So H is strictly increasing at any zero, which means H has at most one zero and crosses from below. This monotonicity makes a crossing

unique, and it remains to show that one exists. By part (i), r_ℓ^* is strictly decreasing in S, hence bounded above by its value at the smallest scale. For the small bank, $w(\pi_2 S/D)$ increases without bound while the payoff per dollar is bounded, $\max_r g \leq E[\max(0, a+X/L)]$, a finite bound that does not depend on S, so past a finite scale the small bank cannot break even at any rate. A bank that declines to lend quotes $r = \infty$. Assume $\max_r g_\ell > 1 + w_\ell$, so the large bank participates at all sufficiently large S, since its participation is monotone in S because F_ℓ/L falls while $\max_r g_\ell$ is constant in S. Then $H > 0$ past the small bank's exit scale immediately. If H is still negative just before exit, the cutoff is the exit scale itself. If $H < 0$ somewhere, continuity then delivers exactly one crossing, which defines $\bar{S}(q, D)$. (ii) Raising D strictly lowers r_s^* at every scale S by Lemma 1(iii) and leaves r_ℓ^* unchanged, so H falls pointwise. Since H crosses zero from below, the crossing moves right. (iii) Holding a fixed, the gap in costs per dollar is $w(\pi_2 S/D) - w_\ell + (F_s - F_\ell)/(\pi_2 S)$, whose derivative in π_2 is $w'S/D + (F_\ell - F_s)/(\pi_2^2 S) > 0$ because $F_\ell > F_s$. The funding-slope condition $w'/D \geq F_s/L^2$ is algebraically equivalent to $w'S/D \geq F_s/(\pi_2^2 S)$, so at any zero of H, $dH/d\pi_2 = \big(w'S/D - F_s/(\pi_2^2 S)\big)/(\partial g_s/\partial r) + \big(F_\ell/(\pi_2^2 S)\big)/(\partial g_\ell/\partial r)$ has a nonnegative first term and a strictly positive second term. H increasing in π_2 at its zero moves the crossing left. □

Figure IA.5 illustrates parts (i) and (ii) of Proposition 2 by plotting both banks' expected profits against the interest rate. The farm borrows from the bank that quotes the lower break-even interest rate. I compare three situations that change one primitive at a time. Panel A considers a farm with 550 acres, which I refer to as the small farm. Its loan is small relative to the small bank's deposit base, so the small bank has low funding costs and its informational advantage is strongest. It breaks even about 0.8 percentage points below the large bank. Panel B increases farm size to 850 acres, which I refer to as the large farm, holding the other parameters fixed. The larger loan increases the small bank's funding costs, while the large bank draws on its internal capital market and spreads its fixed underwriting cost over more dollars. The ranking reverses, and the large bank breaks even about 1.2 percentage points below the small bank. The dotted line holds the small bank's cost of funds fixed at its core interest rate while retaining the differences in information and underwriting costs. Under this counterfactual, the small bank breaks even near 3%, compared with 5.9% for the large bank. Funding costs therefore produce the reversal in this calibration. Panel C doubles the small bank's deposit base, as would occur following a merger between two equally sized small banks. Lower funding costs restore the small bank's advantage. The small bank loses larger farms when their desired loans become large relative to its deposit base. A deposit inflow or merger relaxes this constraint and allows the small bank to retain these borrowers.

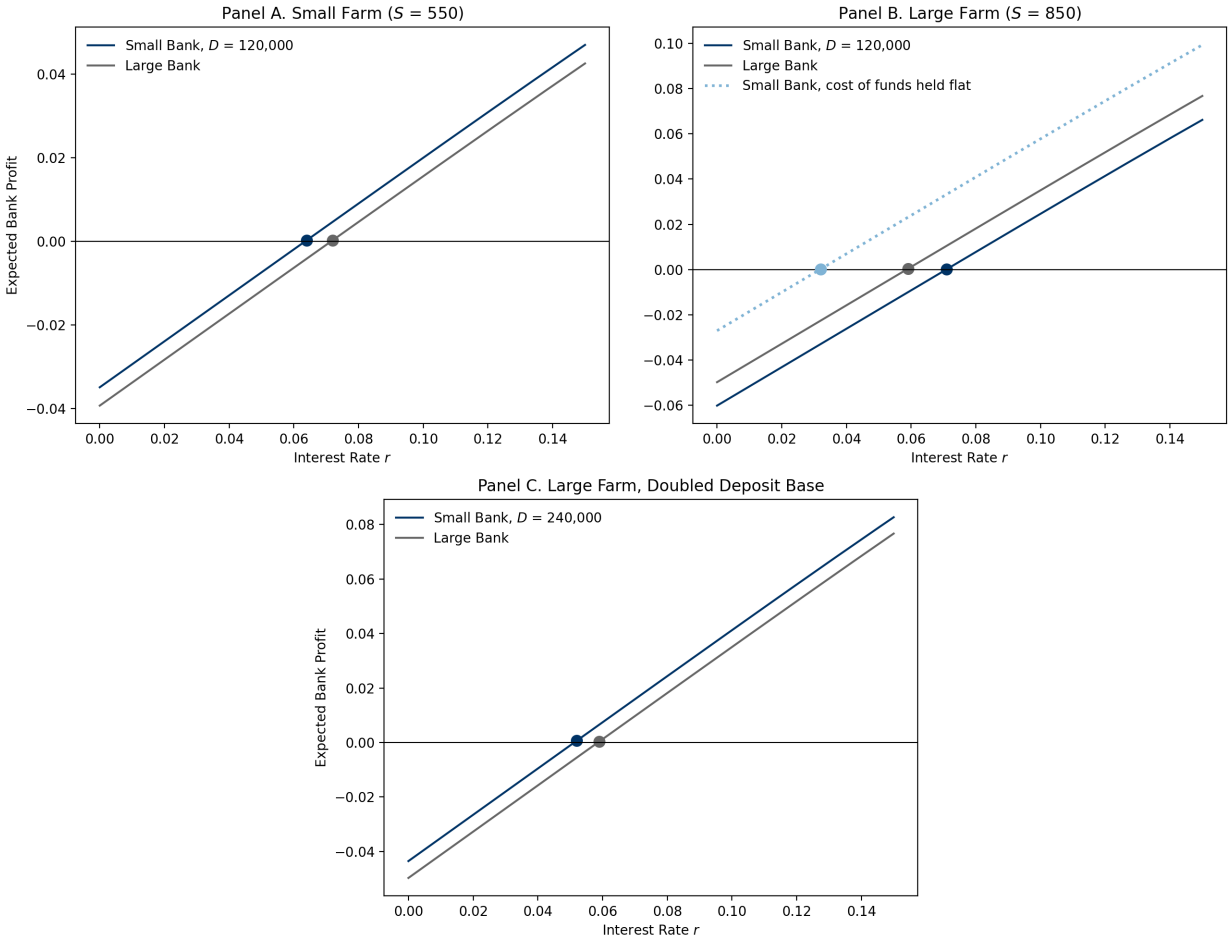


Figure IA.5. Break-Even Rates of Small and Large Banks by Farm Size and Deposit Base

This figure shows expected bank profits, net of funding and fixed underwriting costs, as a function of the interest rate, for the small bank and the large bank. The solid dots mark break-even rates, and the farm borrows from the bank with the lower break-even rate. Panel A considers a small farm ($S = 550$ acres) and a small bank with core deposit base $D = 120{,}000$. Panel B considers a large farm ($S = 850$ acres) with the same deposit base. The dotted line displays the small bank's profits with its cost of funds frozen at the core rate of 2%, shutting off the funding channel while the informational and fixed cost differences remain. Panel C considers the same large farm when the small bank's deposit base has doubled to $D = 240{,}000$, as after a merger of two equal small banks. Parameter values: $\mu = 1.05$, $\sigma = 0.10$, $\pi = 0.15$, $L = \pi_2 S$ with $\pi_2 = \$100$ per acre, collateral per dollar of loan fixed at 0.2, $\delta(S) = 0.02\, e^{-S/500}$, $\ell_s = 0.75$, $\ell_\ell = 0.85$, $F_s = 1{,}000$, $F_\ell = 2{,}000$, $w_s = 0.02 + 0.055\, L/D$, and $w_\ell = 0.035$.

Figure IA.6 illustrates parts (i) and (ii) of Proposition 2 by showing how farm size S and the small bank's deposit base D determine which bank serves the borrower. The figure increases S from 500 to 1,000 acres while holding the land price q fixed, so pledgeable resources per dollar remain constant. Farm size affects break-even interest rates only through average underwriting costs F_j/L, the small bank's funding costs $w(L/D)$, and its informational advantage $\delta(S)$. The small bank's break-even interest rate increases with farm size, while

the large bank's interest rate decreases. When $D = 120{,}000$, the two interest rates cross at a cutoff $\bar{S}$ near 660 acres. Increasing D to 180,000 moves $\bar{S}$ to about 870 acres. When the small bank's cost of funds is instead held fixed, its break-even interest rate decreases from 4.0 to 3.0% and never crosses the large bank's interest rate. The cutoff $\bar{S}$ is the farm size at which higher funding costs offset the small bank's informational and underwriting advantages. A larger deposit base moves this cutoff to a larger farm size.

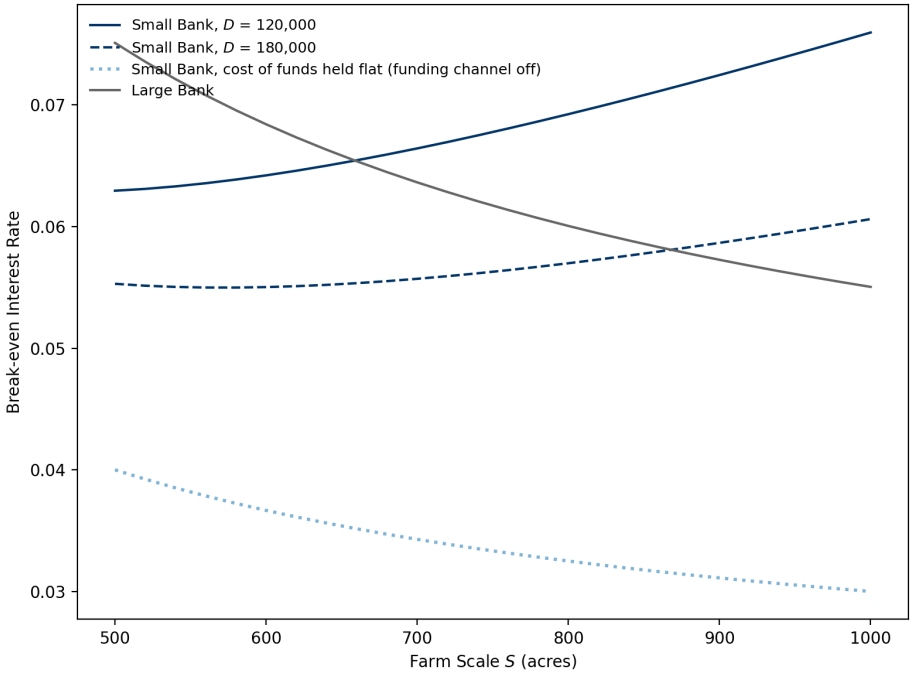


Figure IA.6. Break-Even Interest Rates by Farm Scale

This figure plots each bank's break-even interest rate against farm scale S at a fixed collateral value, for two levels of the small bank's core deposit base D. The crossings mark the cutoff $\bar{S}(q, D)$. The dotted line displays the small bank with its cost of funds frozen at the core rate of 2%, so that the funding channel is shut off while the informational and fixed cost differences remain. Its rate falls with scale and never crosses the large bank's. Parameter values are as in Figure IA.5, with collateral per dollar fixed at 0.2.

The second comparative static varies the price of farmland q instead of the scale of the farm S. Because both the pledged land qS and the loan size L scale with acres, pledgeable resources per dollar of loan, $a = \pi + q/\pi_2$, are the same at every farm size. An increase in the land price q raises pledgeable resources per dollar by the same amount on every loan, large or small. It does not change the deposit base D or the cost of funds w. Its effect on a bank's break-even rate depends on the rate the bank charges, and that rate is higher when the bank's cost of funds is higher.

Lemma 1(i) splits the payoff gain from an extra dollar of pledgeable resources into two effects. The rescue effect is the gain from marginal borrowers pulled out of default, worth

the full repayment net of the forgone recovery, $(1-\ell_j)(1+r)$, per rescued state. It is stronger for the low-recovery bank. The recovery-value effect is the gain of ℓ_j per additional dollar of resources in the default states that remain. It is stronger for the high-recovery bank.

PROPOSITION 3 (Collateral channel): *Hold S, D, and μ fixed and increase the price of farmland q, which raises the farm's pledgeable resources per dollar $a = \pi + q/\pi_2$ for both banks. Then (i) both break-even rates strictly fall and the set of farms that can borrow expands. (ii) Near the cutoff $\bar{S}$, when the small bank's informational advantage is negligible, collateral affects which bank serves the borrower through differences in recovery rates. If the two banks have the same recovery rate, collateral has no effect on the allocation. Locally, the difference between the banks' rate changes is proportional to $\ell_\ell - \ell_s$. The direction of the effect depends on whether the recovery-value effect, which makes the bank with the higher recovery rate more competitive, exceeds the rescue effect, which makes the bank with the lower recovery rate more competitive. (iii) The effect operates at every loan size. For the small bank, the decline in the break-even rate from an increase in q is strictly larger when its deposit base D is smaller.*

Proof. (i) This follows immediately from Lemma 1(i) and (iv). (ii) From Lemma 1(i), the payoff gain from collateral is the sum of two terms whose weights, $1-\ell_j$ and ℓ_j, sum to one. The rescue effect $(1-\ell_j)(1+r)\phi(\bar{z})/\sigma$, the value of pulling marginal borrowers out of default, is weighted by $1-\ell_j$ and is stronger for the low-recovery bank. The recovery-value effect $\ell_j(\Phi(\bar{z})-\Phi(z_0))$, the value of the additional dollar of resources recovered in each default state that remains, is weighted by ℓ_j and is stronger for the high-recovery bank. At a cutoff where $r_s^* = r_\ell^*$ and $\sigma_s \approx \sigma_\ell$, the two banks' payoff derivatives $\partial g_j/\partial a$, and likewise the denominators $\partial g_j/\partial r$, differ only through ℓ_j. Write $\Delta = \ell_\ell - \ell_s$ and view dr^*/dq as a smooth function of the lender's recovery rate, holding a, σ at their common cutoff values. The difference between the two banks' derivatives is then $C\,\Delta + O(\Delta^2)$ by Taylor's theorem, where C is the derivative of dr^*/dq with respect to the recovery rate at the common parameters. It vanishes at $\Delta = 0$, and for small Δ its sign is the sign of C, which depends on the net effect of the recovery-value effect and the rescue effect. (iii) Because q enters both banks' profits only through a, which does not vary with farm size, the rate falls at every loan size. For the small bank, D enters only through the cost per dollar $k_s = 1 + w(L/D) + F_s/L$, which is strictly decreasing in D, so a smaller D raises r_s^* by Lemma 1(iii). By the implicit function theorem, $dr_s^*/da = -(\partial g_s/\partial a)/(\partial g_s/\partial r)$, with both derivatives evaluated at r_s^*. Differentiating the expression for $\partial g/\partial a$ in Lemma 1(i) with respect to r gives

$$\frac{\partial^2 g}{\partial a\,\partial r} = \Big[(1-\ell_s)\Big(1 - \frac{\bar{z}\,(1+r)}{\sigma_s}\Big) + \ell_s\Big]\frac{\phi(\bar{z})}{\sigma_s} > 0,$$

and differentiating g with respect to r gives

$$\frac{\partial g}{\partial r} = 1 - \Phi(\bar{z}) - (1-\ell_s)(1+r)\frac{\phi(\bar{z})}{\sigma_s}, \qquad \frac{\partial^2 g}{\partial r^2} = -\Big[1 + (1-\ell_s)\Big(1 - \frac{\bar{z}\,(1+r)}{\sigma_s}\Big)\Big]\frac{\phi(\bar{z})}{\sigma_s} < 0.$$

Both signs use $\bar{z} < 0$, which holds under the first regularity condition. The numerator of $|dr_s^*/da|$ increases with r, and its denominator, which is positive by the second regularity condition, decreases with r. Hence $|dr_s^*/da|$ is strictly increasing in r_s^* and strictly decreasing in D. Because σ_s, ℓ_s, and a do not depend on D, this comparison holds the farm fixed. □

Figure IA.7 illustrates parts (i) and (ii) of Proposition 3 by showing how collateral affects credit availability and the allocation of borrowers between banks. Panel A plots both banks' break-even interest rates against collateral per dollar. In this calibration, the decrease in both interest rates is an order of magnitude larger than the change in the gap between them. Collateral primarily lowers the price of credit and expands credit availability rather than changing which bank serves the borrower. Part (iii) of the proposition is quantitatively large. At a farm scale of 600 acres and collateral per dollar of loan of 0.05, with the remaining parameter values as in Figure IA.5, an increase in collateral per dollar to 0.10 lowers the small bank's break-even interest rate by 1.90 percentage points when $D = 60{,}000$, 0.70 when $D = 120{,}000$, and 0.45 when $D = 240{,}000$. A small bank with a smaller deposit base quotes a higher rate, at which more farms are close to default, so additional collateral prevents more defaults.

Panel B isolates the recovery channel by giving the two banks the same cost per dollar of lending, 1.05, and the same payoff uncertainty, $\sigma = 0.10$. These conditions isolate differences in recovery technology by holding funding and underwriting costs per dollar and payoff uncertainty equal across banks. Only their recovery rates differ. The resulting gap in break-even interest rates measures the effect of differences in recovery rates. The gap when $\ell_\ell - \ell_s = 0.05$ is half as large as when $\ell_\ell - \ell_s = 0.10$ at every collateral level. When the banks have the same recovery rate, the gap is zero and collateral does not move borrowers between them. The gap also approaches zero as collateral increases. With little collateral, default is costly and the bank with the higher recovery rate has an advantage. With abundant collateral, the loan is safe for either bank and differences in recovery rates matter less. Because pledgeable resources per dollar, a, do not vary with farm size, collateral appreciation increases lending at every loan size and at both bank types. Near the cutoff, where the two banks quote the same rate, any change in which bank serves the borrower depends on differences in recovery rates. Intuitively, collateral makes the loan safer for either bank. It affects the relative position of the two banks only through differences in recovery rates, and greater collateral reduces the importance of those differences.

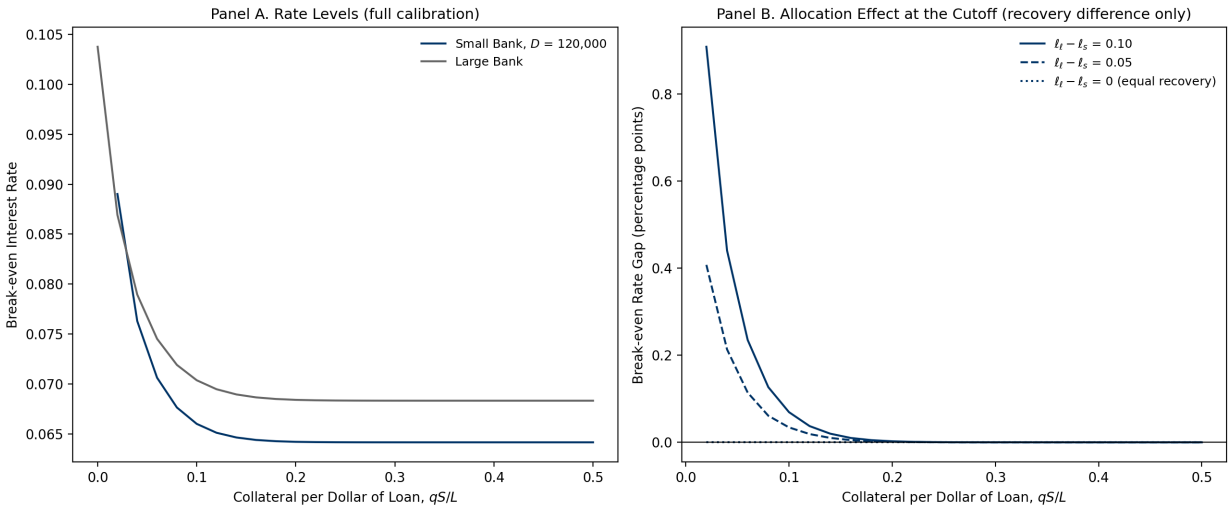


Figure IA.7. The Collateral Channel: Break-Even Rates by Collateral per Dollar

This figure illustrates the collateral channel. Panel A plots the break-even interest rates of the small bank ($D = 120{,}000$) and the large bank against collateral per dollar of loan, qS/L, at a fixed scale $S = 600$ acres, with the remaining parameter values as in Figure IA.5. Panel B plots the gap between the break-even rates of a low-recovery and a high-recovery lender, in percentage points, holding the two lenders' cost per dollar equal at 1.05 and payoff uncertainty equal at $\sigma = 0.10$. The three lines vary the recovery difference, $\ell_\ell - \ell_s = 0.10$ ($\ell_s = 0.75$), 0.05 ($\ell_s = 0.80$), and 0, with $\ell_\ell = 0.85$ throughout. $\mu = 1.05$ and $\pi = 0.15$.

PROPOSITION 4 (Deposit channel): *Hold S, q, and μ fixed, let the small bank's deposit base D vary, and let w be weakly convex. Then (i) an increase in the small bank's deposit base D strictly lowers r_s^*, leaves r_ℓ^* unchanged, and strictly increases the cutoff $\bar{S}(q, D)$. A deposit inflow into small banks expands their credit, allows them to serve previously rationed borrowers, and moves borrowers from large banks to small banks, with no borrower moving in the other direction. (ii) The shift in the small bank's funding cost from a change in its deposit base, $w'(L/D)\, L/D^2$, is increasing in lending relative to deposits L/D, so a given change in the deposit base moves the funding cost most for small banks whose lending is large relative to their deposits. (iii) A decline in D reverses part (i) at every loan size and increases the funding cost reduction, $w(L/D) - w(L/(2D))$, that two small banks obtain by merging and pooling their deposit bases, so the incentive to merge strengthens as D falls. The same comparative static holds as L increases at fixed D: the funding saving from pooling is increasing in lending relative to the deposit base.*

Proof. (i) Lemma 1(iii) and Proposition 2(ii) give the rate and cutoff statements. Participation expands because the cost per dollar $1 + w(L/D) + F_s/L$ falls while $\max_r g$ is unchanged, so farms rationed by the small bank at the previous cost can borrow (Lemma 1(iv)). (ii) Write $x = L/D$. The sensitivity of the funding cost to the deposit base is $|\partial w(L/D)/\partial D| = w'(x)\, x/D$, and $w'(x)\, x$ is increasing in x when w is weakly convex, since both factors are

nonnegative and nondecreasing. (iii) The reversal results from (i) with the sign of the change in D reversed. Regarding the merger claim, the funding saving from pooling two equal deposit bases D into one bank serving the loan is $w(L/D) - w(L/(2D))$, and its derivative in D is $\left[w'(L/D) - \frac{1}{2}w'(L/(2D))\right](-L/D^2) \leq 0$ for weakly convex w, since $w'(L/D) \geq w'(L/(2D)) \geq \frac{1}{2}w'(L/(2D))$. The saving increases as D falls. For the claim in L, the derivative of the saving in L is $w'(L/D)/D - w'(L/(2D))/(2D) \geq 0$ for weakly convex w, since $w'(L/D) \geq w'(L/(2D)) \geq \frac{1}{2}\,w'(L/(2D))$. □

Figure IA.8 illustrates Proposition 4. Panel A holds the large farm from Figure IA.5 fixed and varies only the small bank's deposit base D from 60,000 to 300,000, so any movement in interest rates is attributable to deposits. When deposits determine which bank serves the large farm, three patterns appear. First, the small bank's break-even interest rate decreases from about 11% to below 5% as D increases. Second, it falls below the large bank's interest rate near $D = 180{,}000$, at which point the large farm borrows from the small bank. Third, the curve is steepest at low values of D, where L/D is high, so a given deposit inflow has the greatest effect on small banks closest to their funding limits. Read in the opposite direction, a decrease in D increases the small bank's break-even interest rate and moves the large farm to the large bank.

Panel B illustrates the merger motive in part (iii). It plots the funding savings, $w(L/D) - w(L/(2D))$, from merging two small banks and pooling their deposit bases. The savings increase from less than 1 percentage point when deposits are ample to almost 4 percentage points when deposits are low. A decrease in D increases the benefit of merging. This saving corresponds to the increase in mergers among small banks in the data.

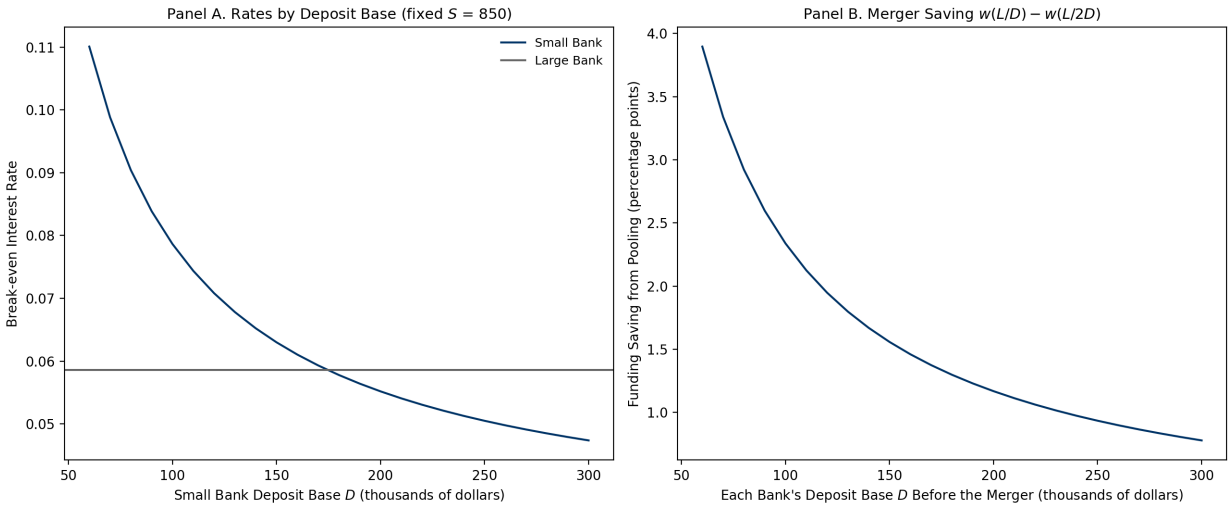


Figure IA.8. The Deposit Channel: Break-Even Rates and Merger Savings by Deposit Base

This figure illustrates the deposit channel. Panel A plots the break-even interest rates of the small bank and the large bank against the small bank's core deposit base D, for the large farm of Figure IA.5 ($S = 850$ acres) with the remaining parameter values as in that figure. Panel B plots the funding saving from pooling two equal deposit bases of size D into one bank serving the same loan, $w(L/D) - w(L/(2D))$, in percentage points, against the pre-merger base D.

PROPOSITION 5 (Production channel): *Hold S, q, and D fixed and increase the mean return μ. Then (i) both small and large bank break-even rates strictly fall and the set of farms that can borrow expands. (ii) Because expected profit depends on μ and a only through the sum $\mu + a$, the effect on the allocation is the same as that of collateral in Proposition 3. It is proportional to the recovery difference $\ell_\ell - \ell_s$, vanishes when the two banks recover at the same rate, and operates at every loan size. As with collateral, an increase in μ lowers the small bank's break-even rate by more when its deposit base D is smaller.*

Proof. Part (i) follows from Lemma 1(i) and (iv). For part (ii), g depends on (μ, a) only through $\mu + a$, so every derivative of the break-even rates in μ equals the corresponding derivative in a, and the result restates Proposition 3(ii) and (iii). □

Figure IA.9 illustrates Proposition 5. It varies only the mean return μ, at a low collateral value so that default risk remains relevant, and shows whether better returns affect the sorting of borrowers between the banks. As μ increases from 1.02 to 1.10, the small bank's break-even interest rate decreases from about 8.9 to 6.6% and the large bank's interest rate decreases from about 8.7 to 7.0%. The gap between the two interest rates never exceeds about 0.45 percentage points and changes sign near $\mu = 1.03$. The large bank quotes the

lower interest rate at low mean returns, while the small bank quotes the lower interest rate at high mean returns.

The effect of μ on which bank serves the borrower is proportional to the recovery-rate difference $\ell_\ell - \ell_s$ and has an ambiguous direction. In this calibration, the rescue effect exceeds the recovery-value effect, so better returns benefit the small bank, which has the lower recovery rate. The production channel predicts a broad expansion of credit but no general reallocation toward either bank type. Because μ and a enter expected profit only through their sum, the production channel shares every lending prediction of the collateral channel. Both break-even interest rates decrease, and any effect on which bank serves the borrower depends on the difference in recovery rates. For the production channel to be empirically relevant, payments must change what farms produce or how. Investment, crop mix, or revenue per acre must change before any lending effect may be attributed to this channel.

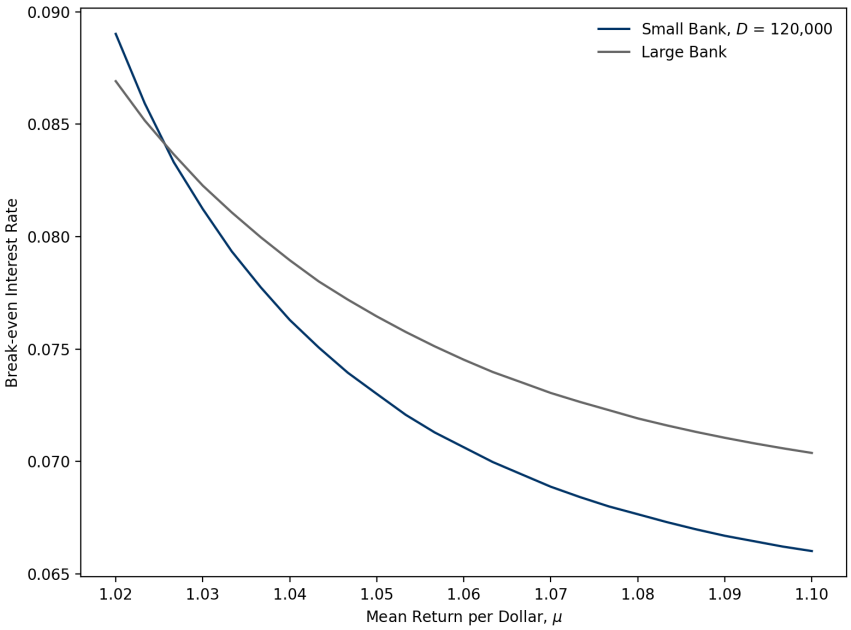


Figure IA.9. The Production Channel: Break-Even Rates by Mean Return

This figure illustrates the production channel. It plots the break-even interest rates of the small bank ($D =$ 120,000) and the large bank against the mean return per dollar μ, at a fixed scale $S = 600$ acres and collateral per dollar of loan fixed at 0.05. The remaining parameter values are as in Figure IA.5.

IA.6 Empirical Implications and Evidence for the Channels

Proposition 2. The Scale and Funding Channel. Proposition 2 generates four empirical implications for the scale and funding channel. First, based on part (i), lending growth should concentrate in large loans. The small bank faces no hard funding limit, but its cost of

funds increases steeply once lending outgrows its core deposit base. Define the corresponding loan-size cutoff as $\bar{L}(q, D) = \pi_2 \bar{S}(q, D)$. Farms whose desired loans exceed $\bar{L}(q, D)$ borrow from the large bank. For typical county deposit bases, originations above \$500,000 are the empirical counterpart of these loans. Second, based on part (ii), within small banks, lending growth should accrue only to banks with the core deposits needed to fund larger loans. Third, based on part (ii) as well as Proposition 4(iii), because $\bar{S}(q, D)$ increases with D, small banks have a reason to merge and pool their deposit bases. Fourth, based on part (iii), large banks should gain the most in counties with a higher π_2, where each acre of farmland requires more credit. The first test below examines the first two predictions together. The second and third tests examine the third and fourth predictions.

The first test compares small banks by their core funding. The test follows naturally from the model, although the model includes only one small bank per market. Both the scale and collateral channels predict that large banks gain when farms consolidate, so the comparison of large and small banks alone does not separate them. Variation in core funding within small banks can separate the two channels. Because the cutoff $\bar{S}(q, D)$ increases with D, consolidation moves the large loans it creates to the small banks with the deposits to fund them, and away from small banks without the deposits to fund them, irrespective of the collateral securing the loan. Under the collateral channel, the ordering within small banks is the opposite. Collateral lowers the break-even rate most at small banks with smaller deposit bases (Proposition 3(iii)), so appreciation should increase lending most at small banks with low-core funding. In the data, a within-county one-standard-deviation increase in farm concentration increases lending above \$500,000 at high-core small banks by about 94% for real estate loans and about 81% for non-real estate loans (Table 9). High-core small banks are those whose pre-subsidy (1990–1995) core deposit share lies above the small bank median. Low-core small banks show no growth, and the two groups do not differ in the smallest origination category. The same effect appears in both real estate and production lending, and in the latter farmland is not pledged, so the effect does not run through collateral. Because this comparison is between small banks that differ only in funding, no force operating through bank size alone can generate it.

The second test concerns bank mergers. Combining two small banks pools their deposit bases, which raises D and with it the cutoff $\bar{S}(q, D)$, while leaving the recovery rate ℓ_s (and likewise the underwriting cost F_s and the information advantage $\delta(S)$) unchanged. Panel C of Figure IA.5 shows that the pooled deposit base lowers the merged bank's break-even rate below the large bank's rate on the large farm. Consolidation gives small banks near the funding cutoff a reason to merge with one another, while under the collateral channel a merger relaxes no binding constraint and so confers no advantage. In the data, mergers

of small banks with other small banks increase by about 23% as farms consolidate, whereas mergers among large banks decline and acquisitions of small banks by large banks do not change (Table 6).

The third test compares markets that differ in capital intensity. The size of the loan that each acre requires, π_2, is a fixed feature of local production and is unaffected by consolidation. For the same increase in S, farms require larger loans where π_2 is higher. The small bank's funding constraint binds at a smaller farm size, and more lending moves to large banks as farms consolidate. I use counties with high and low farming costs to proxy for different values of π_2. The collateral channel predicts the same lending effect in high- and low-cost counties whenever farmland appreciates similarly, because recovery operates through the value of the land rather than through the size of the loan each acre requires. In the data, although high- and low-cost counties experience similar land value appreciation (Figure A.4), large banks gain relative to small banks only in high-cost counties (Table A.18).

Proposition 3. The Collateral Channel. Proposition 3 provides three testable predictions of the collateral channel. First, the channel operates wherever farmland appreciates, whether or not farms consolidate, because the land price q enters bank profits directly through pledgeable resources per dollar a at a fixed farm size. No change in the size distribution is required. Second, its effect appears at every loan size and at both bank types, because a is the same for every farm size. Part (i) shows that collateral appreciation lowers both banks' break-even rates, and part (ii) shows that any difference between the banks is proportional to the recovery difference $\ell_\ell - \ell_s$ and quantitatively small in the calibration. Third, among small banks, the channel lowers break-even rates most at banks with smaller deposit bases, as shown in part (iii), so appreciation should increase lending most at the small banks with the least funding. The tests focus on these three predictions rather than on which bank gains, and none of the three appears in the data.

First, payments have little effect on local banking markets where farms do not consolidate, which is not what the collateral channel predicts. Part (i) implies that both banks' break-even rates fall whenever q increases, so appreciation alone should expand credit and move banking outcomes even at a fixed size distribution. Payments do not increase land values or deposits per capita during the first decade of the program (Figure 3, Panels C and D, and Table A.4), so payments generate no appreciation or deposit inflow through which this prediction could operate. In rugged counties, where payments flow but consolidation is infeasible, payments have little effect on farm concentration, and large banks neither enter nor expand branches or deposits (Table 2). These counties receive payments without borrower consolidation. Second, agricultural lending increases most at large banks (Table 8), not at every loan size

and at both bank types. Third, within small banks, lending above $500,000 increases only at banks with high-core funding (Table 9), whereas the collateral channel predicts the largest gains at small banks with low-core funding. The same split between high- and low-core small banks appears in production lending, where farmland is not pledged and the channel is absent by construction, so the force sorting banks operates through funding rather than through collateral.

Proposition 4. The Deposit Channel. Proposition 4 generates three tests of the deposit channel. First, part (i) predicts that if payment cash accumulates in local accounts, deposits per capita should increase when payments begin, and this cash inflow should strengthen small banks wherever payments arrive. Second, part (ii) implies that a given change in deposits matters most for small banks whose lending is large relative to their deposit base. Third, part (iii) implies that when loan demand increases against a deposit base that does not keep pace, small banks face higher funding costs at every loan size and gain more from pooling deposit bases through mergers.

Deposits per capita do not increase when payments begin. Subsidy payments do not increase deposits per capita during the first decade of the program (Panel D of Figure 3), and the payments are small relative to the base they would need to move: a one-standard-deviation increase in annual payments is $1.15 million, about a quarter of one percent of the median county's deposits. Nor does the predicted inflow appear where it should be easiest to detect. In counties where payments do not consolidate farms, they do not increase the presence, branches, or deposits of either small or large banks (Table 3), and deposits per capita do not increase in any responsiveness tercile (Table A.7). The shift toward large banks occurs only where payments consolidate farms, creating larger credit needs that are more difficult for small banks to fund from local deposits.

The second test jointly considers two funding measures fixed over 1990–1995, core deposit funding and the loans-to-deposits ratio. A lower loans-to-deposits ratio indicates that more deposit funding remains available for additional loans (Acharya and Mora, 2015). Farm concentration increases lending above $500,000 most at small banks with high-core funding and a low loans-to-deposits ratio. Following a one-standard-deviation increase in farm concentration, the difference between these banks and small banks with low-core funding and a high loans-to-deposits ratio is about 140 percentage points for real estate lending and about 322 percentage points for non-real estate lending. The two groups do not differ in real estate lending at or below $100,000 (Table A.17). The same funding effects are visible in prices. A within-county one-standard-deviation increase in farm concentration increases the real rate small banks offer on 12-month certificates of deposit by about 0.30 percentage points, while

large banks lower theirs by about 0.18 percentage points (Table 11). Rates on savings and checking accounts fall at both bank types, consistent with the market power that higher concentration confers on core products. The opposite signs test the model's funding assumption, that small banks fund local lending from local deposits while large banks fund it through internal capital markets. If large banks also funded local lending from local deposits, even at a cost that increases less steeply in L/D than a small bank's, they would still raise their time deposit rates as their lending increases. The model predicts no change in large banks' time deposit rates. That large banks instead lower them indicates that local deposits are not the source from which they fund local lending.

The third test concerns the balance between loan demand and the deposit base. Consolidation increases the size of the loans farms demand while deposits per capita remain flat, so lending needs increase relative to deposits and L/D rises. By part (iii), this raises the small bank's break-even rate on large loans, moves those loans to large banks, and increases the funding saving, $w(L/D) - w(L/(2D))$, that two small banks obtain by merging and pooling their deposit bases. Consistent with this prediction, mergers between small banks increase by about 23% following a within-county one-standard-deviation increase in farm concentration, whereas mergers between large banks decline and acquisitions of small banks by large banks do not change (Table 6). This merger channel does not confound the comparison across small banks in the second test, because that comparison sorts banks on core funding measured over 1990–1995, before payments began. Finally, the dynamics of deposits are inconsistent with the deposit channel itself. If payments increased lending by flowing into local accounts, deposits per capita would increase when payments begin and track the annual payment flow thereafter. Instead, deposits per capita do not increase during the first decade of the program and increase only after 2009, during the farmland boom (Panel D of Figure 3).

Proposition 5. The Production Channel. I find no evidence that subsidy payments operate through the production channel in Proposition 5. This channel requires payments to increase mean returns by changing what farms produce or how they produce it before lending changes. Subsidy payments do not increase crop yields or the share of acreage planted in program crops and decrease machinery stock and tractor investment (Table A.5). Part (ii) implies that a change in mean returns affects which bank serves the borrower only through the recovery difference $\ell_\ell - \ell_s$, which is also how collateral affects bank choice. This effect need not be small. In the calibration of Figure IA.9, the ranking of the two banks flips as μ increases. But in the collateral tests, the effect of recovery differences on bank choice is close to zero. The absence of an increase in production or investment makes it unlikely that

production differences drive the differences between bank types.

Summary. These five propositions and the corresponding evidence point to a single mechanism. Payments increase borrowing capacity and reward consolidation, which the first-stage and event-study results quantify (Proposition 1). As farms consolidate, their larger loans exceed small banks' deposit funding. Within small banks, growth in lending above $500,000 occurs only at banks with greater funding capacity. Small banks merge with one another, and large banks gain most where each acre requires more credit (Proposition 2). Collateral appreciation without farm consolidation does not change local banking market structure (Proposition 3). Deposits per capita do not increase during the first decade of the program. Deposits therefore matter because they limit small banks' funding capacity, rather than because payments generate a deposit inflow (Proposition 4). Last, subsidy payments do not increase production or investment, so the production channel has no positive first stage (Proposition 5).